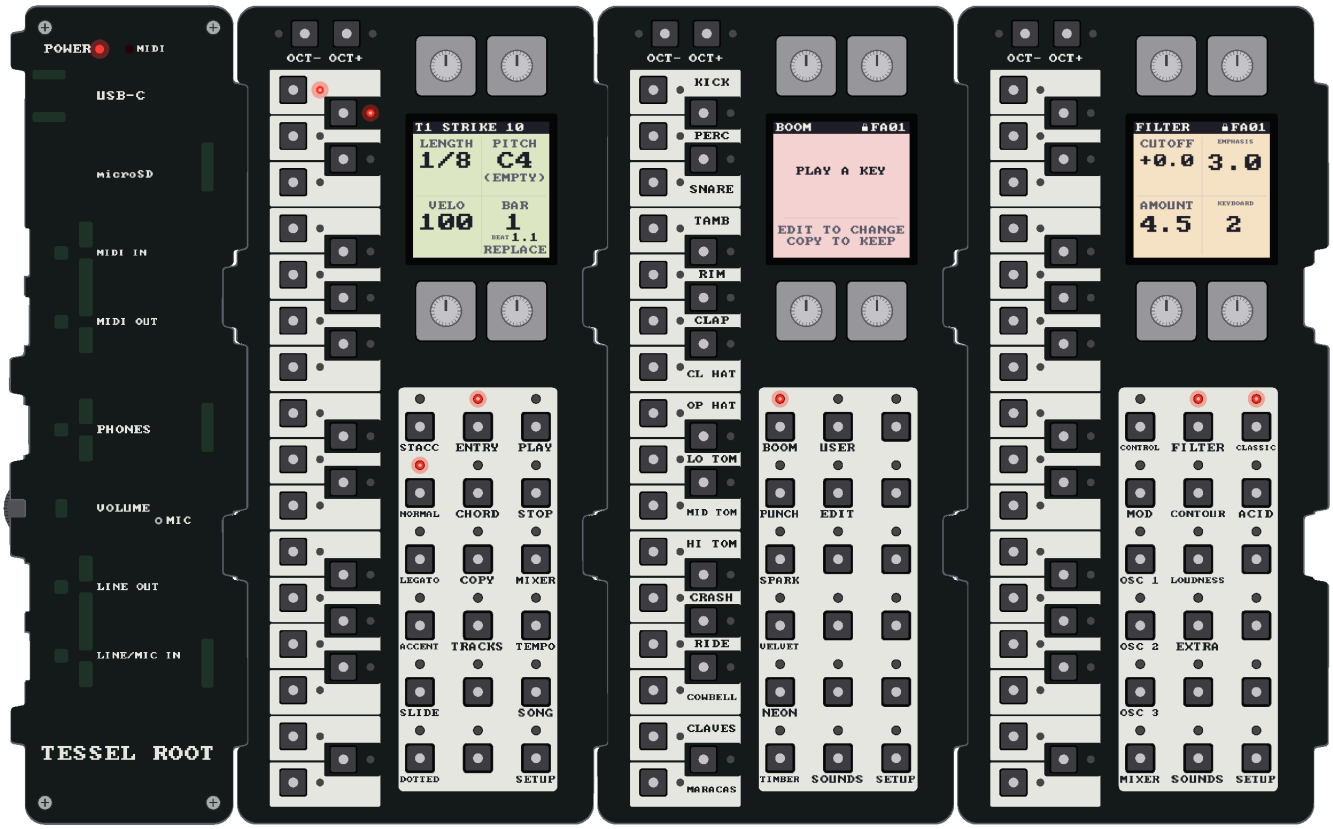
Making Music

The Hidden Craft of Electronic Tracks



UTOPIAN INSTRUMENTS

Draft. Written for firmware 0.9.0.

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What you need

Before you start a chapter, have these to hand.

> **A Root**, the end module: it gives the units their power and mixes their sound together, and your headphones plug into it.

> **Step**, the sequencer. It plays the notes: everything in this book is written on Step and played by it.

> **The units each chapter asks for**, joined in a row to the right of the Root. Each chapter says which near its start, and the table below gathers them all. The first two chapters of music, techno and acid, need only three: Step, Strike (drums) and Volt (bass).

> **Headphones**, or speakers, with an ordinary 3.5 mm plug, into the Root's PHONES socket.

You don't need to own every unit in the table. Every unit is the same box until you choose what it becomes, and you can change it as often as you like: flash a unit as Taste for the hip-hop chapter, then as Grit for ambient. Your sounds aren't lost when you do: they're kept kind by kind, so when you flash it back as Taste, your Taste sounds are there, just as you left them. The Quick Start guide in the box says how to flash a unit.

Chapter	You'll need, with Step and the Root
3 Techno	Strike, Volt. Haze and Glass are optional.
4 Acid	Strike, Volt. Haze is optional.
5 House	Strike, Volt, Ember or Glass, Haze.
6 Hip-hop	Taste, Strike, Volt, and for Track 3 a microphone (or the one built into the Root).
7 Ambient	Glass, Grit, Haze, and Axis for Track 3 (or Glass alone).
8 Jungle and drum and bass	Taste, Strike, Volt, Haze.
9 Synth-pop and synthwave	Glass, Ember, Axis, Volt, Strike, Haze.
10 Step from start to finish	Any one synth to play.
11 Playing live	Pilot, and the units of any chapter before it.

Optional

> **A MIDI keyboard**, for playing parts in by hand and for trying ideas out. Any keyboard with a MIDI OUT socket will do (which cable, below).

> **A microSD card** in the Root, for backing up your songs and sounds, and for loading samples.

> **A computer**, for updating the firmware, and for recording what you make: the Root's LINE OUT goes into a computer's audio input, and its MIDI OUT sends Step's notes and clock to music software.

> **Making Sounds**, the companion to this book, for when you want to build your own sounds instead of choosing the factory ones.

Which keyboard, and which cable

The Root's MIDI IN is a small 3.5 mm socket, like a headphone socket. Plenty of small keyboards have exactly that socket for their MIDI OUT, and a plain 3.5 mm stereo cable joins them. Bigger keyboards have the big round five-pin MIDI OUT socket instead, so you'll need an adapter: a "TRS MIDI Type A" one, which is the commonest kind.

> #### A keyboard with only a USB socket
>
> Look for a keyboard with a MIDI OUT socket. Most small ones get their power from their USB socket, and any phone charger or power bank will do for that: no computer needed.
>
> A keyboard that only has USB can't plug straight into the Root: the Root's USB socket is for a computer (and for the power). Plug the keyboard into a computer, and the computer into the Root, and the computer passes the notes on. Or use a small "USB MIDI host" box, which turns a USB keyboard's notes into an ordinary MIDI OUT.

Before you start - how tracks are made

Most of the music you dance to was made by one person, in a small room, on a few boxes like these.

Put on almost any electronic track and listen to it closely. Underneath, it's surprisingly simple: a drum pattern a bar or two long, a bass line about the same length, perhaps a chord or a hook, all going round and round. What makes it a piece of music rather than a loop is what happens to those few parts over five or six minutes: one drops out and comes back, a filter opens, a delay builds, everything stops for a moment and then crashes back in. That's the hidden craft in this book's title. It isn't writing long tunes; it's knowing what to leave out, and when.

That's also why so much of this music was made by people with no formal training and very little money. Detroit techno and Chicago house were made by teenagers and young DJs on second-hand drum machines nobody else wanted. Jungle was made on cheap samplers in bedrooms. Nobody gave them permission, and few of them had lessons. They learned by doing: they copied the records they loved, worked out by ear how they were made, got it wrong in interesting ways, and kept what worked. Many of them came to know a great deal about music that way, and so will you.

This book does the same thing, with a little more help. Each chapter of music takes one style, tells you where it came from, and then builds three tracks in it with you, each one a step further than the last: the first straight from a recipe, the second with changes, and the third with your own choices. By the end you'll have made techno, acid, house, hip-hop, ambient, jungle and synthwave, and you'll know what makes each one what it is.

> **These tracks are not the point**
>
> Of course, none of the tracks in this book is held up as a great piece of music. In some ways they're the opposite: each one is about the most ordinary, conservative example of its style we could make, so that nothing in it gets in the way of what it's there to show you, and so that your own ideas have a plain, sturdy frame to hang on. Please don't copy them slavishly. That's not because you'd need our permission: you don't, and you never will. It's because you'll come up with much better things yourself.

How to use this book

Chapter 1 is how every unit works, the same whatever it is, and chapter 2 is a short history of electronic music, from the first electronic instruments to the dance floor. Then come the chapters of music, and they're meant to be read in order: each one teaches you something new on Step, and builds on the

chapter before. Techno teaches the basics, acid adds accents and slides, house adds chords and whole songs, and so on.

> ### Two ways through
>
> The chapters of music teach Step a little at a time, as each kind of music needs it, so you're making music from the first page. If you'd rather learn how a tool works before you use it, chapter 10 is Step from start to finish, page by page. Read it first if you like, then come back to chapter 3. Either way works: it's your choice.

Every track uses the units' factory sounds, so you can hear exactly what the book describes. Once a track works, change the sounds: it's the quickest way to make it your own, and Making Sounds, the companion to this book, shows you how to build your own from scratch.

Instrument names in this book (such as TR-808, TR-909 and TB-303, which belong to Roland Corporation, Minimoog, which belongs to Moog Music, and DX7, which belongs to Yamaha Corporation) are trademarks of their owners. They're mentioned for their history. Tessel isn't connected with any of these companies or endorsed by them.

Chapter 1 - Getting around

Everything that's the same on every unit. Learn it once.

Every Tessel unit is the same box. The screen is in the same place, the four encoders are in the same place, the keys run down the same side, and the buttons that matter most, SOUNDS and SETUP, sit in the same spots. What changes from one unit to the next is what the buttons are called and what they do. So this chapter is about the box, and about joining several of them together, which is how music gets made.

If you've read Making Sounds, most of this chapter will be familiar: skim it, and read sections 1.8 and 1.9, which are new. If you haven't, have a unit in front of you as you read: everything here is quicker to learn by doing than by reading.

1.1 How this book tells you what to press

Most of this book is instructions of the "press this, then turn that" kind, and writing them out in full sentences gets tiring for everybody very quickly. So instead they're written as a little strip of coloured shapes, in the order you do them. Here's one:



That reads: press the button marked VCF, then turn encoder 2 (the box on the screen it changes is called RES) until it says 5.0. Every route in the book is built from the same few pieces:

■ VCF	A **button**: press it. White with a small dark square, like a button on the panel's white strip.
■ VECTOR VECTOR ENV	A button with a **page name** on it: press the button until the bar along the top of the screen says that name. Some buttons have two pages, and pressing again flips between them, so you might need one press or two. The name in the pill is the name you'll see at the top of the screen.
■ CHORUS *again*	Press a button **again** to switch something on or off rather than to change page.
■ *hold* OCT+	**Hold** a button down.
❷ RES	An **encoder**, by its number, and the name of the box on the screen it changes: turn it.
push ❶	**Push** an encoder, like a button, and let go.
hold ❹	**Hold** an encoder pushed in until the bar on the screen fills up.
`5.0`	The **value** to set, written exactly as the screen shows it.
C3	A **key**, by its note.

With several units on the go, a route starts by saying which unit it's on, in capitals: on STRIKE, ■ PUNCH.

Apart from the routes, there are three kinds of box, and they always mean the same thing.

> **Try this**
> Something to do with your hands. These are the most important part of the book, and they're short.

> **Watch out**
> A trap, or something that surprises nearly everybody the first time.

> **Keep**
> Something worth remembering after you've closed the book.

1.2 A tour of the box

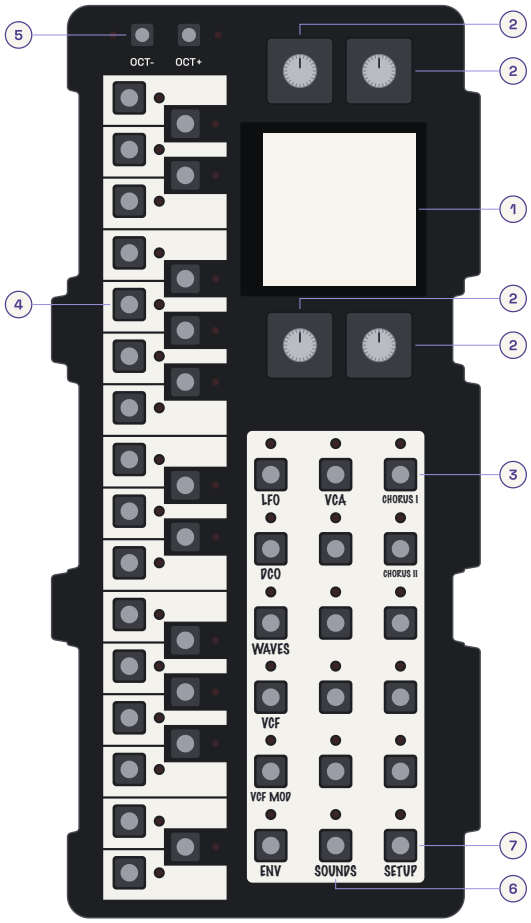


Figure 1.1 Every Tessel unit, from above (this one is Ember). 1: the screen. 2: the four encoders, two above the screen and two below. 3: one of the function buttons (one per page). 4: the keys, down the left. 5: OCT- and OCT+, above them. 6 and 7: SOUNDS and SETUP, the same on every unit.

Everything on the panel is numbered in the picture.

1 **The screen**, at the top right. It shows one page at a time, and every page is laid out the same way, which section 1.5 is about.

2 **The four encoders**, two above the screen and two below. They turn forever in either direction, with no end stop, and they push in like buttons as well.

3 **The function buttons**, the grid of buttons under the screen, three across and six down. Each one takes you to a page: on Ember they're called things like LFO, DCO and VCF, on Step ENTRY, TRACKS and PLAY, and so on.

4 **The keys**, down the left-hand side: white keys along the edge, black keys beside them.

5 **OCT- and OCT+**, the two small buttons above the keys. They move the keys down or up; section 1.4 has the details.

Two of the function buttons are the same on every unit, and they're always in the same place:

6 **SOUNDS**, in the middle of the bottom row, where you choose a sound (section 1.9). Step, which has no sounds of its own, keeps its songs on SONG instead, just above SETUP.

7 **SETUP**, at the bottom right, for the settings that aren't part of any sound (section 1.10).

> ### Your buttons, your labels
>
> If your unit has the plain top panel, the words under the buttons are the ones you wrote on the white strip when you built it. This book uses the names each button has in the firmware, the same ones that appear in the top bar of the screen, so if the book says VCF and your strip says something else, the screen will settle it.

1.3 Two ways to hold it

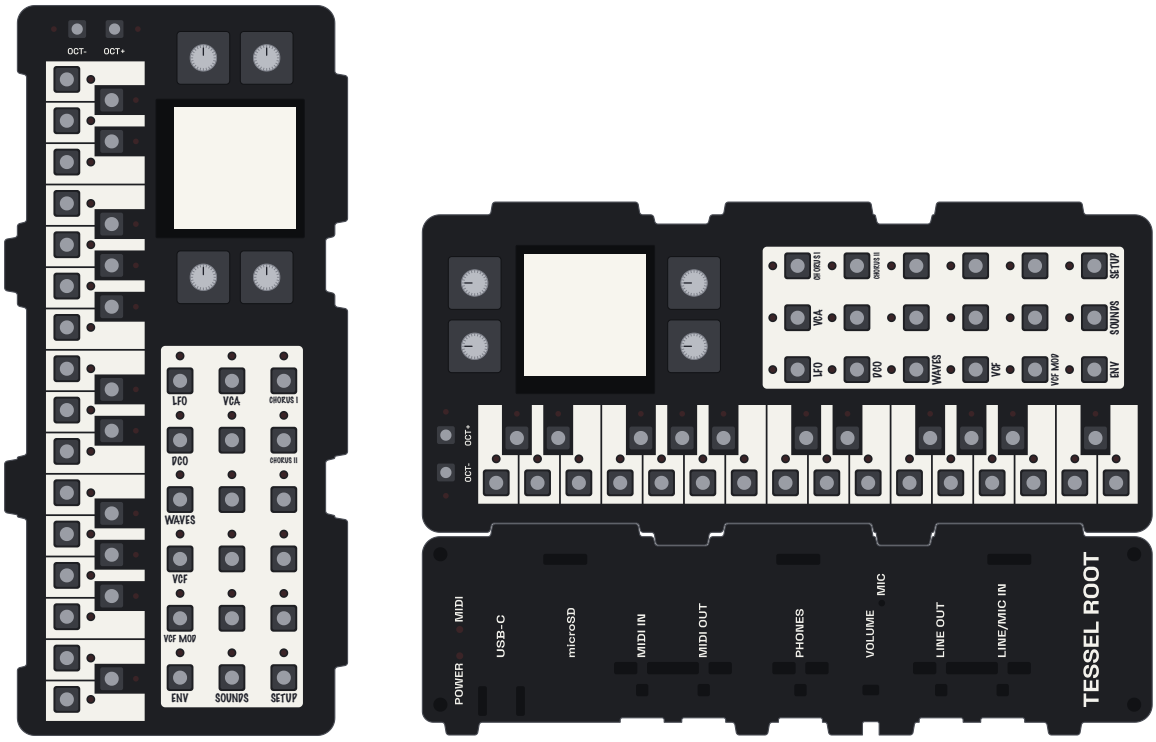


Figure 1.2 Upright, for setting a sound up; turned a quarter to the left, for playing it: the lowest note on the left, the white keys nearest you, like a piano, and the Root along the front.

Upright, the way the pictures in this book show it, is for setting things up: the screen's at the top and the encoders fall nicely under your fingers.

Turned a quarter of the way round to the left, a unit becomes a little piano: the lowest note on the left, the white keys nearest you. That's for playing a sound to hear what it's like. The screen doesn't turn round with it, so it'll be on its side, but you won't be reading it much while you play.

In this book you'll mostly leave your units upright, side by side on the desk, and let Step play them. Their keys are for trying a sound out, and Step's keys are for writing notes, which chapter 3 shows you.

1.4 The keys

There are twenty-seven keys: two octaves and a little bit, from C3 at the top (when the unit's upright) to D5 at the bottom. Middle C, C4, is the eighth white key down, just above halfway. They aren't touch-sensitive, so every note plays at the same loudness however hard you press. A MIDI keyboard plugged into the Root fixes that, and section 1.11 says how.

OCT- and OCT+ move the keys down or up:

- **Tap** one to move the keys an octave, up to three octaves either way. While the keys are moved, the light beside that button is on, so you can always see that you've moved them.
- **Hold** one, and the KEYBOARD page appears for as long as you hold it: encoder 1 moves the keys by octaves, encoder 2 by single semitones, and the screen shows the lowest and highest notes the keys now play.
- **Press both together** and the keys go back to where they started.

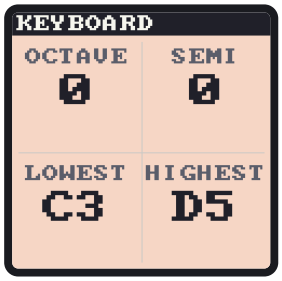


Figure 1.3 Holding OCT+ (or OCT-) brings up the KEYBOARD page: encoder 1 moves the keys by octaves, 2 by semitones, and it shows the lowest and highest notes they now play.

Moving the keys only changes what your fingers play. It isn't part of the sound, and it doesn't change notes coming from Step or a MIDI keyboard. Strike, the drum unit, has no OCT buttons at all, because its keys are drums rather than notes.

1.5 Pages, and the four corners of the screen

Each function button has its own page. Press ■ VCF on Ember and the VCF page comes up, with the filter's settings on it. Press ■ ENV and you're on the envelope page instead. There's no menu to dig through, and nothing to scroll: one button, one page.

A few buttons have a second page, which you get to by pressing the same button again. Press it once more and you're back on the first. You'll meet these as they come up, and they're always written as a route with the page's name in it, like ■ VECTOR VECTOR ENV. Press until the top bar says VECTOR ENV.

Now the most useful thing in this whole chapter. Every page has up to four boxes, one in each corner of the screen, and each box belongs to an encoder. The encoders sit in the same pattern as the boxes: two above the screen, two below.

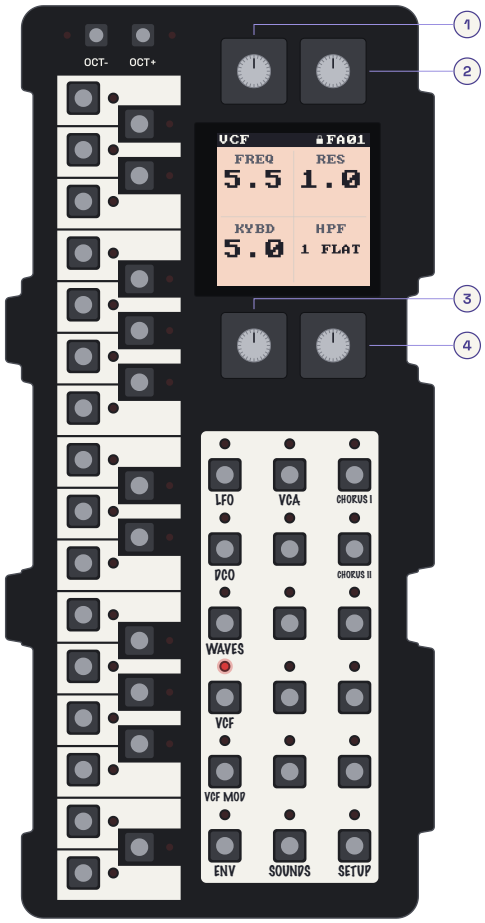


Figure 1.4 The encoders and the screen's four boxes go together: encoder 1 changes the top-left box, 2 the top-right, 3 the bottom-left and 4 the bottom-right. Here, on Ember's VCF page, encoder 2 is RES.

1 Encoder 1, top left, changes the top-left box.

2 Encoder 2, top right, changes the top-right box.

3 Encoder 3, bottom left, changes the bottom-left box.

4 Encoder 4, bottom right, changes the bottom-right box.

That's it. You never have to remember which encoder does what: look at the screen, find the box you want, and turn the encoder in that corner. When you turn one, the name in its box changes colour, so you can see which one you touched last.

> **Try this**
>
> Press any function button. Look at the four boxes, then turn each encoder in turn, a few clicks one way and back again, and watch which box changes. Do it on two or three different pages. After about a minute you'll stop thinking about it, which is the point.

Encoders do three things:

- **Turn.** Slowly, and each click moves the value by one step. Quickly, and it moves several steps a click, so you can cross a big range in one sweep and then slow down for the last bit.
- **Push.** On some boxes, pushing the encoder in does a second job: switching something on and off, or moving to the next letter of a name. The book says when.
- **Hold.** Anything that can't be undone, such as copying over one of your songs or resetting the unit, needs the encoder held in. A bar appears along the bottom of the screen and fills up, and only when it's full does the thing happen.



Figure 1.5 Holding an encoder for something that can't be undone: the bar appears and fills. Let go before it's full and nothing happens.

> **Changed your mind?**
>
> Let go before the bar is full and nothing happens at all. That's what the bar is for.

1.6 The top bar

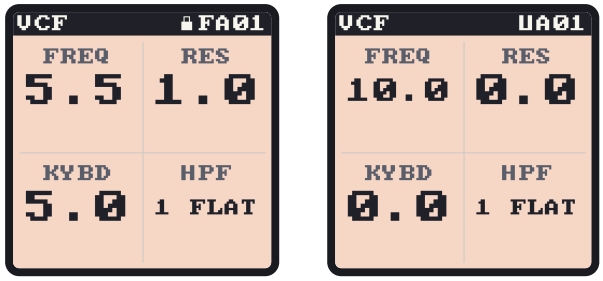


Figure 1.6 The top bar: the page's name on the left, where the sound lives on the right. Left, a factory sound (FA01, with the padlock); right, one of your own (UA01).

Along the top of every page there's a dark strip with two things in it.

On the left, the **name of the page**. It's the same name the button has, and it's the name in the route pills. If you're ever not sure where you are, look here.

On the right, on a synth, **where the sound you're playing lives**, as a short code: FA01, UB07. The first letter is F for one of the factory sounds or U for one of your own; the second letter is which bank (A, B, C…); the number is which sound in that bank. FA01 is the first factory sound, UB07 is the seventh of your own in your second bank.

A small **padlock** next to the code means it's a factory sound: you can change it while you play, but nothing you do is ever kept in it. A small **star** in the padlock's place means the sound has changes that aren't kept anywhere yet. Section 1.9 explains both.

1.7 The lights

Every key and every button has a small red light beside it, and the lights are always telling you something.

- **Bright** beside a function button: that's the page you're on. Beside a key: that note is playing, whether you pressed it or Step is playing it. Beside OCT- or OCT+: the keys are moved that way.
- **Dim** and **blinking** mean different things on different units: on Step, a dim light on a key is a step with a note on it, and a blinking one is the step playing now. Each chapter says what they mean when it matters.
- **Off**: nothing to say.

When you first turn a unit on, the screen shows its name and the firmware version for a moment. Press anything, or turn anything, and it goes away at once.

If the Root carries newer firmware for the unit than the unit has, the next screen says `NEW FIRMWARE`, with the old and new version numbers. Hold encoder 4 until the bar fills to update it: your sounds and songs are kept. Or press or turn anything else to carry on as you were; it will ask again the next time the

unit is turned on.

1.8 Joining units together

A Tessel is a row of units joined side by side, with the Root at the left-hand end. Each unit takes its power from the one on its left and passes it on, and each one's sound travels back along the row to the Root, which mixes them all together for your headphones. A row of joined units is called the **chain**.

- **Joining:** slide a unit up against the right-hand side of the last one, so the edges interlock, and the magnets pull them together. Its screen lights up straight away.
- **Parting:** slide it away. You can do this whenever you like, with everything playing: nothing is lost, because every unit keeps its own sounds and settings, and when you join it on again it comes back exactly as it was.
- **How many:** up to eight units on one Root. How many the Root can give power to depends on what powers the Root: a good USB-C charger runs them all.

Each unit on the chain gets a name made of what it is and a number, such as STRIKE 10, VOLT 1 or GLASS 2. The number is its **MIDI channel**, which is how Step (and a keyboard or a computer) picks which unit it's talking to. Units are given their channels automatically, each its own, and a unit keeps its channel when you take it off and put it back: you don't need to do anything. Strike takes channel 10 by itself, the channel drum machines have used since the 1980s.

The order of the units in the row doesn't matter for anything in this book: Step finds every unit wherever it is.

> ### Try this
>
> Join Step, Strike and Volt to the Root, in any order. Look at each one's screen: SETUP, encoder 1, shows each unit's name and MIDI channel. Now take Volt off and put it back on: it comes back with the same name.

1.9 Choosing sounds

Every synth comes with its own collection of sounds, made at the factory, and room for plenty of your own. They live in **banks** of up to 32 sounds each: one or more factory banks (FAC A, FAC B…) and four banks of your own (USER A to USER D). Every track in this book starts from the factory sounds, so this is the section you'll use most.

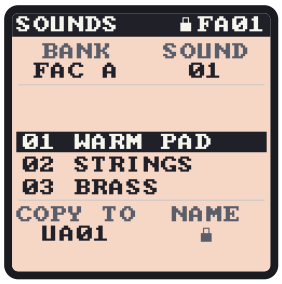


Figure 1.7 The SOUNDS page. Encoder 1 picks the bank, 2 the sound in it, 3 where COPY TO will copy it, 4 its name (in your own banks).

On the SOUNDS page:

1. **BANK**, encoder 1: which bank.
2. **SOUND**, encoder 2: which sound in it. The sound changes the moment you turn it, so you can play a few notes, turn, play again.
3. **COPY TO**, encoder 3: turn it to choose one of your own slots, then hold it in to copy the sound you're playing into that slot.
4. **NAME**, encoder 4: on one of your own sounds, turn it to change a letter of the name, and push it to move on to the next letter.

So choosing a factory sound is always the same route. To choose Volt's DISCO BASS, for instance:

on VOLT, ■ SOUNDS › ❶ BANK › FAC A › ❷ SOUND › DISCO BASS

Two units have quicker ways as well, because they're often changed mid-track:

- **Strike**, the drums: its six factory kits each have a button of their own, down the left-hand column (BOOM, PUNCH, SPARK, VELVET, NEON and TIMBER), so one press changes the whole kit. USER, at the top of the middle column, goes back to the last of your own kits.
- **Haze**, the effects: its white keys are sixteen presets, one press each, and a press sends every synth into that effect at once.

> **Try this**
>
> Go through the first factory bank on Volt, playing a few notes on each sound. Don't be careful about it. Find two or three you like and remember their codes from the top bar.

Changing a sound

Turn something on a factory sound and you'll hear it change. Go ahead: in this book you'll often open a filter or lengthen a sound to make it fit a track. But the factory sound itself never changes. The padlock in the top bar turns into a small **star**, which means "changed, but not kept anywhere". Choose the sound again on the SOUNDS page and it's back as it came: that's your undo.

Your changes go when you choose any other sound, so the first time you turn away from a changed sound the screen says `EDITS NOT KEPT` and stays where it is: copy it if you want to keep it, or turn again to move on. They do survive turning the unit off.

To keep a change, copy the sound into one of your own slots, ■ SOUNDS › 3 COPY TO › `UA01` › *hold* 3. The copy takes your changes with it, and from then on every change you make to it is kept.

> **Songs remember which sound, not the changes**
>
> Step can remember which sound every unit was playing in a song (chapter 5 shows you how), but not changes you haven't kept. If a sound in your song has a star, copy it to one of your own slots first. Step will remind you.

It saves itself

There's no SAVE button anywhere. Every unit, Step included, saves what you've done a moment after you do it. You can take a unit off the chain whenever you like, even halfway through writing a bar, and when you put it back it'll be exactly where you left it.

Your own sounds and songs aren't kept in the unit alone: they're kept on the Root as well, which every unit of the same kind shares. Join a unit to a different Root, a friend's or a new one, and if the sounds aren't the same, the unit asks before anything happens: Making Sounds, section 1.8, has the details.

1.10 SETUP

SETUP, at the bottom right, is where the settings that aren't part of any sound live. It has two parts, and pressing SETUP again (or pushing any encoder) moves on through their pages.



Figure 1.8 SETUP's first page, THIS UNIT: encoder 1 is the unit's MIDI channel, 2-4 its own settings.

THIS UNIT comes first: the settings that belong to this one unit. Encoder 1 is always the unit's MIDI channel (section 1.8), and the others are that unit's own settings, such as how far pitch bend goes, or how much louder Strike plays a step Step has accented. The last THIS UNIT page is **RESET**, which puts things back the way they left the factory, turns the unit into another kind of unit (REFLASH), or tests it. Making Sounds, section 1.9, goes through every choice.



Figure 1.9 The RESET page, the last of THIS UNIT's pages: choose SETTINGS or EVERYTHING, then hold encoder 4 for three seconds.

> ### EVERYTHING means everything
>
> RESET to EVERYTHING wipes your own sounds, and on Step your songs. There's no getting them back, unless you've made a BACKUP to the Root's memory card first (ROOT, encoder 4, below). If you've got a card, do that first.

ROOT comes after THIS UNIT, and it's the settings of the Root, which has no screen or buttons of its own. Every unit's SETUP can reach them, and they're the same whichever unit you use:

- **INPUT** and **GAIN**: which of the Root's inputs units record from (the LINE/MIC socket, or the little microphone built into the Root's top), and how loud. Taste and Grit, the sampling units, have them

on their own recording pages too.

- **MIDI OUT**: what the Root's MIDI OUT sends to a computer or another synth: NOTES (Step's notes, each on the channel of the unit it plays), CLOCK (Step's tempo, with start and stop, so other gear plays in time), BOTH, THRU (a copy of what comes in at MIDI IN), or ALL of them.
- **CARD**: hold encoder 4 to run a job on the Root's memory card. **BACKUP** copies every unit on the chain, your songs included, into a new folder on the card, never over an old one. **RESTORE** puts the newest backup back. **FORMAT** empties the card.

1.11 Playing from a MIDI keyboard

Plug a MIDI keyboard into the Root's **MIDI IN** socket ("What you need", at the start of the book, says which keyboards and which cable) and you can play any unit from it: with the pitch bend and mod wheels, a sustain pedal, and notes that can get louder the harder you play.

On the ROOT2 page of SETUP, MIDI IN decides how the Root passes notes on: **CHANNELS**, where each unit plays only the notes on its own channel (the usual way, with a computer), or **FOLLOW**, where the Root sends your keyboard's notes to whichever unit you last touched. FOLLOW is the easy way to play several units from one keyboard: turn a knob on the unit you want and play.

1.12 When something seems wrong

- **There's no sound.** Check the volume on the Root, and that every unit is joined on properly (its screen will be lit). On Step, check the MIXER page: a unit turned down to OFF or muted plays nothing.
- **A unit plays the wrong notes, or none.** Look at OCT- and OCT+ on it: if one is lit, its keys are moved (Step's notes aren't). Then check which unit Step's track is playing: chapter 3 shows where.
- **There's a star in the top bar.** The sound has changes that aren't kept anywhere. Copy it to one of your own slots to keep them, or choose the sound again to throw them away (section 1.9).
- **You're lost.** Look at the top bar. It always says which page you're on.
- **The screen says `FLASH FULL`.** The unit's memory is full, almost always of samples or songs. Save what you want to keep to the card, then clear what you don't need on SETUP's RESET page, which lists what's using the space.

That's chapter one, and it's the same on every unit, so you'll only ever need to learn it once. Next, a short history of how we got here: from the first electronic instruments to the music in this book.

Chapter 2 - A short history of electronic music

A hundred years of people plugging things in to see what happens.

Every kind of music in this book grew out of the ones before it, and most of them were invented by people misusing machines built for something else. This chapter draws the line from the first electronic instruments to the dance floor. Each chapter of music then picks up its own part of the story in more detail.

2.1 Instruments from electricity

In 1920 a Russian scientist, Léon Theremin, was working on a device for measuring gases when he noticed that the whine it made changed as his hand moved near it. He turned it into an instrument you play without touching: one hand near an upright aerial for pitch, the other near a loop for loudness. The theremin's eerie, swooping voice became the sound of every 1950s film about flying saucers, and the virtuoso Clara Rockmore showed it could play classical music as beautifully as a violin.

In France, Maurice Martenot built the ondes Martenot in 1928: an electronic instrument with a keyboard and a ring on a wire for sliding between notes. Composers took it seriously from the start. Olivier Messiaen gave it a starring part in his huge Turangalîla-Symphonie, first played in 1949, where its voice floats above a full orchestra, and it's still played in concert halls and on film soundtracks today.

These were new instruments, but they were played like old ones: one note at a time, by hand, from a score. The next idea changed what music could be made of.

2.2 Music from recordings

Pierre Schaeffer, an engineer at French radio, recorded steam trains in 1948 and made a piece of music out of the recordings: the Étude aux chemins de fer. He called this **musique concrète**, music made from real recorded sounds, cut up, slowed down, reversed and layered, first on records and then on tape. Any sound could now be music.

In Cologne, Karlheinz Stockhausen went the other way and built music from pure electronic tones, then mixed the two in Gesang der Jünglinge (1956), where a boy's singing voice is woven through electronic sound. At the 1958 World's Fair in Brussels, the Philips Pavilion, designed largely by the composer and architect Iannis Xenakis, played music by Edgard Varèse and Xenakis himself from hundreds of loudspeakers spread across its walls, so the sound moved around the audience.

In Britain the same ideas came out of the BBC Radiophonic Workshop, set up in 1958 to make sound effects and music for radio and television. Its most famous piece is the *Doctor Who* theme of 1963, which Delia Derbyshire built from Ron Grainer's score note by note, out of test-tone oscillators and cut tape, with

no synthesiser at all: there weren't any yet.

Meanwhile, at Bell Labs in America, Max Mathews was teaching computers to make sound. In 1961 an IBM computer there sang "Daisy Bell", the first song ever sung by a computer.

2.3 The synthesiser

The synthesiser as we know it arrived in 1964, when Robert Moog in New York and Don Buchla in California each built a **modular synthesiser**: a cabinet of separate electronic modules, oscillators, filters, envelopes and amplifiers, joined by patch cords, where a voltage could control pitch, brightness or loudness. Making Sounds is about what those modules do; this book is about what people did with them.

Wendy Carlos's *Switched-On Bach* (1968), Bach played on a Moog one painstaking line at a time, became one of the best-selling classical records ever made. Two years later Moog shrank the modular into a portable keyboard, the Minimoog, and synthesisers began moving from studios onto stages and into bands.

The same modulars had sequencers: rows of knobs, one per step, that played a repeating pattern of notes by themselves. In Germany, Kraftwerk built their whole music around machines repeating patterns, on *Autobahn* (1974), *Trans-Europe Express* (1977) and *The Man-Machine* (1978), and Tangerine Dream built long hypnotic pieces from sequenced loops. In 1977 Giorgio Moroder made Donna Summer's "I Feel Love" with a sequenced Moog bass line pulsing under the whole song: a disco hit that sounded like nothing before it, and one many people point to as the moment dance music became electronic.

Brian Eno's *Ambient 1: Music for Airports* (1978) was made partly from tape loops of different lengths drifting in and out of time with each other: music meant to be "as ignorable as it is interesting". Chapter 7 does the same thing on Step.

2.4 Drum machines, and new rules

By the late 1970s cheap electronics put rhythm machines into home organs, and then into boxes of their own. Roland's CR-78 (1978) let you write your own patterns, and in 1980 Roland brought out the TR-808. It was meant to sound like a real drummer, for musicians to practise and make demos with, and it didn't: its kick drum was a long electronic boom, its handclap a burst of hiss, its cowbell like nothing on earth. It sold badly. Then, second-hand and cheap, it fell into the hands of people who loved how it sounded. Afrika Bambaataa and the Soulsonic Force's "Planet Rock" (1982), which borrowed a Kraftwerk tune and laid it over an 808, started **electro**, and the 808's boom has been under hip-hop ever since.

In 1983 the instrument makers agreed on **MIDI**, a common language for electronic instruments, so one keyboard or sequencer could play any synth or drum machine, whoever made it. It's the same language the Root speaks today. In the same years **samplers** arrived, which record any sound and play it back from the keys, and by the late 1980s hip-hop producers were building whole records from short slices of old funk and soul records on machines such as the E-mu SP-1200 and the Akai MPC60. Chapter 6 is about that.

2.5 House and techno

In Chicago in the early 1980s, DJs such as Frankie Knuckles at the Warehouse club played disco, soul and European electronic records, and stretched and re-edited them for the dancers. When disco records dried up, local producers made their own on drum machines and cheap synths: Jesse Saunders' "On and On" (1984) is often counted as the first house record, and the music took its name from the Warehouse. The TB-303, Roland's failed bass-line box, was turned into something new there in 1987, when Phuture turned its knobs while it played and made "Acid Tracks". House is chapter 5, and acid chapter 4.

A few hundred miles away in Detroit, three friends from the suburb of Belleville, Juan Atkins, Derrick May and Kevin Saunderson, took Kraftwerk's cold, mechanical sound, crossed it with funk and electro, and made **techno** on the same cheap machines. In 1988 house and acid reached Britain and set off the "Second Summer of Love": warehouse parties, outdoor raves, and a youth movement the newspapers didn't understand. Techno is chapter 3.

2.6 Everything since

British rave music sped up and broke apart. Producers chopped up old funk drum breaks on samplers, played them at twice the speed of the bass, and around 1992-94 that became **jungle**, and then **drum and bass**: Goldie's *Timeless* (1995) took it into the album charts. That's chapter 8. Ambient came back as the "chill-out room" at raves, where acts such as The Orb played slow, drifting music for dancers to rest to.

Through the 2000s computers became powerful enough to be whole studios, and a new generation made music in bedrooms on laptops: dubstep came out of south London, and electronic music became pop music. Then many of those producers went back to hardware for its feel and its sound, and the synthesisers of the 1980s came back too, in **synthwave**, which is chapter 9.

> ### The thread through all of it
>
> Look back through this chapter and the same thing happens again and again. Someone gets hold of a machine, uses it for something it wasn't meant for, and makes something nobody had heard before: a gas detector that sings, a recording of trains, a practice drum machine, a bass box for guitarists. The machines in this book are made to be used, and misused, in exactly that spirit.

Chapter 3 - Techno

A kick drum on every beat, and the patience to change one thing at a time.

Techno is the simplest music in this book, and one of the hardest to get right. Underneath, a techno track is a kick drum on every beat, a hi-hat between them, a clap, and a bass line a bar or two long, round and round for six minutes. What makes it work is everything around that: which parts play when, how a sound slowly changes, and when to take something away so its coming back hits harder.

That's why it's the first style in the book. You'll learn to write a pattern on Step, to make a short pattern into a long one, and to build a whole track out of a few loops by switching them on and off. Every other chapter uses those three things.

> **What you need for this chapter**
>
> The Root, **Step**, **Strike** and **Volt**, joined in a row, in any order, and headphones. Track 3 can use Haze and Glass as well, but you don't need them.

3.1 Where techno came from

Techno was made in Detroit in the 1980s, by three friends who'd met at school in Belleville, a small town outside the city: Juan Atkins, Derrick May and Kevin Saunderson. Detroit was the Motor City, and by then its car factories were closing and its centre was emptying out. The three of them listened to Kraftwerk and to European synth-pop on the radio, to funk and to Parliament, and to the electro coming out of New York, and they wanted to make music that sounded like the future: cold, mechanical and still funky.

Atkins got there first. With Rick Davis, as Cybotron, he made "Alleys of Your Mind" (1981) and "Techno City" (1984), and then, on his own as Model 500, "No UFOs" (1985). He'd taken the word from Alvin Toffler's book *The Third Wave*, which talked about "techno rebels". Derrick May, as Rhythim Is Rhythim, made "Strings of Life" (1987), all rushing piano chords and strings with no bass line at all, and Kevin Saunderson made "Big Fun" (1988) with Inner City, which went into the pop charts. In 1988 a British record label put their tracks on one album, *Techno! The New Dance Sound of Detroit*, and the music had its name.

They made it on cheap, second-hand machines: Roland's TR-808 and TR-909 drum machines, a few synths, and a sequencer to play them. The 909's kick, short and hard with a click on the front, became the sound of the music, and still is.

The second wave stripped it down. In Detroit, Underground Resistance, with Jeff Mills and "Mad" Mike Banks, made hard, fast, angry records and kept themselves anonymous, and Robert Hood made *Minimal Nation* (1994), which kept only what a track couldn't do without. When the Berlin Wall came down in 1989,

the city was full of empty buildings, and techno moved in: the Tresor club opened in 1991 in an old department store's vault, and Detroit's producers played there and released records on its label. Berlin has been techno's second home ever since.

Today techno is played all over the world, from small basements to festivals, and it comes in every kind, from the deep and dubby to the fast and harsh. What hasn't changed is the kick drum on every beat, and the idea that a few parts, changed slowly, are enough.

3.2 What techno is made of

Put on a techno record and listen for these. Most tracks have every one.

- **The tempo.** Somewhere between about 120 and 140 beats a minute, steady from start to end. Track 1 is 130.
- **A kick on every beat.** Four in a bar, every bar. It's called "four to the floor", and it's the part dancers move to. Everything else fits round it.
- **Hi-hats on the off-beats**, exactly halfway between the kicks: kick, hat, kick, hat. It's the "tss" in "boom tss boom tss", and it pulls the beat forward.
- **A clap or a snare on beats 2 and 4**, with the second and fourth kicks. It gives the bar a shape: one, *two*, three, *four*.
- **A bass line that stays out of the kick's way.** The kick and the bass are both low and heavy, and if they play at the same moment they blur into one. So techno bass lines usually rest on the beat and play in the gaps.
- **Eight bars at a time.** Something changes every eight bars (or sixteen, or thirty-two): a part comes in, a part drops out. Dancers feel it coming even if they've never counted a bar in their lives, and so will you.
- **Slow change.** A filter opening over a minute, a sound getting longer, a part slowly louder. In music this repetitive, a small change sounds big.

Notice what isn't on that list. Most of the music you know moves through **chords**: a hymn, a pop song or a symphony sets off from home, goes somewhere else, and comes back, and the pull to get home is a large part of what carries you along. That journey from chord to chord is a **chord progression**, and it's most of what music theory calls harmony. Techno deliberately throws it out. A track sits on one chord, often on one note, for six minutes, and never goes anywhere in that sense. (Track 1's bass does move, A, F, G, E, but it's a riff going round, not a journey: it never arrives, it just comes round again.) What carries you along instead is the sound itself, changing slowly: parts coming in and dropping out, a filter opening, the same few bars heard a little differently each time round.

That idea didn't start on the dance floor. In the 1960s and 70s a group of American composers, the **minimalists**, came at it from the concert hall. Terry Riley's *In C* (1964) is 53 short phrases that players repeat as many times as they like; Steve Reich built whole pieces from one pattern played against itself as it slowly drifts out of step, as in *Clapping Music* (1972) for two pairs of hands; his *Music for 18 Musicians* (1976) holds each chord for minutes while the patterns shift over it. Philip Glass worked the same way, and John Adams, who came a little later, carried it into opera and song: his "songplay" *I Was Looking at the Ceiling and Then I Saw the Sky* (1995) sets young people's lives in Los Angeles around its 1994 earthquake to numbers built on repeating pop, gospel and funk grooves. Repetition, slow change, and the sense that the music is a place you're in rather than a story being told: listen to any of them next to a techno track and you'll hear the family likeness. The two worlds have met, too: the Orb's "Little Fluffy Clouds" (1990) is built on a sample of Reich's *Electric Counterpoint*.

A word about counting, since the rest of the book does a lot of it. A **beat** is one kick, one tap of your foot, and beats are grouped into **bars**. How many beats a bar has is the music's **time signature**: a waltz counts in threes (3/4), a jig in sixes (6/8), but almost all dance music counts in fours, 4/4, and so does everything in this chapter. In 4/4 each beat is a **crotchet**. A crotchet splits into two **quavers**, and a quaver into two **semiquavers**, so a bar of 4/4 has four crotchets, eight quavers and sixteen semiquavers.

Step's screen writes lengths as fractions, the way sequencers always have: 1/4 is a crotchet, 1/8 a quaver, 1/16 a semiquaver, and 1 a whole bar of 4/4 (a semibreve). The fractions come from the American names, quarter, eighth and sixteenth notes, which you'll meet in plenty of other manuals. Step writes a place in the song as three numbers, `2:4.1`: bar 2, beat 4, the first semiquaver of that beat.

3.3 Track 1 at a glance

Here's the first track, all of it, before you build it. Listen to it first, so you know where you're going.

Sound 3.1 Track 1, as built in sections 3.4 to 3.6: four one-bar loops, brought in and out by muting tracks, and Volt's filter opened by hand from bar 41.

It's made of four one-bar patterns, each on its own Step track:

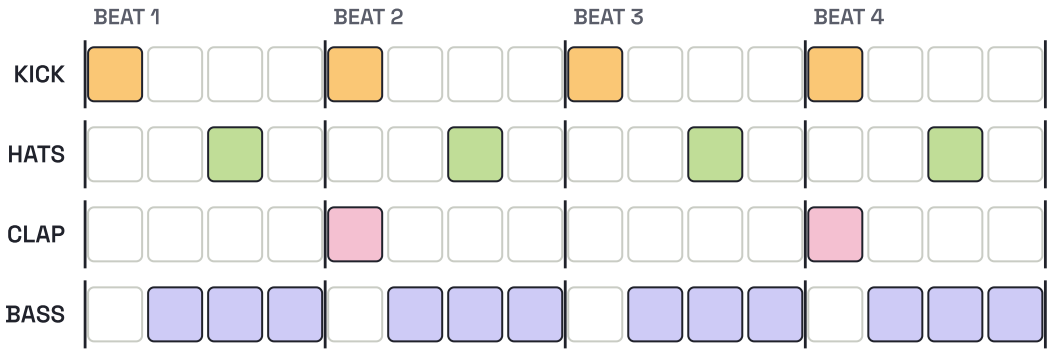


Figure 3.1 Track 1's four parts, one bar each. Each square is a semiquaver; the darker lines are the four beats. The bass rests on every beat, so it never sits on the kick.

That's all the music there is. Sixty-four bars, about two minutes, are made from those four patterns by switching tracks on and off, eight bars at a time:

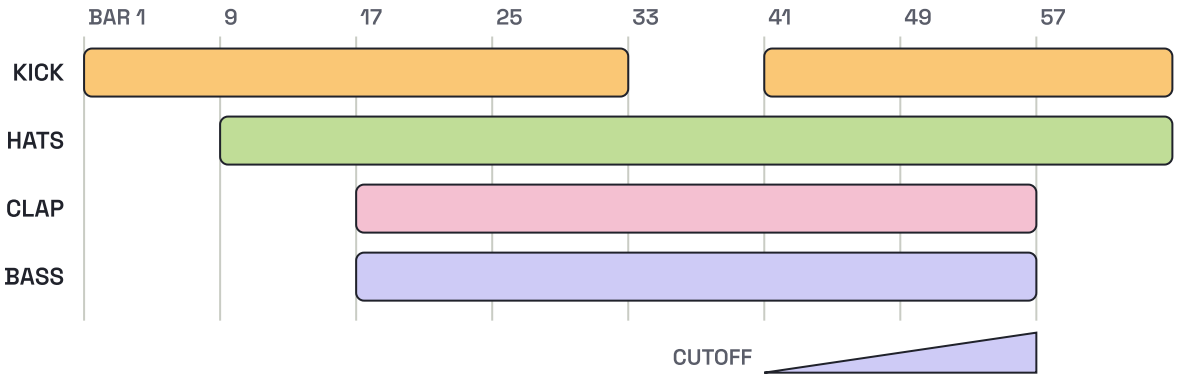


Figure 3.2 How Track 1 plays: sixty-four bars, built from four loops by muting and unmuting tracks. The slope under the bass is Volt's CUTOFF, opened by hand from bar 41.

Track 1

The whole track, as you'll set it up.

STEP	`TEMPO 130`
STRIKE	`PUNCH`
VOLT	`PLUCK BASS (FA09)` `OCT– once`
TRACK 1	`STRIKE` `LOOP 1 BAR` `KICK on every beat`
TRACK 2	`STRIKE` `LOOP 1 BAR` `CL HAT on the off–beats`
TRACK 3	`STRIKE` `LOOP 1 BAR` `CLAP on 2 and 4`
TRACK 4	`VOLT` `LOOP AUTO` `A F G E, 2 bars each`
MIXER	`VOLT –6`

PUNCH is Strike's hard, bright kit, with a short kick that has a click on the front: just what techno wants. PLUCK BASS is a Volt sound whose filter snaps open and shut on every note, so each one is a short, rubbery pluck: if you've read Making Sounds, it's much like the one you built in its section 4.7.

3.4 Setting up

A new song

If Step has a song in it already, start a new, empty one, so nothing old gets in the way. Your old song isn't touched: it stays in Step's list on the SONG page.

on STEP, ■ SONG › 4 ACTION › NEW › *hold* 4

The sounds

1. On STRIKE, press ■ PUNCH.
2. On VOLT, choose PLUCK BASS, ■ SOUNDS › 1 BANK › FAC A › 2 SOUND › PLUCK BASS, then press ■ FILTER, ready for later, and ■ OCT- once. PLUCK BASS is a bass, and with its keys an octave down, the key you'll write it with sits where it should.

The tempo

On STEP, ■ TEMPO › ❶ TEMPO › `130` . A new song starts at 120.

Four tracks

Step has sixteen **tracks**, and each one plays one unit. You can give several tracks to the same unit, and in techno you usually will, one per drum: then each drum is a part of its own, which you can switch off and on by itself, or make a different length. Track 1 will be the kick, 2 the hats, 3 the clap, all on Strike, and 4 the bass, on Volt.

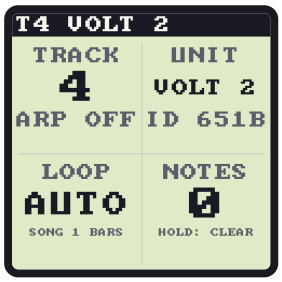


Figure 3.3 TRACKS, with track 4 given to Volt. LOOP AUTO: it plays as long as its notes go (the bass's eight bars). Tracks 1-3 say LOOP 1 BAR.

On the TRACKS page, encoder 1 (TRACK) chooses the track, and encoder 2 (UNIT) chooses which unit it plays, by name:

1. On STEP, ■ TRACKS › ❶ TRACK › `1` › ❷ UNIT › `STRIKE` . The name has a number after it, its MIDI channel: STRIKE 10.
2. ❶ TRACK › `2` › ❷ UNIT › `STRIKE` , and the same for track 3.
3. ❶ TRACK › `4` › ❷ UNIT › `VOLT` .

Step may have given some tracks a unit already: it gives empty tracks to units on the chain that nothing's playing yet. Turn UNIT anyway, until it says what the list says.

How long each track goes round

The third box on TRACKS, LOOP, says how long a track plays before it goes back to its start. **AUTO**, the setting a new track has, means "as long as my notes go": the bass will be eight bars long, so on AUTO it plays all eight bars, then starts again. A set number of bars means "go round every this many bars, whatever else is happening". The drums only need one bar each, so:

1. ❶ TRACK › 1 › ❸ LOOP › 1 BAR
2. The same for tracks 2 and 3. Leave track 4 on AUTO.

Now you write each drum's bar once, and it plays for as long as the song does.

3.5 Writing the patterns

Notes are written on Step's ENTRY page. It works like a typewriter: there's a cursor, the place in the song where the next note goes, and every time you play a note it's written there and the cursor moves on.



Figure 3.4 ENTRY, just after the fourth kick: the cursor (BAR) has moved on to bar 2, beat 1.

1. **LENGTH**, encoder 1: how long the next note is, and so how far the cursor moves on: 1/4 is a crotchet (a beat, here), 1/8 a quaver (half a beat), 1/16 a semiquaver (a quarter of a beat). Push it in for a **rest**: the cursor moves on by that length, leaving a gap.
2. **PITCH**, encoder 2: the note. You won't need it here: the keys choose the note.
3. **VELO**, encoder 3: how hard the note is played. 100 for now; Track 2 uses it.
4. **BAR**, encoder 4: where the cursor is. Turn it to move through the song.

And here's the handy part: you can write with *any* unit's keys, not only Step's. Press KICK on Strike while Step's ENTRY page is showing track 1, and Step writes a kick. The top bar says which track you're writing and which unit it plays: `T1 STRIKE 10`.

The kick

1. On STEP, ■ TRACKS › ❶ TRACK › 1, then ■ ENTRY › ❶ LENGTH › 1/4.
2. On STRIKE, press KICK four times.

Four kicks, one a beat, and the cursor has moved on to the start of bar 2 (BAR says 2, BEAT 1.1). That's bar 1 done, and with LOOP 1 BAR it'll play in every bar of the song.

The hats

The hats go on the off-beats: a rest the length of a quaver, then a hat, four times over.

1. On STEP, ■ TRACKS › ❶ TRACK › 2, then ■ ENTRY › ❹ BAR › 1.
2. ❶ LENGTH › 1/8.
3. Four times: *push* ❶ for a rest, then on STRIKE CL HAT.

> **Back to bar 1 every time**
>
> When you change track, the cursor stays where you left it: at the start of bar 2, after the kick. Turn BAR back to 1 before you write the next track (turn it well anticlockwise: it stops at the start of bar 1), or your hats go into bar 2, and with LOOP 1 BAR, bar 2 is never heard.

The clap

1. On STEP, ■ TRACKS › ❶ TRACK › 3, then ■ ENTRY › ❹ BAR › 1 › ❶ LENGTH › 1/4.
2. Twice: *push* ❶ for a rest, then on STRIKE CLAP.

The bass

The bass plays semiquavers, three to a beat, and rests on the beat itself, where the kick is. Look back at the bottom row of figure figure 3.1: rest, note, note, note, four times.

1. On STEP, ■ TRACKS › ❶ TRACK › 4, then ■ ENTRY › ❹ BAR › 1 › ❶ LENGTH › 1/16.
2. Four times: *push* ❶, then on VOLT A2 three times. With OCT- pressed once, A2 is the sixth white key from the top.

Step's PITCH box says A2. You'll hear it lower, at A1: PLUCK BASS's oscillators are set to sound an octave below the note they're given, as many bass sounds are. Low A is the bottom of a bass guitar, and it's where a techno bass line likes to sit.

Writing from a MIDI keyboard

A MIDI keyboard writes on Step too. On any unit, set SETUP's ROOT 2 page to MIDI IN FOLLOW, turn any encoder on Step so it's the unit you last touched, and play: each note goes in at the cursor, as the keys do, and how hard you played it goes in as its VELO. It's handy for a bass line, where it's quicker to find the notes on a proper keyboard.

One bar of bass into eight

A bass line on one note all the way through would be very minimal techno indeed. Track 1's moves down every two bars, A, F, G, E, then round again: eight bars in all. You could write all eight bars key by key. But it's quicker, and it's how most tracks get built, to copy the bar you've got until there are eight, and then change the notes of the copies. Both are done on the COPY page.

1 On STEP, turn 4 BAR back one click, so the cursor is inside bar 1 again (just now it's at the start of bar 2). Then press ■ COPY.



Figure 3.5 COPY as it opens, with the cursor in bar 1: the range is bar 1 (START 1:1.1 to the end of END's beat, 1:4.1), twelve notes, and TO is bar 2.

COPY opens ready to copy the cursor's bar into the next one. Its boxes say what's copied and where to:

- **START** and **END**, encoders 3 and 4, are the stretch to copy: from the start of START's beat to the end of END's beat. So START 1:1.1 and END 1:4.1 are the whole of bar 1, and the line under END counts its notes: 12.
- **TO**, encoder 2, is where the copy goes: where its first note lands. It starts at the bar just after the cursor's bar, here bar 2 (2:1.1), so the first copy needs no turning. A copy writes over whatever's already at TO, so each time you copy, TO has to move on to the first empty bar after what you've got.
- **TO TRACK**, encoder 1, is which track the copy goes on: this one, 4, unless you turn it. Leave it on 4.

Encoders 2 to 4 move a beat a click, so four clicks is a bar. Now copy, doubling up: one bar makes two, two make four, four make eight.

2 *hold* 4 until the bar fills. Bar 1 is copied into bar 2.

3 Now copy bars 1 and 2 into bars 3 and 4. Make the range two bars long, ❹ END four clicks on to 2:4.1, then move TO past the copy you've just made, ❷ TO four clicks on to 3:1.1, then *hold* ❹.

4 And bars 1 to 4 into bars 5 to 8: ❹ END › `4:4.1`, then ❷ TO › `5:1.1`, then *hold* ❹. That's eight bars of A: played now, it sounds just as one bar did, round and round.

New notes for the copies

Now change bars 3 to 8 to F, G and E. That's called **transposing**: moving notes up or down without changing their rhythm. It's on the COPY page too, because it works on the same kind of stretch of a track, START to END. Push encoder 1 and the page changes job: the top bar says `TRANSPOSE T4` instead of `COPY T4`, and encoder 1's box becomes **SEMITONES**, how far to move the notes. TO isn't needed, so it shows a dash: the notes stay where they are and only their pitch changes. (Push encoder 1 again for DELETE, which empties the stretch, again for FILL, which repeats it, and once more to come back to COPY. FILL comes into its own in chapter 5.)

F is four semitones below A (A, G#, G, F#, F), so bars 3 and 4 move by -4. G is two below A, and E five below.



Figure 3.6 The COPY page after pushing encoder 1: now it transposes instead of copying. The stretch is bars 3 and 4 (START 3:1.1 to the end of END's beat, 4:4.1), and SEMITONES -4 will move them down from A to F. TO isn't used.

5 *push* ❶ for TRANSPOSE. END is still 4:4.1, so turn ❸ START › `3:1.1`: the stretch is now bars 3 and 4. Then ❶ SEMITONES › `−4` and *hold* ❹. Bars 3 and 4 are F.

6 Bars 5 and 6, to G: ❹ END › `6:4.1` first, then ❸ START › `5:1.1`, ❶ SEMITONES › `−2`, *hold* ❹.

7 Bars 7 and 8, to E: ❹ END › `8:4.1`, ❸ START › `7:1.1`, ❶ SEMITONES › `−5`, *hold* ❹.

SEMITONES always counts from where the notes are now, and every stretch here is still A, so each one is counted from A.

> **END first, then START**
>
> START can't go past END. When you move the stretch later in the song, move END on first, then START after it: the other way round, START stops where END is.

Press ■ PLAY and listen. Press ■ STOP when you've heard enough. Everything's playing at once, and it already sounds like techno. It just doesn't go anywhere yet.

3.6 Mixing it and playing it

In techno the kick is the loudest thing in the track, and the bass sits just under it. PLUCK BASS is a full, loud sound, so turn Volt down a little on Step's MIXER, which has a box for every unit on the chain:

on STEP, ■ MIXER, then turn the box that says VOLT to =-6

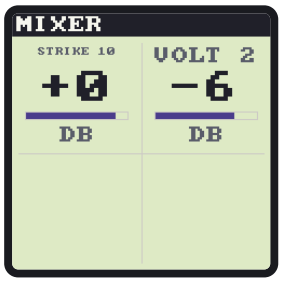


Figure 3.7 MIXER: Strike at 0 dB, Volt 6 dB under it.

The numbers are **decibels** (dB), the way sound engineers measure level, and 0 is the unit as it plays. Every 6 dB down halves the signal, but your ears don't hear it that way: they work in ratios, not steps, so -6 sounds clearly quieter, not half as loud. It takes about -10 to sound half as loud, and about -20 to sound a quarter as loud. So small numbers go a long way: 1 or 2 dB is a nudge, 6 is a real change, and 20 puts a part right at the back. Push a box's encoder and that unit is muted, which you won't need here: you'll switch tracks off instead, because that can take one drum out and leave the rest.

Building the track by taking things away

Now the part that makes four loops into a track. On TRACKS, **pushing encoder 1 mutes the track it's showing**: it stops playing (its notes stay just as they are), the box says **MUTED**, and pushing again brings it back. You'll play the whole of figure figure 3.2 that way, by hand, while Step plays.

First, mute everything but the kick, so the track starts with the kick on its own:

1. On STEP, ■ TRACKS › 1 TRACK › 2 › *push* 1, then the same on tracks 3 and 4. Stay on track 4.
2. Press ■ PLAY.

While Step plays, the top bar shows a small play arrow and the bar the track you're on has got to. Track 4, the bass, is eight bars long, so on track 4 it counts 1 to 8, then 1 again: each time round is one eight-bar phrase. That's your clock. Every change in this track comes as a new phrase starts, when the 8 turns back into a 1.

Phrase	Bars	What you do as it starts
1	1-8	Nothing: the kick on its own.
2	9-16	Unmute track 2: the hats come in.
3	17-24	Unmute tracks 3 and 4: the clap and the bass.
4	25-32	Nothing: let it play.
5	33-40	Mute track 1. The kick drops out, and everything floats.
6-7	41-56	Unmute track 1, and on Volt, turn CUTOFF up three clicks at the start of every bar, so the bass gets brighter and brighter.
8	57-64	Mute tracks 3 and 4: kick and hats to finish. Press STOP at the end of the phrase.

To make a change, turn TRACK to that track during the last bar of a phrase, and push it as the next one begins. You'll be a little early or late the first time, and it doesn't matter: DJs do this by ear every night, and a part coming in a beat late sounds like you meant it. If you lose count, listen for the bass going back to A: that's a new phrase. Once the bass is muted, keep count in eights in your head, four kicks to a bar.

The filter in phrases 6 and 7 is the one that needs two hands. Volt's FILTER page is already showing, from when you set it up: CUTOFF is encoder 1, and it starts at -2.5. Three clicks a bar for sixteen bars brings it to about +2.3, and you'll hear PLUCK BASS go from a dark thud to a bright, buzzing snarl. Turn slowly: it's more exciting when you can hardly hear it change.

> **The star on Volt**
> Once you've turned CUTOFF, Volt's top bar shows a star instead of the padlock: PLUCK BASS is a factory sound, and your change isn't kept in it. That's just what you want here. To play the track again from the start, turn CUTOFF back to -2.5, or choose PLUCK BASS again on SOUNDS.

> **Try this**
> Play it again, and this time make your own changes. Take the hats out for the last bar of a phrase and bring them back on the 1. Bring the bass in eight bars later than the plan. Mute the kick for one bar only, just before it comes back. Turn CUTOFF down instead of up. There's no wrong answer: if it makes you want to move, it's right.

That's a techno track: four one-bar loops and the patience to change one thing at a time. Everything's saved as you go, so the song's still on Step when you come back.

3.7 Track 2 at a glance

Track 1 was four loops and the patience to change one thing at a time. Track 2 keeps that, and adds the things that make a drum pattern feel played rather than typed: some notes softer than others, a pattern two bars long instead of one, an open hi-hat, and a bass line that moves. And instead of a slow climb all the way through, it has a **breakdown**: eight bars where the floor drops away and the tension builds, before everything crashes back in.

Sound 3.2 Track 2, as built in sections 3.8 and 3.9: two-bar hats with an open hat, a bass riff that moves, and soft ghost hats that come in once the groove is settled.

Five tracks this time. The hats come in two kinds: the loud ones on the off-beats, as in Track 1, and soft **ghost hats** between them.

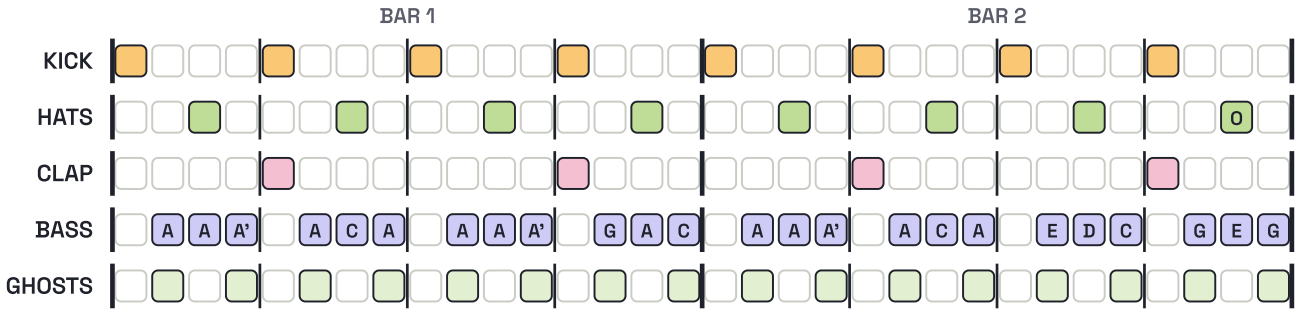


Figure 3.8 Track 2's five parts over two bars. The open hat (O) is the last off-beat of bar 2; the ghost hats, paler, are played softly; the bass riff's notes are written in (A' is the A an octave up).

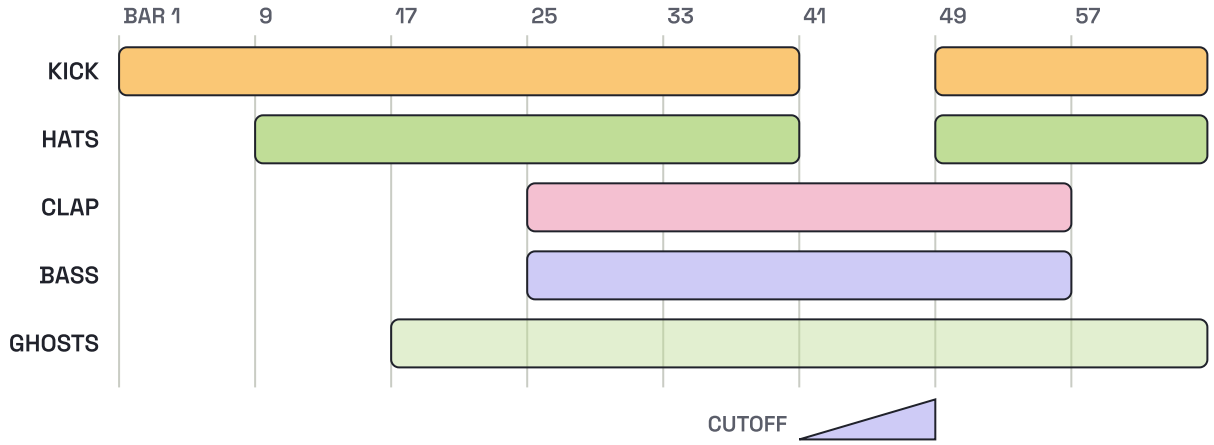


Figure 3.9 How Track 2 plays. The ghost hats come in once the off-beat hats have set the groove. In the breakdown (bars 41-48) the kick and off-beat hats drop out, leaving the ghosts with nothing to lean on, while Volt's CUTOFF opens a little every bar; everything comes back in at bar 49.

Track 2

The whole track, as you'll set it up.

STEP	`TEMPO 132`
STRIKE	`PUNCH`
VOLT	`BOUNCE (FA15)` `OCT- once`
TRACK 1	`STRIKE` `LOOP 1 BAR` `KICK on every beat`
TRACK 2	`STRIKE` `LOOP 2 BARS` `CL HAT on the off-beats, OP HAT on the last`
TRACK 3	`STRIKE` `LOOP 1 BAR` `CLAP on 2 and 4`
TRACK 4	`VOLT` `LOOP AUTO` `a two-bar riff`
TRACK 5	`STRIKE` `LOOP 1 BAR` `CL HAT, VELO 50, between the off-beats`
MIXER	`VOLT -6`

BOUNCE is a Volt bass with a lot of EMPHASIS, so its filter squelches as it opens, and a little glide, so each note slides into the next instead of jumping: it's what makes the riff bounce. Like PLUCK BASS, it sounds an octave below the note it's given.

3.8 Building Track 2

Most of this is what you did for Track 1, so the routes are shorter. The new parts get the explaining.

A new song

On STEP, ■ SONG › 4 ACTION › NEW › *hold* 4. Track 1 isn't touched: it's still there on the SONG page, as song 1. To play it again later, ■ SONG › 1 SONG to it, then *push* 1 to load it.

The sounds and the tempo

1. On STRIKE, ■ PUNCH (it may be on PUNCH already).
2. On VOLT, ■ SOUNDS › 1 BANK › FAC A › 2 SOUND › BOUNCE, then ■ FILTER, ready for the breakdown. OCT- needs to be pressed once, so its light is on: if it's still on from Track 1, leave it alone, as pressing it again would move the keys down another octave.
3. On STEP, ■ TEMPO › 1 TEMPO › 132.

Five tracks

On TRACKS, give each track its unit and its loop, as in section 3.4. The new one is track 2: 3 LOOP › 2 BARS, so it plays two bars before it goes round.

Track	UNIT	LOOP	Plays
1	STRIKE	1 BAR	The kick
2	STRIKE	2 BARS	The off-beat hats, with an open hat
3	STRIKE	1 BAR	The clap
4	VOLT	AUTO	The riff, two bars
5	STRIKE	1 BAR	The ghost hats

Tracks of different lengths play side by side without any trouble: the one-bar tracks just go round twice for every time the two-bar ones go round once.

The kick

Exactly as in Track 1 (section 3.5): on track 1, ■ ENTRY › 1 LENGTH › 1/4, then on STRIKE KICK four times. From here on, remember to turn BAR back to 1 each time you change track.

Two bars of hats, and an open hat

Track 2 is Track 1's off-beat hats, but two bars of them, with the very last one an open hi-hat: the long, splashy "tssss" instead of the short "ts". It comes once every two bars, at the end, and it's a little signal to the dancers that the pattern's going round again.

1. On STEP, ■ TRACKS › 1 TRACK › 2, then ■ ENTRY › 4 BAR › 1 › 1 LENGTH › 1/8.
2. Seven times: *push* 1, then on STRIKE CL HAT.
3. Then *push* 1 once more, and on STRIKE OP HAT. The cursor is at the start of bar 3: two bars written.

Listen to what the open hat does when it plays: it doesn't ring on for long. The next closed hat cuts it short. On a real drum kit the two are one pair of cymbals, opened and closed by the drummer's foot, so the closed sound always stops the open one, and drum machines have done the same ever since. It's called a **choke**, and Strike's closed hat always chokes its open hat.

The clap

As in Track 1: ■ TRACKS › 1 TRACK › 3, then ■ ENTRY › 4 BAR › 1 › 1 LENGTH › 1/4, and twice *push* 1 and on STRIKE CLAP.

The riff

Track 1's bass sat on one note for two bars at a time. Track 2's moves within the bar: it jumps up an octave and back, steps through C, D and E, and only repeats every two bars. It's still built the same way, though: rest on the beat, where the kick is, and three semiquavers after it.

1. On STEP, ■ TRACKS › 1 TRACK › 4, then ■ ENTRY › 4 BAR › 1 › 1 LENGTH › 1/16.
2. Then, beat by beat, a rest (*push* 1) and three keys on VOLT:

Bar	Beat 1	Beat 2	Beat 3	Beat 4
1	rest, A2, A2, A3	rest, A2, C3, A2	rest, A2, A2, A3	rest, G2, A2, C3
2	rest, A2, A2, A3	rest, A2, C3, A2	rest, E3, D3, C3	rest, G2, E2, G2

With OCT- pressed once, Volt's top key is C2, and the white keys go up from there, C D E F G A B, then C again. So, counting white keys from the top: E2 is the 3rd, G2 the 5th, A2 the 6th, C3 the 8th, D3 the 9th, E3 the 10th and A3 the 13th.

A wrong note?

Turn 4 BAR back to it and play the right key. Step is in REPLACE (the word under BAR), so the new note goes in where the old one was. Then turn BAR on again to where you'd got to. If it's gone badly wrong, start the track again: ■ TRACKS › *hold* 4 clears every note on it.

The ghost hats

Now the new idea. Every note Step writes has a **velocity**: how hard it's played, from 1 (as soft as possible) to 127 (as hard as possible). It's called velocity because on a keyboard it's measured by how fast the key goes down. So far every note has been at 100, the VELO box on ENTRY. Turn VELO down before you write, and the notes go in softer: Strike plays them quieter, as a drummer would.

Soft notes between the main ones are called **ghost notes**, and they're how drummers make a beat feel alive. Track 5's ghost hats go on the second and fourth semiquaver of every beat, in between the off-beat hats.

1. On STEP, ■ TRACKS › 1 TRACK › `5`, then ■ ENTRY › 4 BAR › `1` › 1 LENGTH › `1/16`.
2. 3 VELO › `50`.
3. Four times: *push* 1, CL HAT, *push* 1, CL HAT.

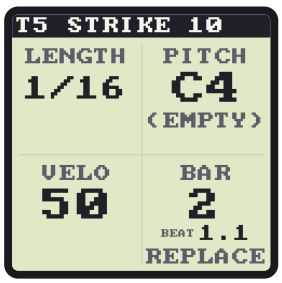


Figure 3.10 ENTRY after the ghost hats: VELO 50, and the cursor at the start of bar 2.

VELO stays where you left it

VELO doesn't go back to 100 by itself. Whatever you write next, on any track, goes in at 50 until you turn it up again. That's why the ghost hats are written last. When you're done, turn 3 VELO › `100`, or the next thing you write will be mysteriously quiet.

VELO works the same whichever unit's keys you write with: Step's own keys, Strike's, Volt's. Volt plays every note at the same loudness unless its VELO setting is on (Volt's SETUP), so on Volt, velocity makes no difference in this book until chapter 4.

3.9 Playing Track 2

Each song keeps its own mix, so on the new song Volt is back at 0. Turn it down again: ■ MIXER, and the box that says VOLT to -6.

Counting bars the way musicians do

In Track 1, the bass track's counter went 1 to 8, which made it a clock for the phrases. Here the longest track is two bars, so the counter only goes 1, 2, 1, 2. So count the bars the way musicians do: instead of "one, two, three, four" over and over, count the bar number in place of the "one":

ONE two three four, **TWO** two three four, **THREE** two three four … **EIGHT** two three four

Each word is a kick. When you get to EIGHT, the next bar is a new phrase. It feels clumsy for about a minute and then you'll never lose count again. Once the off-beat hats are in, the open hat helps too: it comes at the end of every second bar, so the fourth open hat ends a phrase.

The arrangement

As in Track 1: on TRACKS, mute every track but the kick (tracks 2, 3, 4 and 5), press ■ PLAY, and make each change as a phrase begins.

Phrase	Bars	What you do as it starts
1	1-8	Nothing: the kick on its own.
2	9-16	Unmute track 2: the off-beat hats.
3	17-24	Unmute track 5: the ghost hats.
4-5	25-40	Unmute tracks 3 and 4: the clap and the riff.
6	41-48	The breakdown. Mute tracks 1 and 2, the kick and the off-beat hats, and on Volt, turn CUTOFF up six clicks at the start of every bar.
7	49-56	Unmute tracks 1 and 2. Everything's back.
8	57-64	Mute tracks 3 and 4: the drums on their own to finish. STOP at the end of the phrase.

The breakdown

Listen carefully to the ghost hats through the track. When they come in at bar 17, they sit comfortably between the off-beat hats and the beat just feels busier. Then in the breakdown the kick and the off-beat hats go, and the ghost hats have nothing left to lean on. Heard on their own, on the second and fourth semiquavers, they sound as if the beat has slipped sideways: your foot isn't quite sure where to go. That's on purpose. Unsettling the listener is half of what a breakdown is for.

At the same time, Volt's filter is opening: CUTOFF starts at -3.0, and six clicks a bar for eight bars takes it to about +1.8. With BOUNCE's high EMPHASIS, the riff squelches and whistles more and more as it opens. Then at bar 49 the kick and the off-beat hats come back, the beat snaps back into place under the ghost hats, and the riff is wide open. That's the drop, and it works because of the eight bars before it.

> ### The same notes, a different meaning
>
> The ghost hats never change in this track: the same eight soft notes every bar. What changes is what's around them. Notes that sound like decoration in one place sound like tension in another. Much of arranging is exactly this: not writing new parts, but changing what each part has around it.

Turn CUTOFF back to -3.0 when you've finished, ready to play the track again (or choose BOUNCE again on SOUNDS).

> ### Try this
>
> Change a note's velocity after it's written: on track 5, turn BAR to a ghost hat and turn VELO, and that note changes. Try making the hats in the second half of each beat a little louder (60) than the first (40), and listen to the groove lean forward. Then turn VELO back to 100.
>
> Move the open hat: on track 2, turn BAR to another off-beat hat and play OP HAT over it. Try the breakdown without the filter, then with the filter but the off-beat hats left in, and hear how much each part of the trick does.

3.10 Track 3: your own

You've built two tracks from recipes. This one is yours. There's no recipe card: there's a list of things you can change, each only a few presses, and you choose. Change one thing or all of them. Section 3.11 shows one way it might go, if you'd like to hear somebody else's choices first, but treat it as the dullest possible answer rather than the right one: there's no right answer, and the best track is the one that makes you want to move.

Start from Track 2

Most tracks start as a copy of another one. Rather than build everything again, copy Track 2's song and change the copy:

1. On STEP, make sure Track 2 is the song loaded: ■ SONG › ❶ SONG to it, and *push* ❶ if it isn't already.
2. ❹ ACTION › **DUPLICATE** › *hold* ❹. The copy is the song you're working on now; Track 2 is safe as it was.
3. Give it a name, so you can tell them apart: encoder 2, NAME, turns the letter, and pushing it moves on to the next one.



Figure 3.11 SONG after DUPLICATE: the copy is the song you're working on now (the top bar's number), with the same name as the one it came from.

Before you play, check TRACKS: the copy has Track 2's mutes as you left them at the end of playing it. Push encoder 1 on any track that says MUTED and you want to hear.

Choices

- **The tempo.** On TEMPO. Slower, from about 120 to 126, sounds deeper and more hypnotic, and leaves room for echoes; faster, from 135 to 140, sounds harder and more urgent. A few BPM makes more difference than you'd expect.
- **The kit.** Press another kit button on Strike while it plays. The patterns stay as they are, because each key is the same drum in every kit: the kick key is always a kick. BOOM has a long, deep, booming kick; SPARK is lighter and thinner. (NEON has drums but no hi-hats or clap, so those tracks go silent: a breakdown of its own.)
- **The bass sound.** Choose another Volt sound on SOUNDS: PLUCK BASS from Track 1, ROUND BASS, FUNK BASS. The riff stays the same, played by a different instrument.
- **The key.** Move the whole riff up or down on COPY's TRANSPOSE: START at the start of bar 1, END at the end of bar 2, SEMITONES as you like. Up 5 puts it in D, down 2 in G. If you add a stab (below), do it first, and transpose the stab's track by the same amount, so they stay together.
- **The arrangement.** Your own order of tracks coming and going, still eight bars at a time. Try starting with something other than the kick, a breakdown twice as long, or a track that never quite settles.

A stab, if you have Glass

A **stab** is a short chord, played once and left to ring, and it's been in techno since Derrick May's "Strings of Life". You'll need Glass joined to the chain.

1. On GLASS, choose a sound: ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **E.ORGAN 1**. E.PIANO 1 and STRINGS 1, in the same bank, and SYNBRASS 1, in FAC D, all make good stabs too: try each once you've written the chord.
2. On STEP, give the stab a track:
   ■ TRACKS › ❶ TRACK › **6** › ❷ UNIT › **GLASS** › ❸ LOOP › **1 BAR**.
3. ■ ENTRY › ❹ BAR › **1** › ❶ LENGTH › **1/8**, then *push* ❶ three times: the cursor is on the off-beat after beat 2.
4. On GLASS, hold down four keys together, A3, C4, E4 and G4, and then let them all go. Step writes them as one chord, and the cursor moves on when the last key comes up.

Look at where those four keys are: A3 is the sixth white key from the top, and the other three are every second white key after it, A, C, E, G, with one white key skipped between each. Stacking notes that way, skipping one each time, is how most chords are built. This one is A minor seventh, made from the same A the riff keeps coming back to, which is why the two sit together so easily.

A stab is loud for its size. On Step's MIXER, turn the box that says GLASS down to about -13, under the drums and the bass.

An echo, if you have Haze

In the early 1990s, two Berlin producers working as Basic Channel slowed techno down and sent its chord stabs into long echoes, borrowed from Jamaican dub, so each chord repeated and faded away behind the beat. It's called dub techno, and it takes two steps on Haze:

1. On HAZE, press the sixth white key from the top: the screen says **DUB ECHO**. An echo that keeps time with Step, and every synth is now sent into it.
2. Only the stab should echo, so press (■ SENDS): one box per synth. Turn the boxes that say STRIKE and VOLT to 0, and the one that says GLASS to 60.

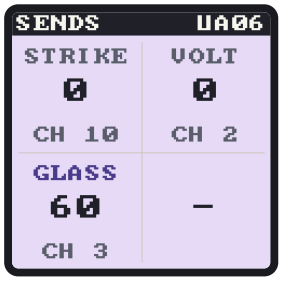


Figure 3.12 Haze's SENDS page, one box per synth: only Glass goes into the echo.

Now each stab is followed by its echoes, quieter and quieter, filling the gaps until the next one. Making Sounds, chapter 10, has everything else Haze can do.

3.11 One way it might go

Here's one set of choices: Track 2 duplicated and slowed to 126, the E.ORGAN 1 stab at -13 on the MIXER, and Haze's DUB ECHO on the stab only. Everything else is as Track 2 left it.

Sound 3.3 Track 3, one way it might go: Track 2 duplicated, slowed to 126, with a chord stab on Glass echoed by Haze.

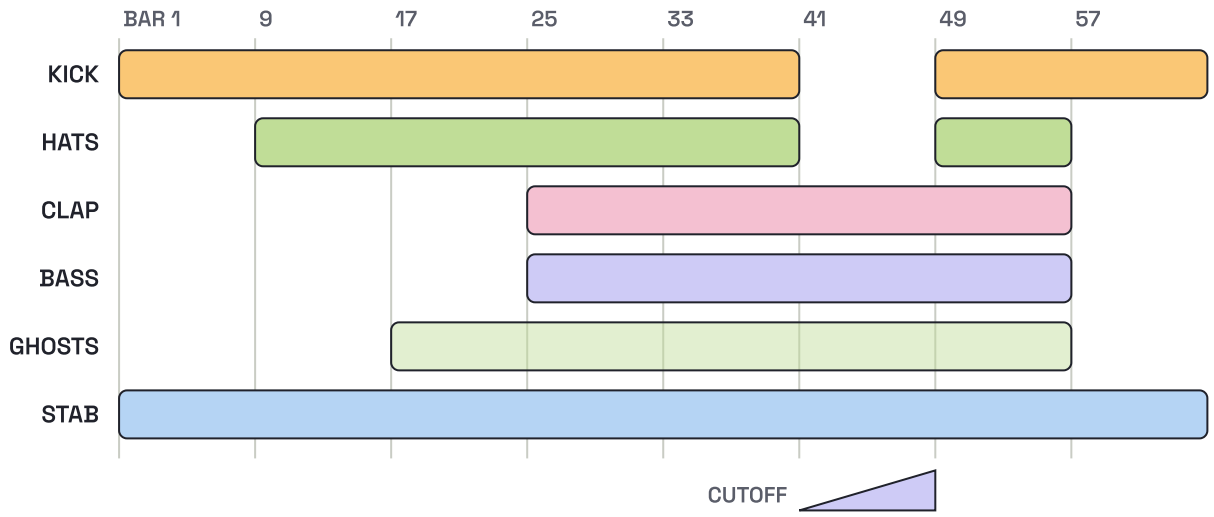


Figure 3.13 Track 3, one way it might go: Track 2's arrangement with the stab and its echo all the way through, starting and ending the track with only the kick for company.

Phrase	Bars	What happens
1	1-8	The kick and the stab, and the stab's echoes: before there's any beat to speak of, the echo sets the mood.
2	9-16	The off-beat hats.
3	17-24	The ghost hats.
4-5	25-40	The clap and the riff.
6	41-48	Track 2's breakdown: no kick, no off-beat hats, and Volt's filter opening.
7	49-56	Everything.
8	57-64	The kick and the stab, as it began.

Ending as it began is a DJ's favourite: the last eight bars sound like the first eight, so the next record can be mixed in over them.

3.12 What you've learned

Three techno tracks, and along the way most of what Step does day to day:

- Tracks, each playing a unit, several on one unit, each with its own loop: one bar, two bars or AUTO.
- Writing on ENTRY: lengths and rests, BAR, REPLACE, and velocity, from any unit's keys.
- COPY and TRANSPOSE: one bar into eight, and new notes for the copies.
- The MIXER, and arranging by muting and unmuting tracks, eight bars at a time.
- Breakdowns and drops: why taking something away is the strongest move there is.
- New songs, and duplicating one to make another.

Next, acid: the same Step and the same Volt, but the other half of Volt, with accents, slides and a filter that squelches. Your techno tracks are safe on the SONG page whenever you want to come back to them.

Chapter 4 - Acid

One bar of bass, sixteen notes long, and a pair of hands on the knobs.

Acid is techno's closest relative, and it's built in the same way: a kick on every beat, hats between them, a clap on 2 and 4, a bar or two going round and round. The difference is the bass. In acid it isn't a bass line sitting under the beat: it's the lead, a squelching, twisting line of sixteen semiquavers, and the person playing the track spends six minutes turning its knobs while it plays, so it never sounds the same twice.

That sound comes from one machine, and Volt has a half made for it: **ACID**. This chapter uses it, with two things on Step you haven't met yet, accents and slides.

> **What you need for this chapter**
>
> The Root, **Step**, **Strike** and **Volt**, as for techno, and headphones. Haze is optional, in Track 3. The tracks start from your techno tracks, so read chapter 3 first.

4.1 Where acid came from

In 1981 Roland brought out the TB-303 Bass Line, a small silver box meant to play bass parts for guitarists practising on their own. It had one oscillator, a filter, and a little sequencer you programmed a note at a time, with two odd extras: **accent**, which made a note louder and brighter, and **slide**, which made a note glide into the next one. It sounded nothing like a bass guitar, and it was hard to program. Roland stopped making it in 1984, and second-hand ones went for almost nothing.

A few people had already heard something else in it. In 1982 the Indian film musician Charanjit Singh made *Synthesizing: Ten Ragas to a Disco Beat*, Indian ragas played over a drum machine with a 303 bubbling underneath, years before anyone called it acid; hardly anyone heard it at the time.

In Chicago, three friends who called themselves Phuture, DJ Pierre, Earl "Spanky" Smith and Herb J, bought a second-hand 303 and, not knowing how it was meant to be used, turned its knobs while it played. The result was "Acid Tracks", a twelve-minute track of a 303 line squealing and growling over a drum machine. The DJ Ron Hardy played it at his club, the Music Box, over and over, until the dancers loved it, and when it came out on record in 1987 it gave the music its name: **acid house**.

A year later acid house reached Britain, and with it the "Second Summer of Love" of 1988: warehouse parties, outdoor raves and the yellow smiley face. Since then the 303's sound has never really gone away. Hardfloor's "Acperience 1" (1992) layered several 303s into one long, building track; Josh Wink's "Higher

State of Consciousness" (1995) took a single screaming line into the pop charts; and acid techno, harder and faster, came out of London's squat parties. Today there are more machines that copy the 303 than there ever were 303s.

4.2 What acid is made of

- **The line.** One bar of semiquavers, sixteen steps, some of them rests. It jumps octaves and repeats, and it's catchy in a way a melody isn't: it's more like a riff that talks.
- **Accents.** A few notes in each bar are accented: louder, and the filter opens wider and snaps back faster, so they bark. Accents in a row build on each other, so a run of them squeals.
- **Slides.** A slid note glides into the next one instead of starting it fresh, a little smear of pitch. Sliding down an octave is the classic acid sound.
- **The filter, and its resonance.** The line's notes barely change, but its sound changes all the time, because the filter is being opened and closed, and its resonance turned up until it whistles.
- **Hands on the knobs.** This is the big one. An acid track is played, not just written: the knobs are turned slowly through the whole track, and that's where the music is.

Acid is techno's idea pushed even further. In techno, you'll remember (section 3.2), there's no chord progression: the music doesn't travel from chord to chord, it stays put and the sound changes. In acid even the notes barely change. The same bar of line goes round for six minutes, and everything you hear happening is the filter, the resonance and the accents, under someone's hands. It's as close as dance music comes to the minimalists' slow-changing patterns, and Track 2 borrows one of Steve Reich's favourite tricks, a pattern going round against another of a different length.

4.3 Track 1 at a glance

Acid Track 1 is techno Track 1's drums with a new bass: Volt's ACID half, playing a one-bar line with four accents and four slides, and played by turning four of its knobs, one stretch of the track at a time.

Sound 4.1 Acid Track 1: one bar of Volt's ACID half with accents and slides, over techno Track 1's drums on BOOM, played by turning its knobs.

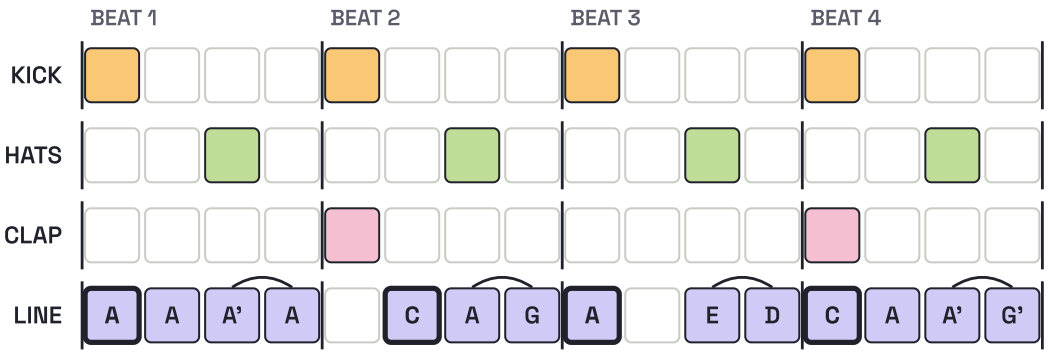


Figure 4.1 Acid Track 1: the drums, as techno Track 1's, and the line. A heavy outline and bold letter is an accent; an arc is a slide into the next note. A' and G' are an octave up.

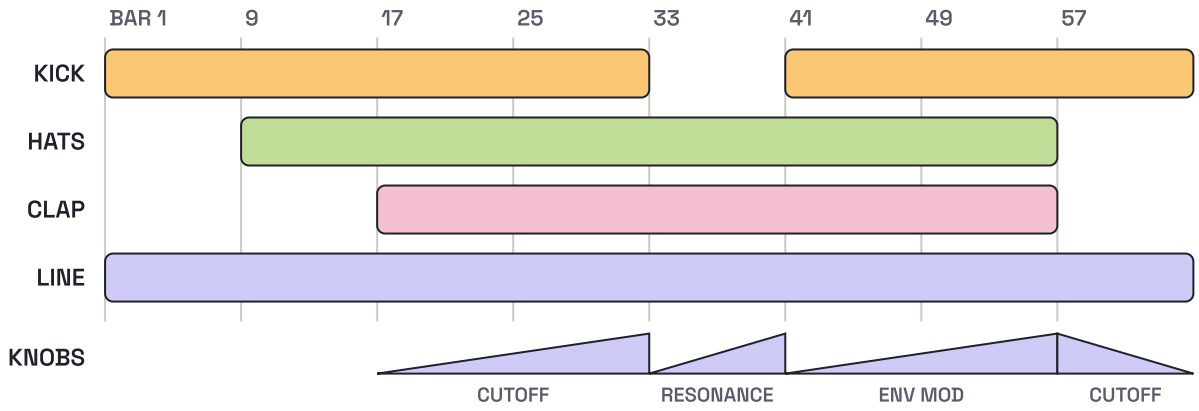


Figure 4.2 How acid Track 1 plays: the drums come and go as in techno, but what really moves is underneath: the knobs on Volt's ACID page, turned a little every bar.

Acid Track 1

Techno Track 1, duplicated, with:

STEP	`TEMPO 122`
STRIKE	`BOOM`
VOLT	`ACID (FB01)` `OCT- once` `CUTOFF 2.0`
TRACKS 1-3	`as techno Track 1`
TRACK 4	`VOLT` `LOOP AUTO` `the line, one bar`
MIXER	`VOLT -9`

BOOM is Strike's deep, round kit, with a long kick that booms: the other sound of early dance music, alongside PUNCH's harder one. Acid house was made on both.

4.4 Setting up

Techno Track 1, copied

1. On STEP, load techno Track 1: ■ SONG › ❶ SONG to it (song 1, unless you've moved things about), then *push* ❶.
2. ❹ ACTION › **DUPLICATE** › *hold* ❹. Then name the copy, on encoder 2, so you can find it again.
3. ■ TEMPO › ❶ TEMPO › **122**. Acid house is a little slower than techno.
4. On TRACKS, check each track: push encoder 1 on any that says MUTED, so all four play.

The sounds

1. On STRIKE, press ■ BOOM.
2. On VOLT, ■ SOUNDS › ❶ BANK › **FAC B** › ❷ SOUND › **ACID**. FAC B is Volt's bank of ACID sounds, and the ACID button lights. Leave OCT- as it was for techno, pressed once.
3. Press ■ ACID, for the ACID page: TUNING, CUTOFF, RESONANCE and ENV MOD. These four are the instrument you'll play.
4. On STEP, ■ MIXER, and turn the box that says VOLT down from -6 to -9: in acid the line squeals, and the kit needs a little more room than in techno.

A clean track for the line

Track 4 still has techno Track 1's bass in it. Clear it: ■ TRACKS › ❶ TRACK › 4 › *hold* ❹, until the bar fills and the screen says **TRACK CLEARED**. The track keeps its unit, Volt, and its loop, AUTO: only the notes go. (Clearing the copy doesn't touch techno Track 1 itself, which is a separate song.)

4.5 Writing the line

The old bass machines were programmed in passes: first the notes, then which of them were accented and slid. That's a good way to do it on Step too. It splits one hard job into two easy ones, and you get to hear the line on its own before you dress it up.

First, the notes

1 On STEP, ■ ENTRY › ❹ BAR › 1 › ❶ LENGTH › 1/16.

2 Then, beat by beat, on VOLT (a rest is *push* ❶ on Step):

Beat 1	Beat 2	Beat 3	Beat 4
A2, A2, A3, A2	rest, C3, A2, G2	A2, rest, E3, D3	C3, A2, A3, G3

The same keys as techno Track 2's riff, with one new one, G3: the 12th white key from the top. Unlike techno's bass lines, this one plays on the beat as well, right on top of the kick. A 303 line doesn't stay out of the kick's way: it fights with it, and that's half its energy.

Press ■ PLAY and listen to it plain, then ■ STOP.

Then, the accents and slides

On Step, ACCENT and SLIDE are the fourth and fifth buttons down the left-hand column. Press one and it changes the note under the cursor, on or off; the light beside it shows whether the note under the cursor has it. So walk the cursor back through the line with BAR, one semiquaver a click, and press them where the line needs them.

1 Turn ❹ BAR all the way back to 1. The cursor is on the first note.

2 Then, a click of BAR at a time, press ACCENT on four notes and SLIDE on four:

Step	Note	Press
1	A2	■ ACCENT
3	A3	■ SLIDE: it glides down the octave to the A2 after it
6	C3	■ ACCENT
7	A2	■ SLIDE: down to G2
9	A2	■ ACCENT
11	E3	■ SLIDE: down to D3
13	C3	■ ACCENT
15	A3	■ SLIDE: down to G3

A slide belongs to the note it glides *from*: the A3 on step 3 is held on, and its pitch slides down into step 4's A2, which doesn't start again. Play it now and listen for the difference: the accents bark, the slides smear, and the line suddenly sounds like acid.

> **VELO doesn't make accents**
>
> On Volt's ACID half, a note is accented only by ACCENT, the way bass machines of this kind have always worked, however high or low VELO is. (From a MIDI keyboard, a note played hard, velocity 100 or more, is accented instead.) On other units, ACCENT just plays a note a little harder.

4.6 Playing Track 1

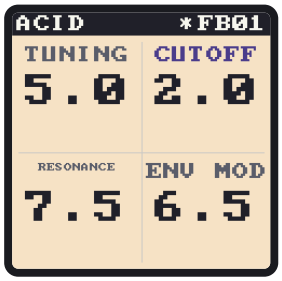


Figure 4.3 Volt's ACID page with CUTOFF turned down to 2.0, ready to play. The star: the factory sound's changes aren't kept.

Volt's ACID page has the four knobs you'll play. Each turns in tenths, a click at a time.

1. **TUNING**: the pitch. Leave it at 5.0.
2. **CUTOFF**: how open the filter is. Low is dark and muffled, high is bright and buzzing.
3. **RESONANCE**: how much the filter rings at its cutoff. Up high, the line whistles and squeals.
4. **ENV MOD**: how far each note opens the filter and snaps it back. Up high, every note goes "yow".

Turn CUTOFF down to 2.0 before you start, so the line begins dark. Then, on TRACKS, mute tracks 2 and 3, the hats and the clap, and press (■ PLAY). Count the phrases as in techno Track 1 (with only one bar in the song, count the bars in your head: ONE two three four, TWO two three four…).

Phrase	Bars	On Step, as it starts	On Volt, every bar
1	1-8	Nothing: the kick and the line.	Nothing yet: let it sit dark.
2	9-16	Unmute track 2: the hats.	
3-4	17-32	Unmute track 3: the clap.	CUTOFF up 3 clicks, to about 6.8.
5	33-40	Mute track 1: no kick.	RESONANCE up 2 clicks, to about 9.1.
6-7	41-56	Unmute track 1.	ENV MOD up 2 clicks, to about 9.7.
8	57-64	Mute tracks 2 and 3: kick and line. STOP at the end of the phrase.	CUTOFF down 5 clicks, closing it to about 2.8.

Step on one side, Volt on the other, a hand on each: that's how acid has always been played. Keep the knob moves small and steady. A click or two a bar sounds like nothing while you're doing it, but over sixteen bars the line becomes a different animal, and the listener can't say when it happened.

> ### The star on Volt
> Every knob you've turned is a change to the factory ACID sound, so Volt's top bar shows a star: nothing's kept. To play the track again from the start, choose ACID again on SOUNDS, and turn CUTOFF down to 2.0.

Try this

Play it again and follow your ears instead of the table. Push RESONANCE right up with CUTOFF low, then sweep CUTOFF slowly: the squeal moves up through the line. Try ACID 2, Volt's second ACID page (press ACID again): DECAY makes each note's "yow" longer or shorter, and ACCENT how hard the accents bark. Then move the accents: walk BAR through the line and press ACCENT on different notes. Three accents in a row squeal; one on its own barks.

4.7 Track 2 at a glance

Acid Track 2 starts from techno Track 2, with its fuller kit: two-bar hats with the open hat, the clap and the ghost hats. Its line is new, and it does something no line in this book has done yet: it's three beats long, in a bar of four.

Sound 4.2 Acid Track 2, as built in sections 4.8 to 4.10: a three-beat line on Volt's square wave going round against techno Track 2's drums, with an accented kick and a snare holding every bar's first beat.

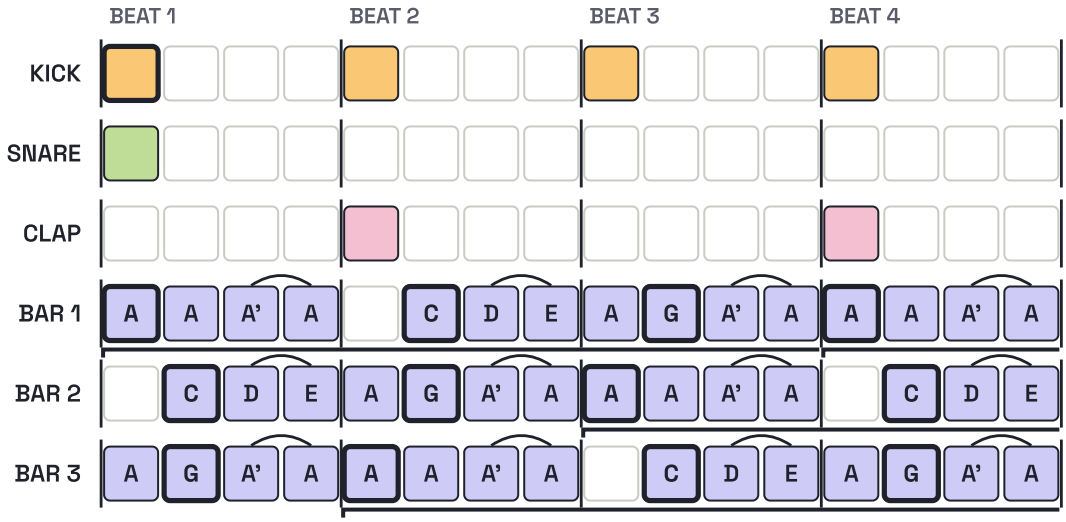


Figure 4.4 Acid Track 2: one bar of the beat (the hats and ghost hats as in techno Track 2), and under it the three-beat line as it falls in bars 1, 2 and 3. Each bracket is one time round the line: it starts a beat earlier every bar, and only in bar 4 lines up with bar 1 again. The heavy outline on the kick is its accent.

Follow the line down figure figure 4.4. It starts with the kick, on beat 1 of bar 1. Three beats later it starts again, on beat 4; then on beat 3 of bar 2, then on beat 2 of bar 3: a beat earlier every bar. Only at bar 4 does it start on beat 1 again: four times round the line is three times round the bar, twelve beats either way. In between, its accents land somewhere new every bar, and the line seems to tumble over the beat without ever falling. Playing a pattern of three against a beat of four like this is a **polyrhythm**, and acid producers have loved it from the start: one short loop makes a line that takes three bars to repeat.

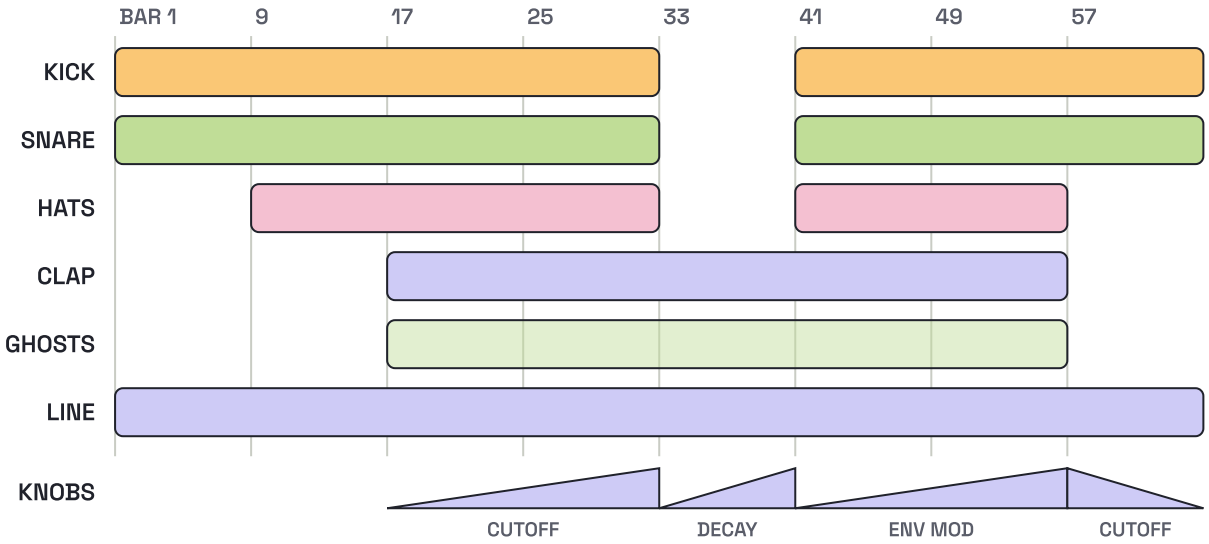


Figure 4.5 How acid Track 2 plays. The kick and the snare hold the bar from the start; the line plays all the way through. In the breakdown (bars 33-40) only the ghost hats, the clap and the line are left, while DECAY, on the ACID 2 page, lengthens every note.

Acid Track 2

Techno Track 2, duplicated, with:

VOLT	`SQ SQUELCH (FB07)`	`OCT– once`	`CUTOFF 2.0`
TRACK 1	`as techno Track 2`	`beat 1 ACCENT`	
TRACKS 2, 3, 5	`as techno Track 2`		
TRACK 4	`VOLT`	`LOOP 0+3`	`the line, three beats`
TRACK 6	`STRIKE`	`LOOP 1 BAR`	`SNARE on beat 1`
MIXER	`STRIKE –4`	`VOLT –19`	

4.8 Setting up

Techno Track 2, copied

1. On STEP, load techno Track 2: ■ SONG › ❶ SONG to it, then *push* ❶.
2. ❹ ACTION › **DUPLICATE** › *hold* ❹, and name the copy on encoder 2.
3. Leave the tempo at 132. Acid that leans towards techno runs faster than acid house.
4. On TRACKS, push encoder 1 on any track that says MUTED, so all five play.

The sounds

1. Strike stays on PUNCH, as in techno Track 2: hard, bright, and made for this.
2. On VOLT, ■ SOUNDS › ❶ BANK › **FAC B** › ❷ SOUND › **SQ SQUELCH**, then ■ ACID for the ACID page.

SQ SQUELCH plays a square wave where ACID played a sawtooth. A sawtooth has every harmonic, so it buzzes; a square has only every other one, the odd ones, and it sounds hollower and woodier, like a clarinet next to a trumpet. Bass machines of this kind had a switch for just these two, and so does Volt: WAVEFORM, on the ACID 2 page.

Making room for the kit

On STEP, ■ MIXER. Turn STRIKE down 4, to -4, and VOLT down from -6 to -19. That's a long way down, and the reason is the line itself. A loop of three beats is so insistent that if it's as loud as the drums, your ear follows it instead of them, and the kick seems to be the thing that's out of step. Whatever is loudest and repeats most strongly is what a listener hears as the beat, so the beat has to be the loudest thing here. (STRIKE comes down too because of what you'll do to the kick in a moment: it gets louder.)

A three-beat loop for the line

1. Clear track 4, as in Track 1: ■ TRACKS › ❶ TRACK › 4 › *hold* ❹, until it says **TRACK CLEARED**.
2. Push encoder 3, LOOP. Its title now says **LOOP BEATS**: it turns a beat a click instead of a bar.
3. Turn it 3 clicks, to 0+3: no bars and three beats.

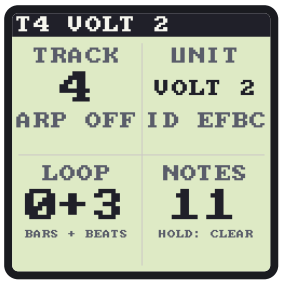


Figure 4.6 TRACKS on track 4: LOOP 0+3, no bars and three beats. The line runs on its own, three beats round, whatever the drums are doing.

A loop like this runs on its own: from PLAY it goes round every three beats, whatever the drums are doing, and Step never starts it again with the song. (When you turn to another track, LOOP counts in bars again.)

4.9 Writing the line, and holding the bar

The line

Notes first, then accents and slides, as in Track 1. On STEP,
■ ENTRY › ❹ BAR › 1 › ❶ LENGTH › 1/16, then on VOLT (a rest is *push* ❶ on Step):

Beat 1	Beat 2	Beat 3
A2, A2, A3, A2	rest, C3, D3, E3	A2, G2, A3, A2

Twelve semiquavers, and the cursor stops at the end of the loop. VELO still says 50, left from the ghost hats. Leave it: on ACID it makes no difference, as you saw in Track 1.

Then ❹ BAR back to 1, and through the line a click at a time:

Step	Note	Press
1	A2	■ ACCENT
3	A3	■ SLIDE: down the octave to A2
6	C3	■ ACCENT
7	D3	■ SLIDE: up to E3
10	G2	■ ACCENT
11	A3	■ SLIDE: down the octave to A2

Three accents, one a beat, each a semiquaver later than the last: on the beat, just after it, then later again. That's what makes the line feel as if it's leaning forward.

Holding the bar

Against a line like this, the drums need to say clearly where each bar starts, its first beat, the "one". Two small things do it.

1. **An accent on the kick.** ■ TRACKS › ❶ TRACK › `1`, then ■ ENTRY › ❹ BAR › `1`: the cursor is on the first kick. Press ■ ACCENT. On Strike, an accent plays a drum 6 dB harder (SETUP sets how much), so every bar's first kick now hits harder than the other three.

2. **A snare with it.** ■ TRACKS › ❶ TRACK › `6`, ❷ UNIT › `STRIKE`, and LOOP one click, to 1 BAR. Then ■ ENTRY › ❹ BAR › `1` › ❸ VELO › `100` › ❶ LENGTH › `1/4`, and on STRIKE, press SNARE once.

A snare on beat 1 is an odd place for one: in most dance music it's on 2 and 4, where the clap is here. But on its own, once a bar, it isn't a backbeat. It's a marker, and with the accented kick it plants a flag on the one. The line can tumble as much as it likes; you always know where you are.

4.10 Playing Track 2

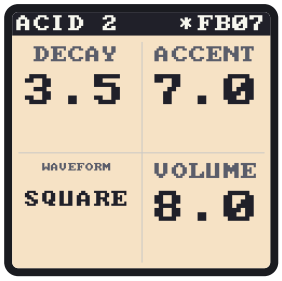


Figure 4.7 Volt's ACID 2 page (press ACID again): DECAY, ACCENT, WAVEFORM and VOLUME. SQ SQUELCH's WAVEFORM is the square wave.

This time you'll also play a knob on Volt's second ACID page. Press ■ ACID again for ACID 2: DECAY, ACCENT, WAVEFORM and VOLUME. **DECAY** is how quickly each note's filter closes after it opens: short is a tight, plucky blip; long lets each note bloom and run into the next. Press ACID again to go back to the first page.

Turn CUTOFF down to 2.0 before you start. On TRACKS, mute tracks 2, 3 and 5: the hats, the clap and the ghost hats. Then press ■ PLAY: the kick, the snare and the line.

Phrase	Bars	On Step, as it starts	On Volt, every bar
1	1-8	Nothing: kick, snare and line.	Nothing yet.
2	9-16	Unmute track 2: the hats.	
3-4	17-32	Unmute tracks 3 and 5: the clap and the ghost hats.	CUTOFF up 3 clicks, to about 6.8.
5	33-40	Mute tracks 1, 2 and 6: no kick, no hats, no snare.	Press ACID for ACID 2. DECAY up 5 clicks, to about 7.5.
6-7	41-56	Unmute tracks 1, 2 and 6.	Press ACID, back to the first page. ENV MOD up 1 click, to about 8.6.
8	57-64	Mute tracks 2, 3 and 5: kick, snare and line. STOP at the end of the phrase.	CUTOFF down 5 clicks, to about 2.8.

Listen to the breakdown. With the kick and the snare gone, nothing says where the one is any more: the ghost hats, which unsettled the beat in techno Track 2, and the clap, now don't know where they stand either, and the three-beat line takes over. Everything floats. Then at bar 41 the kick and the snare land on

the one together and the whole track snaps back into four.

Counting the phrases is harder this time, because the line keeps pulling you off the bar. Count with the kick, not the line: the accented kick is ONE, every bar.

> **Try this**
>
> Mute track 6 and turn STRIKE and VOLT back to where techno Track 2 had them, 0 and -6, and listen to the line drag the drums into three. Then try other loop lengths on track 4: five beats (LOOP 1+1) comes round every five bars; seven beats every seven. Change WAVEFORM on ACID 2 from the square to the sawtooth while it plays. And move the accents: one in every three semiquavers makes a second, faster three on top of the first.

4.11 Track 3: your own

As in techno, the third track is yours: a copy of Track 2, and a list of things to change. Section 4.12 has one way it might go; the moment yours stops sounding like it, you're doing it right.

Start from Track 2

1. On STEP, load acid Track 2: ■ SONG › ❶ SONG to it, and *push* ❶.
2. ❹ ACTION › **DUPLICATE** › *hold* ❹, and name the copy.
3. On TRACKS, unmute any track you want to hear.

Choices

- **The tempo.** Acid house sits around 120 to 125; acid techno, which grew up harder and faster in the 1990s, from about 135 to 145.
- **The sound.** FAC B has sixteen ACID sounds, each a different starting point for the same knobs: DEEP ACID is darker and rounder, BITE snappier, WAH has a long, vocal sweep, and RUBBER and MELLOW SQ are square waves like SQ SQUELCH.
- **The wave.** On ACID 2, WAVEFORM swaps the square for the sawtooth and back. The sawtooth is brighter and buzzier, and it's the one that takes distortion best.
- **The kit.** BOOM for acid house's deeper kick, PUNCH for acid techno's harder one.
- **The line.** Change a note or two, move the accents and slides, or try another LOOP length: four beats for a straight line, five or seven for a stranger one. Or go back to Track 1's line, sixteen semiquavers, and see how much calmer it sounds over this kit.

- **The key.** COPY's TRANSPOSE on track 4, START at the start of bar 1, END at the end of beat 3: up 5 puts the line in D.

Distortion, if you have Haze

A 303 line pushed through distortion stops squelching and starts snarling, and acid techno has used it that way ever since. Haze does it in its COLOUR half: the effects it puts on one synth only, the one chosen on its SOURCE page. Join Haze to the chain, then:

1. On HAZE, press the 13th white key from the top: the screen says **ACID DRIVE**. Distortion, a little echo and a little reverb, and every synth is sent into the echo and reverb.
2. The drums should stay dry, so ■ SENDS and turn the box that says STRIKE to 0. Leave VOLT's as it is.
3. ■ SOURCE. Turn encoder 1, SYNTH, a click at a time until it says VOLT: the one synth whose sound Haze changes.
4. Turn encoder 2, MIX, all the way down to 0%. MIX is how much of Volt's sound goes through the distortion: at 0% none of it, at 99% all of it. You'll play it.

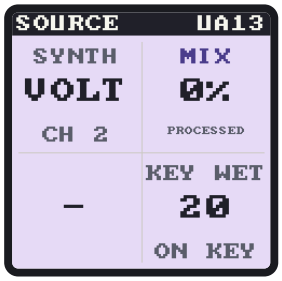


Figure 4.8 Haze's SOURCE page, ready to play: SYNTH is VOLT, the one synth Haze changes, and MIX 0%, so none of it is changed yet.

MIX is the knob to play here. Turn it up slowly and the distortion creeps in under the clean line until it's all there is. For other kinds of dirt, try Haze's other drive presets, WARM DRIVE (gentler) and FUZZ BASS (fuzzier), or the DRIVE page, where DRIVE sets how hard the sound is pushed.

4.12 One way it might go

Here's one set of choices: acid techno. Track 2 duplicated and sped up to 138, WAVEFORM back to the sawtooth, and Haze's ACID DRIVE on Volt, with MIX at 0% to start. Everything else is as Track 2 left it, the knob moves too, except one: the breakdown is played on Haze instead of on DECAY.

Sound 4.3 Acid Track 3, one way it might go: Track 2 duplicated, at 138 with the sawtooth, and Haze's ACID DRIVE on Volt, played up through the breakdown.

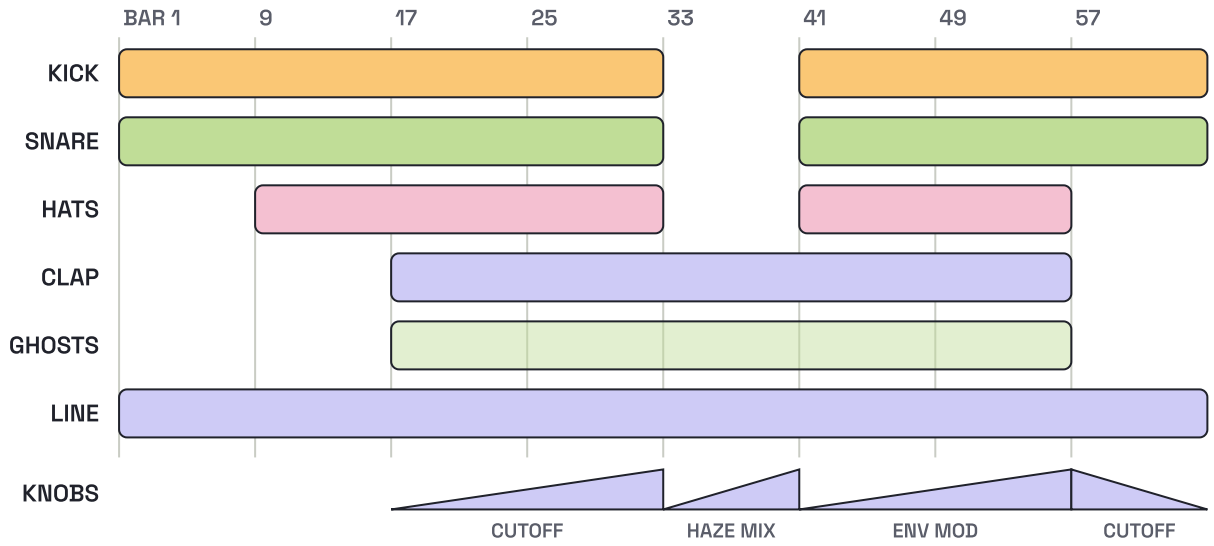


Figure 4.9 Acid Track 3, one way it might go: Track 2's arrangement at 138, with Haze's MIX played up through the breakdown, so the line comes back at bar 41 distorted.

Phrase	Bars	What happens
1-2	1-16	Kick, snare and line, clean, then the hats.
3-4	17-32	The clap and the ghost hats, and Volt's CUTOFF opening, 3 clicks a bar.
5	33-40	The breakdown. Mute tracks 1, 2 and 6, then a hand on Haze: MIX up 12 clicks a bar, to about 96%. The line turns from squelch to snarl with nothing else to hide behind.
6-7	41-56	Everything back, the line now fully distorted, and Volt's ENV MOD up a click a bar.
8	57-64	Kick, snare and line, CUTOFF closing. STOP at the end of the phrase.

Notice what the breakdown does this time. In Track 2 it took the kick away and let the line float; here it takes the kick away and changes the line while nobody's looking, so when the kick comes back the track is a harder thing than the one that left. A breakdown doesn't only let the listener breathe: it's the best place in a track to change something for good.

4.13 What you've learned

- Writing in passes, notes first and then accents and slides, as on the old bass machines; ACCENT and SLIDE, and why a slide belongs to the note it glides from.
- Volt's ACID half, and playing a track on its knobs: CUTOFF, RESONANCE, ENV MOD, DECAY.
- Clearing a track, and loops shorter than a bar: three against four, and a line that takes three bars to repeat.
- Balance: the loudest, most insistent part is what the listener hears as the beat, so the beat has to win. Accenting a drum, and marking the one.
- Distortion on one synth with Haze's COLOUR half, played in with MIX.

Next, house: the same four to the floor, but with chords, and whole songs built from sections.

Chapter 5 - House

The same four to the floor, with chords, a shuffle and a bass line that dances.

House is techno's older cousin, from Chicago rather than Detroit, and it's built on the same floor: a kick on every beat, eight bars at a time. But where techno is cold and mechanical, house is warm. It came out of disco and soul, and it kept their chords, their swing and their joy. This chapter brings in the things that make that warmth on Step: chords, and swing.

> **What you need for this chapter**
>
> The Root, **Step**, **Strike**, **Volt** and **Ember**, and headphones. If you have Glass and not Ember, Glass can play the chords: try E.PIANO 1. Haze comes in later in the chapter.

5.1 Where house came from

In 1977 a New York DJ, Frankie Knuckles, moved to Chicago to play at a new club, the Warehouse. He played disco, soul and gospel records, and European electronic music, and he re-edited them on tape so the best parts lasted longer, the drums were harder, and the dancing never stopped. In New York his old friend Larry Levan was doing the same at the Paradise Garage. When disco fell out of fashion around 1980 and the records dried up, Chicago's DJs and their fans started making their own, on cheap drum machines and synths, and the records were sold as the music they played at the Warehouse: house.

Jesse Saunders' "On and On" (1984) is often counted as the first house record. Two small Chicago labels, Trax and DJ International, put out dozens more, made quickly and pressed cheaply. Ron Hardy played them at the Music Box, rawer and wilder than the Warehouse. Marshall Jefferson's "Move Your Body" (1986), with its piano, called itself "The House Music Anthem", and it was; Larry Heard, as Mr. Fingers, made "Can You Feel It" (1986), soft chords floating over a drum machine, and started deep house. In January 1987 Steve "Silk" Hurley's "Jack Your Body" went to number one in Britain, and house went round the world.

It has never stopped since. Every few years it changes its clothes, garage, deep house, French house, tech house, and comes back. The Daft Punk and Stardust records of the late 1990s ("Music Sounds Better With You", 1998) are house, and so is a good part of what plays in clubs and shops today.

5.2 What house is made of

- **The tempo.** A little slower than techno: about 118 to 128. Track 1 is 122.
- **Four to the floor**, as in techno, and a clap or snare on 2 and 4.
- **Open hi-hats on the off-beats**, the "tss" that rings, where techno usually has a closed one that ticks.
- **Swing.** A slight shuffle, which comes from disco's and soul's drummers, who never played quite in a straight line.
- **A bass line that moves**: often on the off-beats, jumping octaves, as in disco.
- **Chords.** This is the big difference.

In section 3.2 you saw that techno throws out the chord progression: it sits on one chord, or one note, and keeps you moving with slowly changing sound instead. House brings the chords back, from the gospel, soul and disco it grew out of, and they're rich ones: sevenths, chords of four notes, warm and a little bittersweet. But listen to what it does with them. Most house tracks use two chords, or four, round and round, never going on a journey the way a song's verse and chorus do. It's halfway between the two worlds: the harmony of soul music, held in techno's loop. Those chords colour the loop rather than going anywhere, and that's why a house track can last eight minutes and still feel like it's arriving.

5.3 Track 1 at a glance

House Track 1 is two bars long: two chords, a bar each, over a kit with open hats, a bass on the off-beats and a little swing.

Sound 5.1 House Track 1, as built in sections 5.4 to 5.8: two chords on Ember, a bar each, over open hats, a disco bass and a little swing.

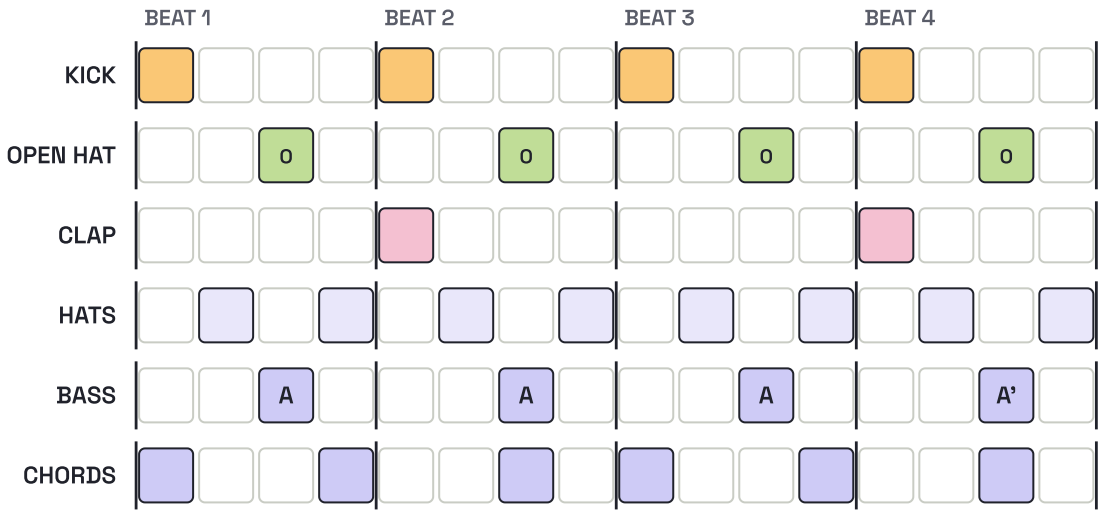


Figure 5.1 House Track 1, one bar. The open hats (O) sit on the off-beats; the soft closed hats on the 2nd and 4th semiquaver of each beat, where the swing moves them. The bass is the same in bar 2 a fourth higher, on D; the chords are A minor 7 in bar 1 and D minor 7 in bar 2, in the same rhythm.

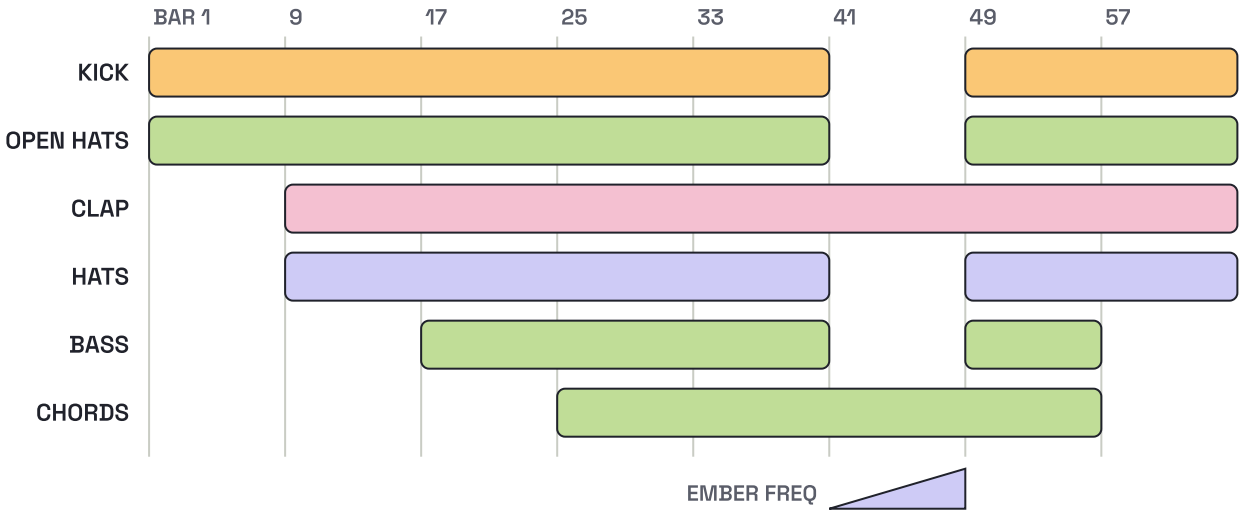


Figure 5.2 How house Track 1 plays. The chords arrive at bar 25; in the breakdown they're left alone with the clap, their filter shut and opened again by hand.

House Track 1

A new song, with:

STEP	TEMPO 122	SWING 58% 1/16	
STRIKE	PUNCH		
VOLT	DISCO BASS (FA04)	OCT- once	
EMBER	CHORD STAB (FA07)		
TRACKS 1-3	STRIKE	LOOP 1 BAR	kick, open hats, clap
TRACK 4	VOLT	LOOP AUTO	the bass, two bars
TRACK 5	EMBER	LOOP AUTO	Am7, Dm7
TRACK 6	STRIKE	LOOP 1 BAR	soft closed hats
MIXER	VOLT 0	EMBER -18	

5.4 Setting up

A new song, and the sounds

1. Join Ember to the chain, on the right of Volt.
2. On STEP, ■ SONG › 4 ACTION › **NEW** › *hold* 4, and name it.
3. On STRIKE, ■ PUNCH.
4. On VOLT, ■ SOUNDS › 1 BANK › **FAC A** › 2 SOUND › **DISCO BASS**, and press ■ OCT- once.
5. On EMBER, ■ SOUNDS › 1 BANK › **FAC A** › 2 SOUND › **CHORD STAB**, then press ■ VCF, ready for later.

DISCO BASS is one of the most famous sounds of its kind of synth, the bubbling, plucked bass of late-1970s disco, and house took it over with the rest of disco's furniture.

Tempo, and swing

1 On STEP, ■ TEMPO › TEMPO › 122.

2 Turn encoder 3, SWING, up 8 clicks, to 58%. Underneath, it says 1/16: it swings semiquavers.

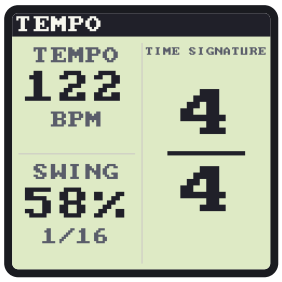


Figure 5.3 TEMPO with SWING at 58%, on the semiquavers (1/16). The time signature, 4/4, is drawn on the right.

Played straight, every semiquaver is exactly as long as the next. Swing makes them uneven, in pairs: the first of each pair a little longer, the second a little later. At 58% the first takes 58% of the pair's time instead of half of it, and the second comes in that much late. At 66%, two-thirds, the pair becomes a triplet with its middle note missing, the lilting "shuffle" of blues and jazz. House sits much nearer straight: just enough to make the beat lean back and walk.

Notice which notes move: only the *second* of each pair, the second and fourth semiquaver of each beat. A note on the beat, or on the off-beat halfway between, never moves at all. So swing does nothing to a part that doesn't play on those semiquavers, which is why this track will put something there for it to swing.

Six tracks

On TRACKS, give each track its unit with encoder 2: tracks 1, 2, 3 and 6 STRIKE, track 4 VOLT, track 5 EMBER. Then LOOP 1 BAR for tracks 1, 2, 3 and 6, the drums; leave 4 and 5 on AUTO.

5.5 The groove

The drums

On STEP, ■ ENTRY, then, a track at a time, from bar 1 (BAR back to 1 each time), on STRIKE:

Track	LENGTH	Write
1	1/4	KICK four times
2	1/8	rest, OP HAT, four times over: the open hats on the off-beats
3	1/4	rest, CLAP, twice

The bass

1. Track 4, LENGTH 1/8, on VOLT: rest, A2, rest, A2, rest, A2, rest, A3. The off-beats again, with the hats, and the octave jump at the end of the bar.
2. 4 BAR back one, into bar 1, then ■ COPY › *hold* 4: bar 1 is copied to bar 2.
3. Push encoder 1 for TRANSPOSE. Turn END on 4 beats, then START on 4, so the range is bar 2 alone, then SEMITONES to +5, and *hold* 4. Bar 2 is now on D.

Up five semitones from A is D: up a **fourth**, as musicians say, because D is the fourth note counting up from A in A's scale (A, B, C, D). A bass line that steps up a fourth and back is one of the oldest moves in music, and you'll hear why when the chords arrive: they make the same move.

The swung hats

Track 6 is the part the swing works on. On TRACKS, turn to track 6 (it's STRIKE, LOOP 1 BAR), then on ENTRY, BAR to 1, VELO down to 70, LENGTH 1/16, and on STRIKE, four times over: rest, CL HAT, rest, CL HAT.

Soft closed hats on the second and fourth semiquaver of every beat: exactly where the swing moves notes. And each one does something else as well. A closed hi-hat stops an open one, as the real thing does, so the closed hat just after each open hat cuts it short: the open hat's "tssss" becomes a tight "tss-t". It's the house hi-hat sound, made of two parts that fit together.

5.6 The chords

There are two ways to write a chord on Step, and you've already met the first.

By hand. With track 5 open on ENTRY, hold down a chord on Ember's keys and let them all go: Step writes the notes as one chord, as Glass's stab did in techno Track 3. That's the way when you're finding chords by ear.

By name, on the CHORD page. Choose a chord by its name, A minor 7, and Step works out the notes. That's the way when you know what you want, and it's how this track's chords are written.

1 On TRACKS, turn to track 5. Then press ■ CHORD: the button under ENTRY.

2 Press Step's ■ OCT- once. The CHORD page starts its chords at C4, which is high for chords like these; OCT- moves them down an octave.

3 ❹ BAR › 1 › ❸ LENGTH › 1/16.

4 Turn encoder 1, ROOT, to A. Push it, and encoder 1 turns the TYPE instead (the one underlined is the one it turns): turn it to m7. The screen says Am7, with its four notes under it: A3, C4, E4, G4.



Figure 5.4 The CHORD page, ready to place the first chord: A minor 7, its four notes underneath, LENGTH 1/16, at the start of bar 1. Encoder 1 is turning the TYPE (m7 underlined), after its push; push again for ROOT. INVERT says ROOT: the chord stands on its root, not turned over.

Every time you turn ROOT or TYPE, Ember plays the chord, so you can hear what you're choosing. m7 is a minor seventh, the chord of techno Track 3's stab, built the same way: A, then every other note up from it, A, C, E, G.

The rhythm

Pushing encoder 4 places the chord at the cursor, and the cursor moves on a semiquaver. Turning encoder 4 moves the cursor without placing anything. Place six chords in bar 1, on these semiquavers:

Semiquaver	1	4	7	9	12	15
Do	push 4	turn 2, push 4	turn 2, push 4	turn 1, push 4	turn 2, push 4	turn 2, push 4

Count it: three semiquavers, three, then two, and the same again. That 3+3+2 is one of the most travelled rhythms in the world. In Cuban music it's called the **tresillo**, and you'll hear it under salsa, reggaeton, rock and roll and a good share of pop. Against the straight four of the kick it pulls the bar sideways, which is exactly what the chords should do.

The second chord

1. Turn encoder 4 on once more, to the start of bar 2.
2. Push encoder 1, back to ROOT, and turn it to D: Dm7, D3, F3, A3, C4.
3. Turn encoder 2, INVERT, to 2ND: A3, C4, D4, F4.
4. Place the same rhythm in bar 2.

INVERT turns a chord over: the lowest note moves up an octave, once for the first inversion, twice for the second. The notes are still D, F, A and C, so it's still D minor 7, but look at them next to Am7's: A3 and C4 are in both, and the other two each move down one step, E to D and G to F. Moving from chord to chord as little as possible, each note taking the shortest way, is what musicians call good **voice leading**, and it's why the two chords seem to melt into each other instead of jumping. Turn INVERT back to ROOT (not turned over) and listen to the difference: the whole chord drops, and it sounds clumsier.

And the bass: A in bar 1, D in bar 2, the same move up a fourth as the chords. The bass plays each chord's root, the note it's named after, and that's what holds a chord down.

5.7 Balance, and remembering the sounds

On STEP, ■ MIXER. Leave VOLT at 0, level with the drums: house is a dance music, and its bass is half of the dancing. Turn EMBER down to -18. Chords take up a lot of room, four notes at once across the middle of the sound, and a little goes a long way; they're the colour, not the beat.

Then one more thing. This song uses a kit, a bass sound and a chord sound, and next time you load it you want them all back. On STEP, ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4. Step asks every unit which sound it's playing, and keeps the answers with the song: whenever you load it again, or press PLAY from the start, every unit goes back to its sound.

> **Remember before you play**
>
> REMEMBER SOUNDS keeps the sound you chose, not changes you've made to it while playing. You're about to turn Ember's filter by hand, which puts a star in its top bar, so remember the sounds first, while everything is as chosen.

5.8 Playing Track 1

On TRACKS, mute tracks 3, 4, 5 and 6, so only the kick and the open hats play, and press ■ PLAY.

Phrase	Bars	As it starts
1	1-8	Kick and open hats.
2	9-16	Unmute tracks 3 and 6: the clap and the swung hats.
3	17-24	Unmute track 4: the bass.
4-5	25-40	Unmute track 5: the chords.
6	41-48	Mute tracks 1, 2, 4 and 6: the chords and the clap alone. On EMBER (on its VCF page), turn FREQ right down to 0, then up 4 clicks a bar, to about 3.2.
7	49-56	Unmute tracks 1, 2, 4 and 6: everything.
8	57-64	Mute tracks 4 and 5: the drums to finish. STOP at the end of the phrase.

Listen to bars 25 to 40, once the chords are in. Two chords, nothing more, and yet it already sounds warmer and sadder and happier than anything in the techno chapter. That's what harmony does, even when it doesn't go anywhere.

> **Try this**
>
> Turn SWING up while it plays, a click at a time, to 66%, and listen to the hats start to skip; then down to OFF, and hear the groove stiffen. Try the other chord sounds on Ember: HOUSE ORGAN and E PIANO. And on the CHORD page, try another TYPE for the second chord: m6, or 7sus4. Some will sound wrong, some just different; your ears are the judge.

5.9 Track 2 at a glance

House Track 2 is Track 1 with a longer journey round: four chords instead of two, a bar each, so the loop is four bars long. The drums, the rhythm of the chords and the arrangement are Track 1's.

Sound 5.2 House Track 2, as built in sections 5.10 to 5.12: Track 1 with four chords instead of two, A minor 7, F major 7, D minor 7 and E minor 7, and the bass on their roots.

House Track 2

House Track 1, duplicated, with:

TRACK 4	VOLT	A, F, D, E: a bar each
TRACK 5	EMBER	Am7, Fmaj7, Dm7, Em7
MIXER	VOLT +3	

5.10 Setting up

1. On STEP, load house Track 1 (■ SONG › ❶ SONG to it, *push* ❶), then ❹ ACTION › **DUPLICATE** › *hold* ❹, and name the copy. The remembered sounds come with it.
2. On TRACKS, unmute any track that says MUTED.
3. Clear the chords and the bass: ■ TRACKS › ❶ TRACK › **5** › *hold* ❹, then the same on track 4.
4. ■ MIXER, and turn VOLT up to +3. With four chords moving over it, the bass has more to do, and it can come forward a little.

5.11 Four chords

Where chord progressions come from

You don't have to invent a chord progression to use one. Musicians have always borrowed them: a progression isn't owned by anybody, and the same few have carried thousands of songs. Search for something like **deep house chord progressions** and you'll find lists of them, written as chord names you can dial straight into the CHORD page. That's a fine way to start.

You'll often find them written in Roman numerals instead, like this track's: **i, VI, iv, v**. The numerals count up the scale from the key's home note. In A minor the scale is A, B, C, D, E, F, G, so A is i, D is iv, E is v and F is VI, a capital numeral for a major chord, a small one for a minor. The point of writing them that way is that the same list works in any key: start on D instead, count up D's scale, and you have the same progression, higher. Lists of progressions are a way in, and every bit of theory you pick up along the way, why one chord leads to another, why some sound sad and some hopeful, makes you freer. More theory is only ever a good thing.

Just don't let it tell you what you're not allowed to do. The Beatles' "Because" (1969), Led Zeppelin's "The Crunge" (1973) and Madness's "Our House" (1982) all go places, harmonically (and in "The Crunge" rhythmically too), that would make a less imaginative composer tut and shake their head, and they're all the better for it. Theory describes what people have found works; it doesn't forbid anything. If it sounds good, it is good.

Writing them

On STEP, on TRACKS turn to track 5, then press ■ CHORD. 4 BAR › 1; LENGTH is still 1/16 from Track 1, and Step's OCT- still pressed (if its light is off, press it once). Then, for each bar, set the chord and place it in Track 1's rhythm: push encoder 4, then turn 2, push, turn 2, push, turn 1, push, turn 2, push, turn 2, push, and turn on once more to the next bar. Remember that encoder 1 turns whichever of ROOT and TYPE is underlined, and a push swaps them.

Bar	ROOT	TYPE	INVERT	The notes
1	A	m7	ROOT	A3 C4 E4 G4
2	F	maj7	1ST	A3 C4 E4 F4
3	D	m7	2ND	A3 C4 D4 F4
4	E	m7	1ST	G3 B3 D4 E4
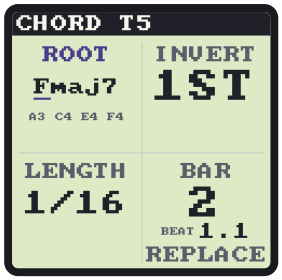


Figure 5.5 The CHORD page at the start of bar 2: F major 7, turned over once (INVERT 1ST), so its notes are A3, C4, E4 and F4: only one note different from A minor 7's.

Look down the last column. Each chord is turned over (INVERT) until it sits as close to the one before as it can: from A minor 7 to F major 7 only one note moves, G down to F; to D minor 7, one more, E down to D; to E minor 7, three notes step down while D stays put. That's the voice leading of Track 1 again, all the way round, and it's why four chords can change under the music without anyone hearing them jump. Try bar 2 at INVERT ROOT and hear the difference: F major 7 drops to F3 and the progression lurches.

Listen to the second chord, too. Three of the four are minor, and F major 7 is the only major one. That's the lift in bar 2, the moment the loop seems to look up, before D minor 7 and E minor 7 bring it back down to A. Major and minor, bright and dark, in four bars, and it never stops going round.

5.12 The bass

The bass plays the same bar as Track 1, on each chord's root: A, F, D, E. Write bar 1 and copy it out to four bars, then move bars 2 to 4 to their roots.

1. On TRACKS, turn to track 4; on ENTRY, BAR to 1, LENGTH 1/8, and on VOLT, as in Track 1: rest, A2, rest, A2, rest, A2, rest, A3.
2. 4 BAR back one, then ■ COPY › *hold* 4: bar 1 to bar 2.
3. Turn END on 4 beats and TO on 4, and *hold* 4: bars 1 and 2 to bars 3 and 4.
4. Push encoder 1 for TRANSPOSE, then a bar at a time:

Bar	Turn	SEMITONES	Becomes
2	START on 4	-4	F
3	END on 4, then START on 4	+5	D
4	END on 4, then START on 4	-5	E

hold 4 after each. (When you open COPY again later, it starts on COPY, whatever you used last.)

Most of the time the bass plays the root, the note a chord is named after, because that's what holds the chord down and tells your ear what it is. But it doesn't have to. The CHORD page names every note of the chord, so you have a choice of four. Try bar 2's bass on A instead of F, SEMITONES 0 instead of -4: F major 7 over an A sounds softer, less settled, almost like A minor again. Musicians write that as F/A, "F over A", and call it a slash chord. Any of a chord's notes can go in the bass, and each one gives it a different colour.

5.13 Playing Track 2

Play it as you played Track 1 (section 5.8, figure figure 5.2): kick and open hats, then the clap and the swung hats, the bass, the chords at bar 25, the breakdown with Ember's filter, everything, and the drums to finish. This time, in the breakdown, the chords are on their own long enough to hear the whole progression come round twice as the filter opens.

> **Try this**
>
> Find a progression of your own. Search for one, or take one from a song you love, and write it on track 5 in place of these four, a bar each. Keep the bass on the roots to start with, then try other notes of the chords under it. If a chord sounds too far from its neighbours, try INVERT until it sits close.

5.14 Track 3: your own, and a whole song

So far, every third track has been the second one with your own choices on top. This one has those too, but it also brings in one more new thing, because it belongs with house: building a whole song, written out from the first bar to the last, so that it plays itself. Section 5.17 has one way it might go, written out by us to show the method, not to be the song: the song is the one you write out.

Start from Track 2

1. On STEP, load house Track 2, then ■ SONG › ❹ ACTION › **DUPLICATE** › *hold* ❹, and name the copy.
2. On TRACKS, unmute any track that says MUTED.

Choices

- **The tempo**, between about 118 and 128: deep house sits at the slow end, the jumpier kinds at the fast end.
- **The swing.** A click or two either way on TEMPO's SWING changes the whole feel.
- **The chords.** Another sound on Ember (HOUSE ORGAN, E PIANO), or a progression of your own on track 5, found by ear or borrowed.

- **The bass.** Other notes of the chords under them (section 5.12), or another Volt sound: FUNK BASS, ROUND BASS.
- **The kit.** BOOM for a deeper, rounder kick under the same patterns.

Make your choices first. The next section writes everything out, and it's easiest to change a sound or a chord while each part is still four bars long.

5.15 Writing the whole song out

Until now you've arranged tracks by muting and unmuting them as they play. That's how a lot of dance music is made, and it's live and fun, but the arrangement is in your hands, not in the song: play it again and you have to do it all again. The other way is to write the song out, every part in every bar where it should play, and then it plays itself, the same every time, and you can change one bar without touching the rest.

Step is built for that. A song on Step is one long timeline of bars, like a score: it doesn't hold "patterns" or "parts", only notes at their places. A track's own loop (LOOP 1 BAR) is the exception, repeating by itself for the whole song, and that's why a written-out song starts by giving those tracks up.

Every track four bars long

The drums are one-bar loops on tracks 1, 2, 3 and 6. Make each one an ordinary four-bar track, like the bass and the chords. For each of those four tracks:

1. On TRACKS, turn to the track and turn encoder 3, LOOP, down one click, from 1 BAR to AUTO.
2. ■ ENTRY › 4 BAR › 1, then ■ COPY: the range is bar 1.
3. Push encoder 1 three times, past TRANSPOSE and DELETE, to **FILL**.
4. Turn encoder 2, UNTIL, on 12 beats, to 4:4.1, the last beat of bar 4. Encoder 1's box says 3 TIMES. Then *hold* 2.

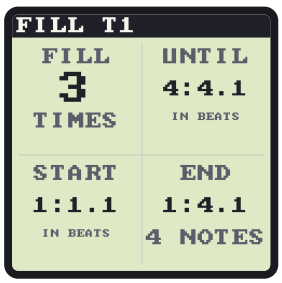


Figure 5.6 The COPY page in FILL, on track 1, the kick: bar 1 (START to END) will go round 3 more times, UNTIL the end of bar 4.

FILL repeats the range, end to end, after itself, until UNTIL: bar 1 copied into bars 2, 3 and 4 in one go. Now all six tracks are four bars long, and together they're one four-bar block of everything.

Sixty-four bars

SECTIONS is COPY's work on every AUTO track at once. On STEP, press ■ SONG, then ■ SONG again: the top bar says SECTIONS. Then:

1. The range starts as bar 1. Turn encoder 4, END, on 3 bars, to bar 4.
2. Push encoder 1 three times, to FILL.
3. Turn encoder 2, UNTIL, until it says 64:1.1, bar 64. (It turns a bar a click; turn it quickly and it speeds up.)
4. *hold* 2. The screen says how many notes it wrote.



Figure 5.7 SECTIONS in FILL: bars 1 to 4, every AUTO track, repeated UNTIL the end of bar 64.

That's the four-bar block sixteen times over: 64 bars, about 1,800 notes, every part playing all the way through.

Carving the arrangement out

Now take away what shouldn't be there, the way a sculptor takes away stone. On each track below, on ENTRY move BAR to 1, press ■ COPY, push encoder 1 twice to DELETE, turn END on to the last bar's last beat, then START on to the first bar, and *hold* 4. (Turn END first: START can't go past it. They move a beat a click, so turn quickly.)

Track	Bars	END on	START on	Why
3 clap	1-16	60 beats	0	The intro: drums only
4 bass	1-16	60 beats	0	
5 chords	1-16	60 beats	0	
1 kick	41-48	188 beats	160 beats	The breakdown: chords and clap
2 open hats	41-48	188 beats	160 beats	
4 bass	41-48	188 beats	160 beats	
6 swung hats	41-48	188 beats	160 beats	

Play it now, from the start, and don't touch anything: the intro, the bass and chords arriving at bar 17, the breakdown at 41 and the return at 49, all on their own. To change the arrangement later, delete some more, or copy a bar back from somewhere else. The song is yours to edit bar by bar.

> **Room for a whole song**
> A song has room for 8,192 notes, shared by all its tracks, so a busy track can take thousands. If you ever fill it, COPY and FILL say SONG FULL, and how many notes went in.

5.16 The pump, if you have Haze

Listen to a French house record from the late 1990s and you'll hear the music breathe: everything but the kick dips every time the kick lands and swells back between, so the whole track seems to pump. In a studio it's done with a compressor on the music, keyed by the kick, called sidechain compression or ducking. Haze does it with PUMP: it turns one synth down whenever another plays.

1. Join Haze to the end of the chain.
2. On HAZE, press the 12th white key from the top: the screen says **PUMP**.
3. The preset sends every synth into its reverb. Keep it on the chords only: ■ SENDS, and turn STRIKE and VOLT to 0.
4. ■ SOURCE, and turn encoder 1, SYNTH, a click at a time until it says EMBER: the synth that pumps.
5. ■ PUMP, and turn encoder 1, KEY, to STRIKE: what it pumps to.

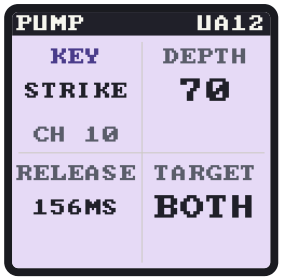


Figure 5.8 Haze's PUMP page: KEY is STRIKE, so every drum Strike plays turns the chosen synth, Ember, down by DEPTH, and RELEASE lets it back up.

DEPTH is how far the chords dip, RELEASE how quickly they come back. The pump follows Strike, so the claps and hats push the chords down too, but it's the kick, four times a bar, that you hear breathing.

5.17 One way it might go

Here's one set of choices: Track 2 as it was, written out to 64 bars and carved as in section 5.15, with Haze's PUMP on the chords. The only thing played by hand is Ember's filter in the breakdown, as in Tracks 1 and 2.

Sound 5.3 House Track 3, one way it might go: Track 2 written out as a whole song that plays itself, with Haze's PUMP making the chords duck under every kick.

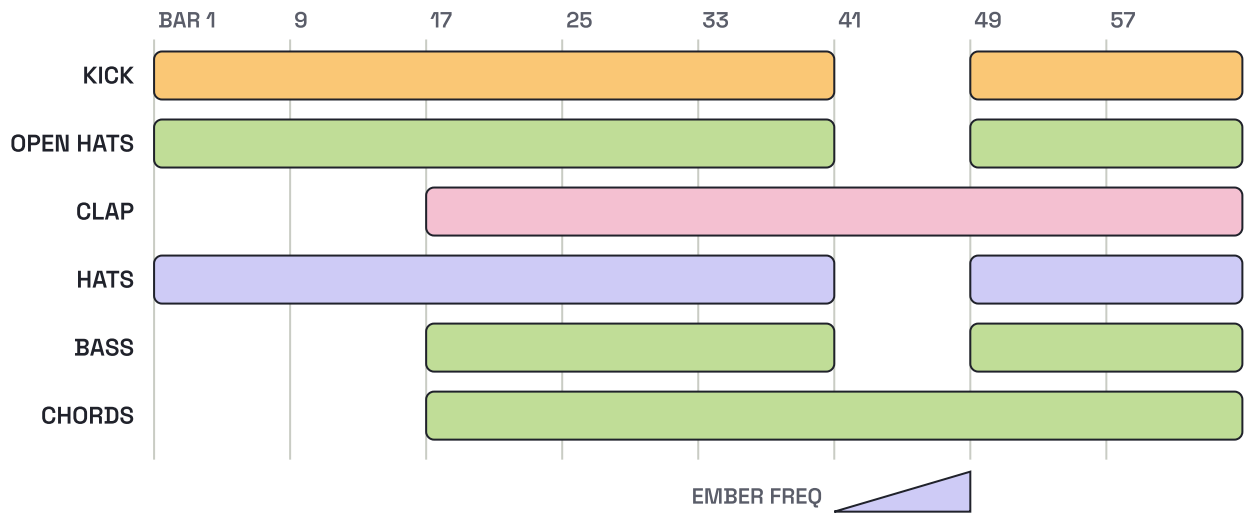


Figure 5.9 House Track 3 written out: every part in its place for the whole 64 bars, carved out with DELETE, so the song plays itself. Ember's filter is still played by hand in the breakdown.

Phrase	Bars	What happens
1-2	1-16	Kick, open hats and the swung hats.
3-5	17-40	The clap, the bass and the chords, all at once, pumping.
6	41-48	The breakdown: the chords and the clap. On EMBER, FREQ right down at bar 41, then up 4 clicks a bar.
7-8	49-64	Everything. STOP at the end.

Press PLAY, and for once you have nothing to do but listen, and turn one knob. That's the difference a written-out song makes.

5.18 What you've learned

- Chords two ways, by hand and by name on the CHORD page; sevenths, inversions, and voice leading.
- Chord progressions: borrowing them, Roman numerals, major and minor, and that if it sounds good, it is good.
- The bass on the roots, and slash chords when it isn't.
- Swing, and why it needs notes on the semiquavers it moves; open and closed hats together.
- REMEMBER SOUNDS.
- A song written out: FILL, SECTIONS, and carving an arrangement with DELETE.
- Haze's PUMP: one synth breathing with another.

Next, hip-hop: samples, chopped and replayed, and in its third track, your own voice.

Chapter 6 - Hip-hop

A drum break cut into pieces and put back together, a bass that nods, and room left for a voice.

Every other kind of music in this book was made by people playing instruments, even if the instruments were machines. Hip-hop began with people playing records. Its first instrument was two turntables, its first sound the best four bars of someone else's song, and it grew into a way of making music out of music: finding a moment on an old record, cutting it out, and building something new on it. This chapter does that on Taste, the sampler.

> **What you need for this chapter**
>
> The Root, **Step**, **Taste**, **Strike** and **Volt**, joined in that order, and headphones. For Track 3, a microphone in the Root's LINE/MIC IN socket, or the little microphone built into the Root's top.

6.1 Where hip-hop came from

On 11 August 1973, in the recreation room of a block of flats at 1520 Sedgwick Avenue in the Bronx, New York, a young Jamaican-born DJ called Kool Herc played at his sister's back-to-school party. He'd noticed that the dancers went wildest in the **break**: the few bars of a funk record where the singing and the band stop and only the drummer plays. So he bought two copies of the same records, and played the break on one, then the same break on the other while he spun the first back to its start, over and over, so a ten-second break lasted as long as the dancers wanted. The dancers were the b-boys and b-girls, and the breaks gave breakdancing its name.

Over those breaks, the DJ's friends took the microphone to hype the crowd, and the shouts became rhymes: the MC, the rapper. Grandmaster Flash turned Herc's trick into a precise art, and DJs learned to scratch, cut and juggle records like instruments. For six years it lived only at parties, on street corners and on cassettes. Then in 1979 the Sugarhill Gang's "Rapper's Delight", fifteen minutes of rhymes over a studio band replaying Chic's "Good Times", sold millions, and the world heard rap.

Chapter 2 told how the 808 arrived under hip-hop in 1982. The bigger change came a few years later. Marley Marl, a producer in Queens, found that a sampler could record a single drum hit off a record, and play it back in a new rhythm. Now the break didn't have to be played as it was: it could be cut into pieces and rebuilt. Two machines made that the way hip-hop was made. The E-mu SP-1200 (1987) sampled at 12 bits and 26 kHz, rougher than a CD, with only ten seconds of memory, and its gritty, dark sound became the sound of New York. The Akai MPC60 (1988), designed with Roger Linn, put sixteen rubber pads under the producer's fingers, and a swing setting from his drum machines.

The late 1980s and early 90s are called the golden age. Public Enemy's producers, the Bomb Squad, piled dozens of samples into a single track; De La Soul's *3 Feet High and Rising* (1989) borrowed from everything from Steely Dan to French lessons; Pete Rock, DJ Premier and the Wu-Tang Clan's RZA chopped soul and jazz records into beats that are still copied today. KRS-One named that hard, swung sound on an album title in 1993: *Return of the Boom Bap*, boom for the kick and bap for the snare. Later, J Dilla in Detroit let his drums fall slightly out of time, lazy and loose, as if the beat were leaning on a wall, and changed how a generation hears rhythm; DJ Shadow's *Endtroducing.....* (1996) was a whole album of beats, made from samples on an MPC60, with no rapping at all.

> ### Whose record is it?
>
> A record belongs to the people who made it, and a sample of it, however short, is a piece of their work. In 1991 a court ruled that Biz Markie had broken the law by sampling a Gilbert O'Sullivan song without asking, and since then hip-hop producers have had to get permission for their samples, and pay for it, before a record comes out, or play the part again themselves. Sample what you like to learn and to practise; ask before you put someone else's recording into anything you share or sell. Everything in this chapter is safe: Taste's starter break was played on Strike and made for Taste, and Track 3's sample is your own voice.

6.2 What hip-hop is made of

Before anything else, the most important thing about a hip-hop beat: it's usually only half of the record.

Most hip-hop is made to go under a rapper. The beat is the stage; the rapper is the show. That's why a beat repeats, why it leaves gaps in the middle of the sound, and why the best ones seem almost too simple on their own: there's a voice still to come. The rapping is the heart of hip-hop, and it belongs to the person doing it, out of their own life, their own streets and their own way of talking. We're a small company who make synthesisers. We know our limits, and we aren't going to try, under any circumstances, to tell anyone how to express themselves in rap. So this chapter builds the stage, and only the stage. If you rap, these are beats for you to rap over. If you don't, they stand up on their own, as plenty of instrumental hip-hop does, and somewhere there's a rapper looking for a beat.

- **The tempo.** Slow: about 85 to 95, and that's the head-nodding speed. These tracks are at 90.
- **A break.** The drums come from a recording, sampled and chopped, instead of being played on a drum machine: a kick and a snare, the boom and the bap, with the life of a real kit around them.
- **Swing**, as in house, but here you feel it more, because the beat is slow enough to hear every semiquaver.
- **An 808 kick** under the break's own, for a weight you feel in your chest more than hear.
- **A short bass line**, a bar or two, round and round.

- **Room for the voice.** The arrangement is built for rapping: verses of sixteen bars, hooks (the choruses) of eight, and moments where the beat drops out so a line can land on its own.

6.3 Track 1 at a glance

Hip-hop Track 1 is boom bap: Taste's starter break, cut into sixteen pieces and played back in a new order at a new tempo, with an 808 kick under it and a two-bar bass line on Volt.

Sound 6.1 Hip-hop Track 1, as built in sections 6.4 to 6.8: Taste's break, chopped into sixteenths and played in a new order at 90, an 808 kick under it, and a two-bar bass on Volt.

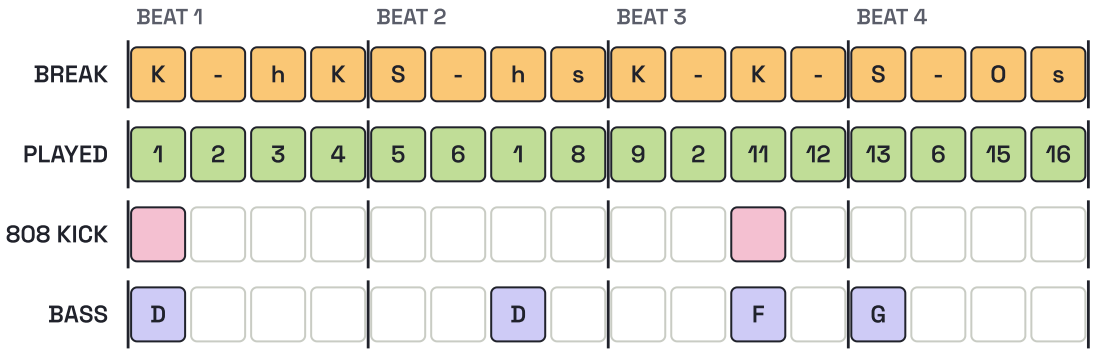


Figure 6.1 Hip-hop Track 1, one bar. BREAK is what's on each of Taste's sixteen keys, the break's sixteenths in order (K kick, S snare, s a ghost snare, h a hi-hat, O the open hat, - only the ring of the hit before). PLAYED is the order track 1 plays them in: slice 1, the kick, again on the 7th semiquaver, and slices 2 and 6 in place of 10 and 14. The bass is bar 1's; bar 2 is A, A, G, F.

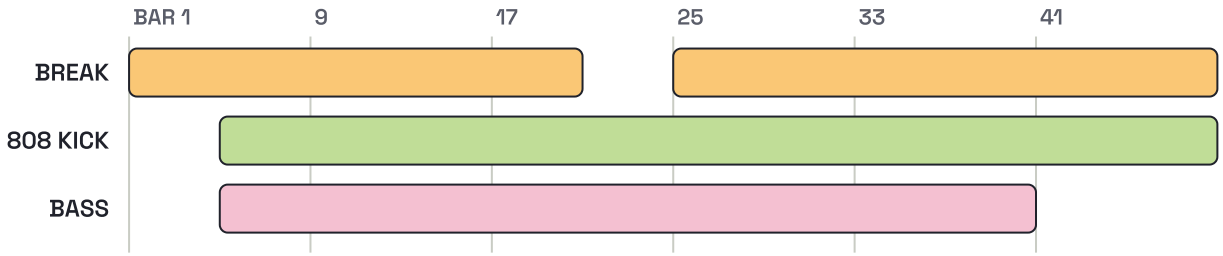


Figure 6.2 How hip-hop Track 1 plays: forty-eight bars. The break drops out for bars 21 to 24, leaving the 808 and the bass, and the bass leaves the beat alone for the last eight.

Hip-hop Track 1

A new song, with:

STEP	`TEMPO 90`	`SWING 57% 1/16`	
TASTE	`BREAK CHOP (FA02)`		
STRIKE	`BOOM`		
VOLT	`FUNK BASS (FA11)`	`OCT– once`	
TRACK 1	`TASTE`	`LOOP 1 BAR`	`the break, rebuilt`
TRACK 2	`STRIKE`	`LOOP 1 BAR`	`KICK, VELO 80`
TRACK 3	`VOLT`	`LOOP AUTO`	`the bass, two bars`
MIXER	`TASTE 0`	`STRIKE –6`	`VOLT –3`

6.4 Setting up

A new song, and the sounds

1. Join the units in this order from the Root: Step, Taste, Strike, Volt.
2. On STEP, ■ SONG › ❹ ACTION › `NEW` › *hold* ❹, and name it.
3. On TASTE, ■ SOUNDS › ❶ BANK › `FAC A` › ❷ SOUND › `BREAK CHOP`.
4. On STRIKE, ■ BOOM.
5. On VOLT, ■ SOUNDS › ❶ BANK › `FAC A` › ❷ SOUND › `FUNK BASS`, and press ■ OCT- once.

Tempo, swing and three tracks

1. On STEP, ■ TEMPO › ❶ TEMPO › `90`, then turn encoder 3, SWING, up 7 clicks, to 57%.
2. On TRACKS, give each track its unit with encoder 2: track 1 TASTE, track 2 STRIKE, track 3 VOLT.
3. LOOP 1 BAR for tracks 1 and 2; leave track 3 on AUTO.

Swing at 57% is a little less than house's 58%, but at 90 beats a minute every semiquaver lasts much longer, so the same lean is easier to hear. Those percentages, by the way, come from Roger Linn's drum machines and the MPC: hip-hop producers talk about "a 57 swing" or "a 62 swing" the way guitarists talk about strings.

6.5 The break, chopped

Sixteen pieces

BREAK CHOP is one bar of drums, recorded at 95 beats a minute, and already cut into sixteen pieces, one for each semiquaver, laid out on sixteen keys in order from the top: the first semiquaver on C3, the next on C#3, and so on, black keys and white, to D#4. Press them one after another, slowly, and you'll hear the bar go by a piece at a time. Each piece is called a **slice**, or a chop.

Key	C3	C#3	D3	D#3	E3	F3	F#3	G3
Slice	1	2	3	4	5	6	7	8
Key	G#3	A3	A#3	B3	C4	C#4	D4	D#4
Slice	9	10	11	12	13	14	15	16

To see one, press ■ TRIM and a key: the screen shows the whole bar with that slice marked out, and its name at the top.



Figure 6.3 Taste's TRIM page on key E3: the whole break, and the fifth of its sixteen slices, the snare, marked out. START and END are where that slice begins and ends, in seconds.

Why chop it?

Here's the old problem with samples. Play a recording faster and it goes up in pitch; slower, and it goes down, like a record on the wrong speed. The break was played at 95 and this song is at 90. Slow the whole bar down to fit, and every drum drops in pitch and goes woolly.

Chopped, it doesn't have to. Step plays each slice at its own place in the bar, at 90, and each one plays at its own speed and pitch, as recorded. The slices are a little shorter than a semiquaver at 90, so each one finishes a moment before the next begins, and you never hear the join. The break now plays at any tempo you like, and it can play in any order you like too. That's the trick the SP-1200's owners found.

A new order

On STEP, ■ ENTRY, LENGTH 1/16, BAR at 1. On TASTE, press these sixteen keys in turn, the first four in each row being a beat:

Beat 1	C3	C#3	D3	D#3
Beat 2	E3	F3	**C3**	G3
Beat 3	G#3	**C#3**	A#3	B3
Beat 4	C4	**F3**	D4	D#4

It's the break almost as it was, slices 1 to 16, with three changed (in bold). The one you'll notice is the third semiquaver of beat 2: slice 1, the first kick of the bar, played again, so a kick lands just before beat 3 and pushes the beat forward. The other two swap a slice that's only the ring of a hit for another one, so they change the colour of the hi-hats, not the rhythm. Try your own: anything that sounds good is right, and figure figure 6.1 shows what's on every key.

Musicians count the four semiquavers of a beat "one-e-and-a", "two-e-and-a"; that extra kick is on the "and" of two. You'll hear producers talk like that.

6.6 The 808, and the bass

Under the break

The break's kick is a real-sounding kick, tight and quick. Hip-hop often puts an 808 underneath it, for the long low boom no recorded kit has.

1. On TRACKS, turn to track 2; on ENTRY, BAR back to 1, and VELO down to 80.
2. LENGTH is still 1/16. On STRIKE: KICK, nine rests, KICK, five rests: semiquavers 1 and 11, the first kick of the bar and the one on the "and" of three.
3. VELO back up to 100.

Two kicks a bar, on two of the break's own, and soft: that's **layering**, two sounds played together as one. The break gives the kick its click and its life, the 808 its weight. Listen with track 2 muted and unmuted: you'll hardly hear the 808 as a separate drum at all, only the floor dropping lower.

The bass

Track 3, LENGTH 1/8, BAR at 1, on VOLT, two bars:

Bar 1	D2	rest	rest	D2	rest	F2	G2	rest
Bar 2	A2	rest	rest	A2	rest	G2	F2	rest

D, F, G and A are four notes of D minor's scale, and the line climbs up through them in bar 1 and falls back in bar 2, a slow nod, like the listeners' heads. FUNK BASS is round and short, so the rests between the notes are real silences, and the break shows through them.

6.7 Balance, and remembering the sounds

On STEP, ■ MIXER: TASTE at 0, STRIKE down to -6, VOLT down to -3. The break leads, as the drums lead in every kind of hip-hop; the 808 is felt under it; the bass sits just behind.

Then, as in house, ■ SONG › 4 ACTION › `REMEMBER SOUNDS` › *hold* 4, so the song brings Taste's break, Strike's kit and Volt's bass back whenever you load it.

6.8 Playing Track 1

On TRACKS, mute tracks 2 and 3, so only the break plays, and press ■ PLAY.

Bars	As it starts	A rapper would hear
1-4	The break alone.	The intro.
5-20	Unmute tracks 2 and 3: the 808 and the bass.	The first verse, sixteen bars.
21-24	Mute track 1: the 808 and the bass alone.	A drop-out: the beat holds its breath.
25-40	Unmute track 1: everything.	The second verse.
41-48	Mute track 3: the beat alone to finish. STOP at the end of bar 48.	The outro.

Four bars at a time, not eight: hip-hop's phrases follow the rapper's, and a rhyme scheme moves in fours. And listen to bars 21 to 24. Taking the drums out for a moment is how a beat makes space for one line to hit on its own, and when the break comes back at bar 25 it lands twice as hard.

Try this

Write your own order on track 1 (hold NOTES on TRACKS to clear it first). Try a snare slice, E3 or C4, on a semiquaver where there wasn't one, or end the bar with the open hat, D4, twice. Then turn SWING up to 62% while it plays and feel the beat start to limp, in the best way, and back down to OFF to hear how stiff it was.

6.9 Track 2 at a glance

Hip-hop Track 2 is Track 1 made dustier and longer: Taste's sound made gritty like the old samplers, a two-bar beat with a fill at the end, a four-bar bass line, and soft swung hats.

Sound 6.2 Hip-hop Track 2, as built in sections 6.10 to 6.12: the break at 12 bits and 26 kHz, two bars with a fill, a four-bar bass going from D minor to G minor, and soft swung hats.

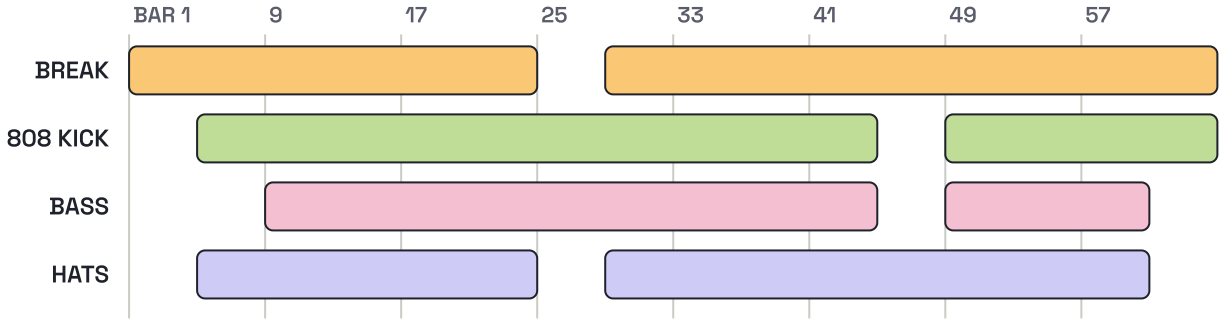


Figure 6.4 How hip-hop Track 2 plays: the hats and the 808 come in first, the bass at bar 9, and the beat steps out twice, for four bars at bar 25 and four more at bar 45 when the bass does.

Hip-hop Track 2

Hip-hop Track 1, duplicated, with:

TASTE	`CHARACTER 12-BIT 26K`	`copied to USER A 01`	
TRACK 1	`TASTE`	`LOOP 2 BARS`	`the break, with a fill`
TRACK 3	`VOLT`	`four bars: D minor, then G minor`	
TRACK 4	`STRIKE`	`LOOP 1 BAR`	`CL HAT, VELO 60, swung`

6.10 Setting up

1. On STEP, load hip-hop Track 1, then ■ SONG › 4 ACTION › **DUPLICATE** › *hold* 4, and name the copy.
2. On TRACKS, unmute any track that says MUTED.

6.11 Twelve bits

On TASTE, press ■ CHARACTER, the top button on the right, and turn encoder 1 two clicks, to **12-BIT 26K**.



Figure 6.5 Taste's CHARACTER page on 12-BIT 26K: twelve bits at 26 kHz, the old hip-hop samplers' way of playing a sound back. The star: a change to the factory sound, not kept.

Listen to the break change: darker, grainier, a little crunchy round the snare. That's Taste playing its sound back the way the SP-1200 did. A sampler stores a sound as a long row of numbers, thousands a second, each one the height of the wave at that moment. A CD uses 16 bits for each number, so the heights come in 65,536 steps; 12 bits give only 4,096, and the coarser steps add a faint fizz. And 26,000 numbers a second instead of 44,100 can't hold the very top of the sound, so the cymbals go dull and a metallic edge creeps in. On paper it's worse. To a generation of listeners it was the sound of hip-hop, and producers with far better machines still choose it. The middle choice, 12-BIT, is the MPC60's: gritty, with more of the top left in.

Keeping it

There's a star in Taste's top bar: you've changed a factory sound, and a factory sound is never changed for good. That matters here, because when you press PLAY from the start Step chooses each remembered sound again, and choosing a sound drops any changes to it. So make the gritty break a sound of your own:

1. On TASTE, ■ SOUNDS, and *hold* 3, COPY TO. It copies what you're hearing, 12 bits and all, into USER A 01, and Taste is now playing your copy.
2. On STEP, ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4, so the song remembers your copy instead of the factory break.

6.12 Two bars of beat, four of bass

The beat, with a fill

A beat that's the same every bar starts to sound like a machine. A drummer keeps the groove but adds something at the end of a phrase, a little roll or an extra snare, to say "here comes the next one": a **fill**.

1. Clear track 1: ■ TRACKS › ❶ TRACK › 1 › *hold* ❹. Then turn encoder 3, LOOP, up one click, to 2 BARS.
2. On ENTRY, BAR to 1, LENGTH 1/16, and on TASTE write bar 1 as in Track 1 (section 6.5), then carry straight on into bar 2:

Bar 2, beat 1	C3	C#3	D3	D#3
Beat 2	E3	F3	C3	G3
Beat 3	G#3	A3	A#3	**G3**
Beat 4	C4	**D#4**	**C4**	D#4

The first half of bar 2 is bar 1 again. Then slice 8, a ghost snare, comes early, and the last beat is snare, ghost, snare, ghost: the snare doubled, tumbling over the bar line into bar 1.

Four bars of bass

1. On TRACKS, turn to track 3; on ENTRY, BAR to 1, and press ■ COPY: the range is bar 1.
2. Turn END on 4 beats and TO on 4: bars 1 and 2 will be copied to bars 3 and 4. *hold* ❹.
3. Push encoder 1 for TRANSPOSE. Turn END on 8 beats, then START on 8, so the range is bars 3 and 4, SEMITONES to +5, and *hold* ❹.

Bars 3 and 4 are now up five semitones, a fourth, from D to G, the same move as house Track 1's bass. Two bars in D minor, two in G minor: the line goes away and comes home, and it takes four bars to come round, so a verse of sixteen goes round it four times.

Swung hats

1. On TRACKS, turn to track 4, give it STRIKE, and LOOP 1 BAR.
2. On ENTRY, BAR to 1, VELO down to 60, LENGTH 1/16, and on STRIKE, four times over: rest, CL HAT, rest, CL HAT. Then VELO back to 100.

Soft hats on the "e" and the "a" of every beat, the semiquavers the swing moves, as in house Track 1. The break has hats of its own, and these sit between them, very quietly: you'll feel the beat lean back further before you hear them.

6.13 Playing Track 2

On TRACKS, mute tracks 2, 3 and 4, and press ■ PLAY.

Bars	As it starts	A rapper would hear
1-4	The beat alone.	The intro.
5-8	Unmute tracks 2 and 4: the 808 and the hats.	
9-24	Unmute track 3: the bass.	The first verse.
25-28	Mute tracks 1 and 4: the 808 and the bass.	A drop-out.
29-44	Unmute tracks 1 and 4: everything.	The second verse.
45-48	Mute tracks 2 and 3: the beat and the hats.	A breath before the last verse.
49-60	Unmute tracks 2 and 3: everything.	The third verse.
61-64	Mute tracks 3 and 4: the beat and the 808. STOP at the end of bar 64.	The outro.

> **Try this**
>
> Try the other CHARACTER choices while it plays, CLEAN and 12-BIT, and decide which you like. Put the fill somewhere else in bar 2, or make the loop four bars with the fill only in the fourth. And try another of Taste's beats: on SOUNDS, DUSTY BEAT and BREAK BACK are the same break cut other ways.

6.14 Track 3: your own voice

Every third track so far has been yours to choose. This one is yours more than any: you'll record your own voice on Taste, cut it into words, and play the words as a hook, the catchy line that comes round between verses. Not a rap: a few words, chopped and played as an instrument, the way producers have chopped voices off records since the 1980s. Section 6.18 has one way it might go, and an apology.

Start from Track 2

1. On STEP, load hip-hop Track 2, then ■ SONG › 4 ACTION › **DUPLICATE** › *hold* 4, and name the copy.
2. On TRACKS, unmute any track that says MUTED.

Taste is still playing USER A 01, the copy you made in section 6.11. That matters: a recording always goes into a sound of your own, never a factory one. (If you try to record onto a factory sound, the screen says `COPY IT TO RECORD`.)

What to say

Think of four to six short words, or a short phrase, and say them with a clear gap after each one, about half a second: the gaps are how Taste will find the words. Anything you like. A name, a place, a line of your own, the sound of a word you like saying.

6.15 Recording

1. On TASTE, press E4, the key just after the break's sixteen. That's where the recording will go.
2. Press ■ SAMPLE, the top button on the left.
3. Turn encoder 1, SOURCE, one click to **MIC IN**, for a microphone in the Root's LINE/MIC IN socket, or two to **BUILT-IN MIC** for the one in the Root's top. (SOURCE moves one choice a click, however fast you turn it.)
4. Turn encoder 2, THRESH, up eleven clicks, to -30 DB.



Figure 6.6 Taste's SAMPLE page on key E4, ready to record: SOURCE MIC IN, THRESH -30 DB (the red mark on the level bar), so recording starts only when you make a sound louder than that.

THRESH is the threshold: how loud a sound must be to start the recording. At -30 DB, Taste waits through the hum of the room and starts the moment you speak, so the recording begins exactly on your first word. Say something now and watch the bar under the middle of the screen: it's how loud the microphone hears you. The red mark is the threshold, and your words should go well past it.

5 Push encoder 4, REC. It says `WAITING`.

6 Say your words, with the gaps.

7 Push encoder 4 again, STOP.

Your voice is on E4: press it and you'll hear the whole recording. If it's too quiet, or cut off, or you just want another go, push REC and record again: each take replaces the last.

6.16 Chopping, and loudness

A word on each key

The break was cut into even sixteenths, because drums play in time. Words don't, so Taste cuts them where they start.

1 Press ■ CHOP, the fourth button down on the left.

2 Turn encoder 2, MODE, one click to **AT HITS**.

3 Turn encoder 3, SENS, right down to 0.0. (It moves a tenth a click: turn quickly.)

4 *hold* 4, CHOP.



Figure 6.7 CHOP on key E4 after the recording is chopped AT HITS: one slice a word. The darker one is the slice on this key, YEAH; the five after it are on the next five keys. The line at the bottom is what chopping again from this key would make.

Taste listens through the recording for each sudden jump in loudness, the start of a word, and cuts there: one slice on E4, the next on F4, the next on F#4, and on up the keys, black and white. Press them in turn and your words come back one at a time. SENS is how small a jump counts. At 0.0 only the big ones do, the starts of words after a gap; turn it up and Taste cuts at smaller jumps too, the hard sounds inside a word, a K or a T, so the words split into pieces. If two words have come out on one key, leave a longer gap between them; if a word has split in two, try SENS a little lower. Either way, press E4, record again on SAMPLE, and chop again: CHOP always cuts from the key you're on.

As loud as it goes

A voice into a microphone usually comes out quieter than the break. Press ■ MIX, the fifth button in the middle column, and *hold* 4, NORMAL. Taste turns the recording up until its loudest moment is as loud as a sound can be, and says how much louder it made it. The screen then says FULL.

6.17 The hook

1. On STEP, on TRACKS, turn to track 5, give it TASTE, and turn LOOP up two clicks, to 2 BARS.
2. On ENTRY, BAR to 1, LENGTH 1/8, and play your words in on Taste's keys, with rests between, across two bars.

Our six words went like this, a key each:

Bar 1	E4	rest	rest	rest	F4	F#4	rest	rest
Bar 2	G4	rest	G#4	rest	A4	rest	rest	rest

One word, a gap, two together, then the last three spaced out, each word starting on a quaver, in time with the break. That's what chopping a voice gives you that a recording doesn't: the words fall exactly on the beat, in any order you choose, as often as you like. Play one word three times in a row, or the last word first.

Then on TRACKS mute tracks 2, 3 and 4, so the song starts with the beat and the voice alone.

6.18 One way it might go

Here's ours, and we owe you an explanation.

To make an example for the book, we needed a voice, and the only one in the building that would say the words on demand, take after take, without getting embarrassed, was a computer's: the speaking voice built into an ordinary laptop, reading "Yeah. Check it. Make some noise." in the tone of a railway station announcement. It was recorded exactly as section 6.15 tells you to record yours, through the Root's mic input, then chopped AT HITS and played on track 5. If you remember the BBC's *Look Around You* and its "Music 2000" competition, you'll know what's coming. We're sorry. We did say we knew our limits.

Sound 6.3 Hip-hop Track 3, one way it might go: Track 2 with a voice recorded on Taste, chopped into words and played as a hook. The voice is a laptop's built-in speech standing in for yours, and we apologise.

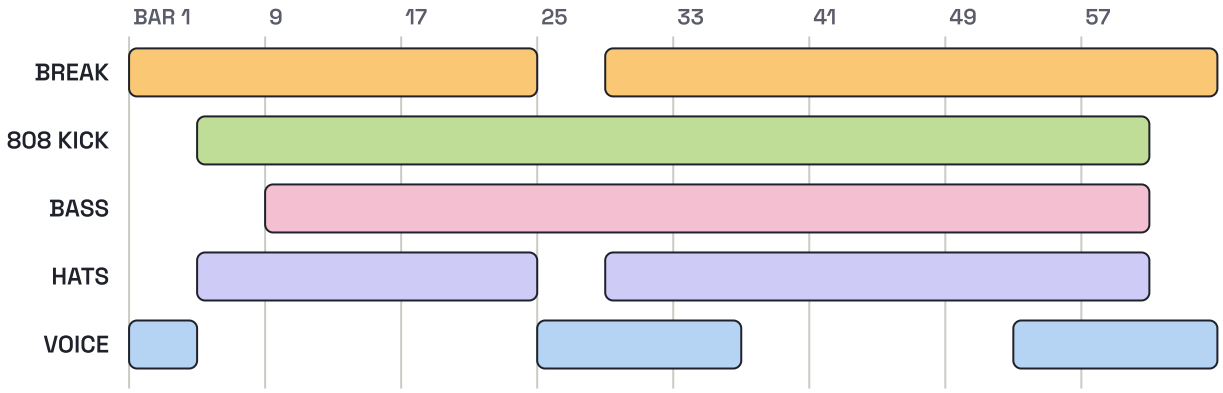


Figure 6.8 Hip-hop Track 3, one way it might go: Track 2's arrangement with the voice on top at the start, through the hook at bars 25 to 36, and again from bar 53 to the end.

Bars	As it starts
1-4	The beat and the voice.
5-8	Mute track 5; unmute tracks 2 and 4: the 808 and the hats.
9-24	Unmute track 3: the first verse.
25-28	Mute tracks 1 and 4; unmute track 5: the 808, the bass and the voice.
29-36	Unmute tracks 1 and 4: the hook, everything.
37-52	Mute track 5: the second verse.
53-60	Unmute track 5: the hook again.
61-64	Mute tracks 2, 3 and 4: the beat and the voice, as it began. STOP at the end of bar 64.

Listen anyway, because it makes the point of this chapter better than we could. Everything else in this book can be written down: every press, every setting, every track, which is why your Track 1 sounds like ours. Your voice can't be. Once it's on Taste, saying your words in your way, you're off the edge of our map, and that's where the music starts. You'll be more creative off the book than this book could ever be. Put your voice where our laptop's is, and you'll hear the difference straight away.

> **Try this**
>
> Record other sounds onto the keys after your words: a hand clap, a cough, a tap on a mug, a friend saying one word. Chop them AT HITS and drop them into the beat on a track of their own. And if you know someone who raps, you have a beat for them now.

6.19 What you've learned

- Where hip-hop came from: breaks, DJs, MCs, and the samplers that cut records into pieces; and that a sample of someone's record needs their permission.
- That a beat is a stage, and its arrangement leaves room for a voice: verses, hooks and drop-outs, in fours.
- Chops: a break sliced into semiquavers, played at a new tempo without changing its pitch, and in a new order.
- Layering an 808 under a recorded kick; swing on a slow beat; a two-bar beat with a fill.
- Bits and sample rates, and why 12 bits at 26 kHz still sounds right.

- Keeping a changed factory sound as a sound of your own, so REMEMBER SOUNDS keeps it too.
- Recording on Taste: SOURCE, THRESH, REC; chopping AT HITS with SENS; NORMAL.

Next, ambient: music with no beat at all, made of slowly drifting sound.

Chapter 7 - Ambient

No beat, no verse and no chorus: chords that drift past each other, a cloud of grains, and a lot of space.

Every chapter so far has had a beat to hold on to. This one lets go of it. Ambient music doesn't march, or dance, or tell a story with a beginning and an end. You can listen to it closely, or leave it playing at the edge of the room while you do something else, and it's meant to work both ways. It's made of the things the rest of this book used as decoration: long notes, slow change, and space. On Step it's made of loops that never line up.

> **What you need for this chapter**
>
> The Root, **Step**, **Glass**, **Haze** and **Grit**, joined in that order, and headphones. Haze hears every unit wherever it is in the row, with one exception: a unit that is itself listening to the chain, as Grit does, adds its sound on the way back, so Haze hears it only if it's on Haze's right. Haze goes to the left of Grit so its reverb takes in Grit's grains as well as Glass. For Track 3, **Axis** too, on the end of the row; if you don't have one, Glass can play Track 3 on EVOLUTION (in its FAC D bank).

7.1 Where ambient came from

In 1917 the French composer Erik Satie wrote what he called *musique d'ameublement*, furniture music: short pieces to be repeated over and over in the background, like wallpaper, while people talked. At its first performance, in 1920, the audience politely stopped talking to listen, and Satie walked round the room telling them to carry on chatting. He was fifty years early.

In 1975 Brian Eno was lying in bed recovering from a road accident when a friend put on a record of harp music for him. One speaker wasn't working and the volume was very low, and he was too sore to get up and fix it. So he lay there hearing the harp mixed with the rain outside, barely there, and realised he liked it that way: music as part of the room, as much a colour as a tune. His records *Discreet Music* (1975) and *Ambient 1: Music for Airports* (1978) gave it a name. *Music for Airports* is made from tape loops of different lengths, each with a single sung note or a short piano phrase on it, all playing at once, so the notes drift together and apart and never meet the same way twice. That's exactly what this chapter's tracks do on Step. The second record of the series, *The Plateaux of Mirror* (1980), was made with the pianist Harold Budd, all slow chords soaked in echo.

In the 1990s ambient came back through the dance floor, as the chapter 2 history said: in the chill-out rooms of raves, and on records such as the KLF's *Chill Out* (1990) and Aphex Twin's *Selected Ambient Works 85-92* (1992), made by a teenager in Cornwall on home-built and second-hand synths. It has never gone away since: long, slow pieces that play for hours, and apps that make music which never ends.

7.2 The room next door

Ambient has a family on the other side of the wall, in the concert hall and the studio, and you'll hear more of what's possible on these units by listening to them than to almost anything else. None of this is background music. Most of it asks you to listen harder than any dance record does. But it's where long sounds, slowly changing textures, grains, drones and strange harmony were first worked out, and the crossover with ambient, and with everything in this chapter, is real. Here are ten places to start, with what each one has to do with your units.

> **Edgard Varèse** called music "organised sound", and meant any sound at all. Chapter 2 met the Philips Pavilion at the 1958 Brussels World's Fair; the piece he made for its hundreds of loudspeakers was *Poème électronique*. Listen to *Ionisation* (1931) too, for percussion alone.

> **Iannis Xenakis** designed that pavilion, working as an architect in Le Corbusier's office, and made the piece played between the shows in it, *Concret PH* (1958): the crackle of burning charcoal, recorded, cut into tiny pieces and piled up into clouds. He was one of the first to think of sound as grains, which is what Grit does. His orchestral *Metastaseis* (1954) has dozens of strings sliding, each on its own line, into great moving masses.

> **György Ligeti**'s *Atmosphères* (1961) has no melody and no beat at all, only enormous chords of notes packed side by side, shifting so slowly you notice only that something has changed. If you've seen Stanley Kubrick's *2001: A Space Odyssey*, you've heard it. Track 2's clusters come from here.

> **Karlheinz Stockhausen**'s *Stimmung* (1968) is six singers on a single chord for about seventy minutes, singing its overtones, the notes hidden inside a note, so the one chord shimmers with colours. It might be the most ambient piece ever written for a concert hall.

> **Éliane Radigue** worked in Paris with the inventors of musique concrète, then spent decades making long, slow drones on a modular synthesiser and tape, pieces in which a sound changes so gradually that you can't hear it change, only find that it has. Start with *Trilogie de la Mort* (1988-93). It's the closest thing there is to this chapter played with infinite patience.

> **Pauline Oliveros** made tape music in San Francisco in the 1960s and then turned to listening itself: she called her practice Deep Listening, listening to everything, all the time. Her album *Deep Listening* (1989) was recorded with two friends at the bottom of an empty underground water tank, whose echo lasted 45 seconds. When you turn Haze to CAVE in Track 3, think of her.

> **Laurie Spiegel** made *The Expanding Universe* (1980) on a room-sized computer system at Bell Labs: slow, glowing music in which simple rules let the notes find their own way, as Step's loops do here. A piece of hers is on the gold records the Voyager spacecraft carried out of the solar system in 1977.

> **Hildegard Westerkamp** makes **soundscape** compositions from recordings of real places. In *Kits Beach Soundwalk* (1989) she walks you along a beach in Vancouver, talking quietly, filtering the city's roar until you can hear the tiny clicks of barnacles. The storm in Track 3 is a very small step in her direction.

> **Trevor Wishart**, from Leeds, transforms sounds into other sounds: in *Red Bird* (1977) words melt into birdsong and the noises of machines. His *Vox* cycle (1980-88) does it with voices, and his book *On Sonic Art* is one of the best ever written about sound. If you liked chopping your voice in chapter 6, he's where that road leads.

> And for the clusters, the first of them: **Henry Cowell**, who in *The Tides of Manaunaun* (1917) told the pianist to press the low keys with a whole forearm.

Put one on, turn the lights down, and don't do anything else for a while. Then come back and make something.

7.3 What ambient is made of

- **No beat you can dance to.** Step still has a tempo, and it still counts bars, but nothing plays on the beat. These tracks are at 70 and 60.
- **Long notes, and space between them.** Silence is part of the music, and a sound is allowed to fade all the way out.
- **Loops that never line up.** Each part goes round on a loop of its own length, so the parts keep meeting in new ways. It's acid Track 2's three beats against four, made into the whole method.
- **Reverb and echo as instruments.** Here Haze isn't the finishing touch: it's half of the sound.
- **Harmony that floats.** With no chord progression to follow, the chords are free to be strange: notes rubbing against each other, packed together, or from no ordinary scale at all. Nothing needs to resolve, because nothing is going anywhere.

7.4 Track 1 at a glance

Ambient Track 1 is five chords on Glass's electric piano, each on a loop of a different length, echoing into a hall.

Sound 7.1 Ambient Track 1, as built in sections 7.5 to 7.8: five chords on Glass's E.PIANO 1, each on a loop of its own length, 17 to 31 beats, so they never meet the same way twice.

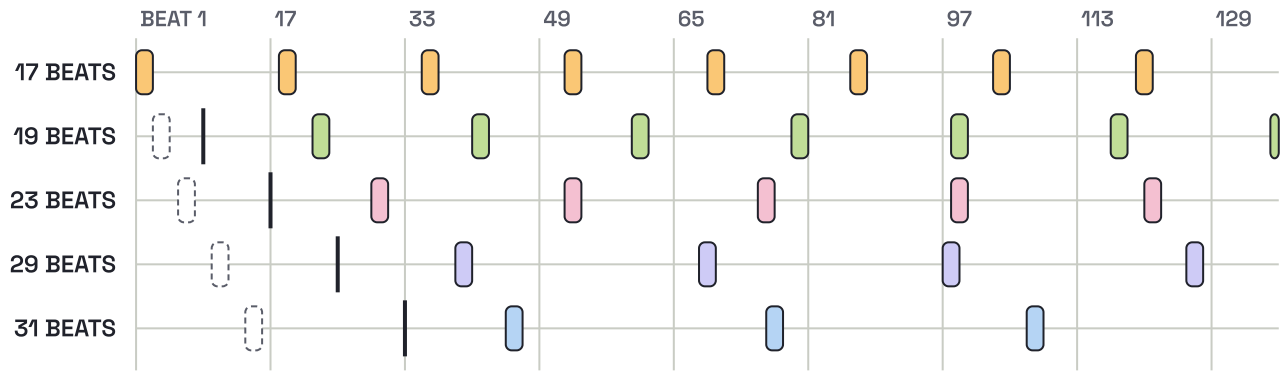


Figure 7.1 Ambient Track 1's five chords over the first two minutes. Each track plays its chord once a time round its own loop, 17, 19, 23, 29 or 31 beats long, so the chords meet in a different way every time: look down any column to see which are sounding together. Tracks 2 to 5 start muted and come in at bars 3, 5, 7 and 9 (the dark lines); a muted loop keeps going round, so the dashed chords before each line are silent, but every chord after it lands where its loop says.

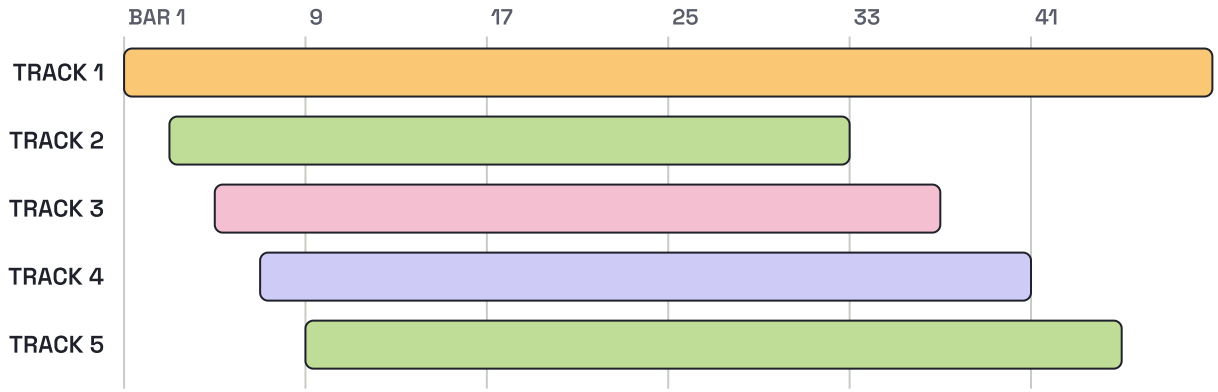


Figure 7.2 How ambient Track 1 plays: one chord, then the other four arriving two bars apart, and leaving in the same order near the end, four bars apart, until only the first is left.

Ambient Track 1

A new song, with:

STEP	`TEMPO 70`
GLASS	`E.PIANO 1 (FA11)`
HAZE	`ECHO HALL` `GLASS 70`
TRACKS 1-5	`GLASS` `LOOPS of 17, 19, 23, 29, 31 beats` `a chord each`
MIXER	`GLASS -6`

7.5 Setting up

1. Join the units in this order from the Root: Step, Glass, Haze, Grit.
2. On STEP, ■ SONG › 4 ACTION › `NEW` › *hold* 4, and name it.
3. On GLASS, ■ SOUNDS › 1 BANK › `FAC A` › 2 SOUND › `E.PIANO 1`.
4. On STEP, ■ TEMPO › 1 TEMPO › `70`.

Five loops of their own

On TRACKS, give tracks 1 to 5 to GLASS with encoder 2. Then, on each track, push encoder 3, LOOP, so it counts in beats (the screen says BARS + BEATS underneath), and turn it on to the track's length:

Track	1	2	3	4	5
Beats	17	19	23	29	31
LOOP says	4+1	4+3	5+3	7+1	7+3

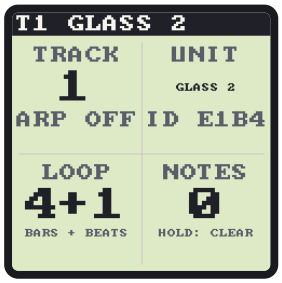


Figure 7.3 TRACKS on track 1, LOOP pushed for beats: 4+1, four bars and one beat, seventeen beats round.

Why those numbers? They're all **prime numbers**, which nothing divides into but themselves and 1. Two loops of 16 and 8 beats line up every 16 beats, so you'd hear the same meeting over and over. Loops of 17 and 19 beats line up only every 323 beats, and all five of these only every 6.6 million beats, which at this tempo is about two months. So in practice the chords never meet the same way twice. Eno's tape loops in *Music for Airports* were cut to lengths like that for the same reason.

7.6 Five chords

Each track has one chord, written by holding its four keys on Glass together and letting them all go, as you did for techno Track 3's stab. A few crotchet rests before each one start the loops off at different moments, so they don't all begin together.

On STEP, (■ ENTRY), then, a track at a time, BAR back to 1 each time:

Track	Rests (LENGTH 1/4)	Then LENGTH 1/2, the chord	What it is
1	none	D3 A3 E4 F4	D minor, with E and F a semitone apart at the top
2	2	E3 A3 D4 G4	fourths, stacked
3	5	F3 C4 E4 G4	F major 9, with no third
4	9	B3 C4 E4 A4	A minor, with B and C rubbing underneath
5	13	G3 B3 C#4 F#4	G Lydian: the odd one out

Play each one on its own before you write it, and listen to what makes it strange. Track 1's chord is D minor, but its top two notes, E and F, are side by side on the keyboard, a semitone apart, and they rub against each other like two strings slightly out of tune. Track 4 has its rub at the bottom. Track 2 isn't built in thirds, every other note, like most chords, but in fourths: E, A, D, G, open and hollow, the sound of a lot of jazz and film music. Track 3 leaves out F major's third, A, the note that says major or minor, so it's neither. And Track 5 brings in C# and F#, notes none of the other chords have. It's built on the Lydian scale, a major scale with its fourth note raised, which sounds bright and slightly unearthly; but the moment it meets track 1's F or track 4's C, C# and F# clash with them a semitone apart, and for a few seconds the harmony goes somewhere nobody planned.

That's the point. You've written five chords, but you'll hear dozens, made by whichever ones happen to be sounding together. Some meetings are sweet, some sour, some odd; none of them were written. Figure figure 7.1 shows a couple of minutes of them.

7.7 Space, balance and remembering

1. On HAZE, press the 8th white key from the top: the screen says **ECHO HALL**, an echo that repeats into a big hall.
2. ■ SENDS, and turn encoder 1, GLASS, up from 30 to 70: most of Glass goes into the echo.
3. On STEP, ■ MIXER, and turn GLASS down to -6. Four-note chords piling up get loud. (MIXER's boxes are the units in order: GLASS, HAZE, GRIT.)
4. ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4.

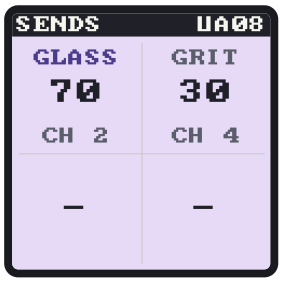


Figure 7.4 Haze's SENDS page in Track 1: GLASS at 70, sent into ECHO HALL; GRIT at the preset's 30, though it isn't playing yet.

7.8 Playing Track 1

On TRACKS, mute tracks 2 to 5, so only track 1 plays, and press ■ PLAY. A bar at 70 lasts nearly three and a half seconds, so there's time; follow the bar number on Step's screen.

At bar	Do
1	Track 1 alone.
3, 5, 7, 9	Unmute tracks 2, 3, 4 and 5, one at each.
33, 37, 41, 45	Mute them again, in the same order.
48	STOP at the end of the bar, and let the echo die away.

Then just listen. There's no chorus coming, and no drop. Notice how differently the same chord sounds depending on what it meets: track 1's D minor alone, then under the fourths, then against the Lydian chord's C#. You'll find yourself waiting for particular meetings, and they won't come when you expect.

Try this

Change a loop's length while it plays, a beat at a time, and hear the meetings change. Make your own chords: start from three notes that sound good and move one of them a semitone, then listen to the rub. Try a sixth track with one low note on a very long loop, 37 or 41 beats. (Glass plays up to sixteen notes at once; if more are asked for, the oldest stop, so long chords on many tracks can cut each other short.)

7.9 Track 2 at a glance

Ambient Track 2 keeps Track 1's loops and gives them denser chords, held for a whole bar on Glass's voices. Grit listens to it all and trails a cloud of grains behind it, and halfway through, the cloud freezes and the chords stop.

Sound 7.2 Ambient Track 2, as built in sections 7.10 to 7.13: clusters on Glass's VOICE 1, a cloud of grains from Grit following them, and eight bars frozen in the middle.

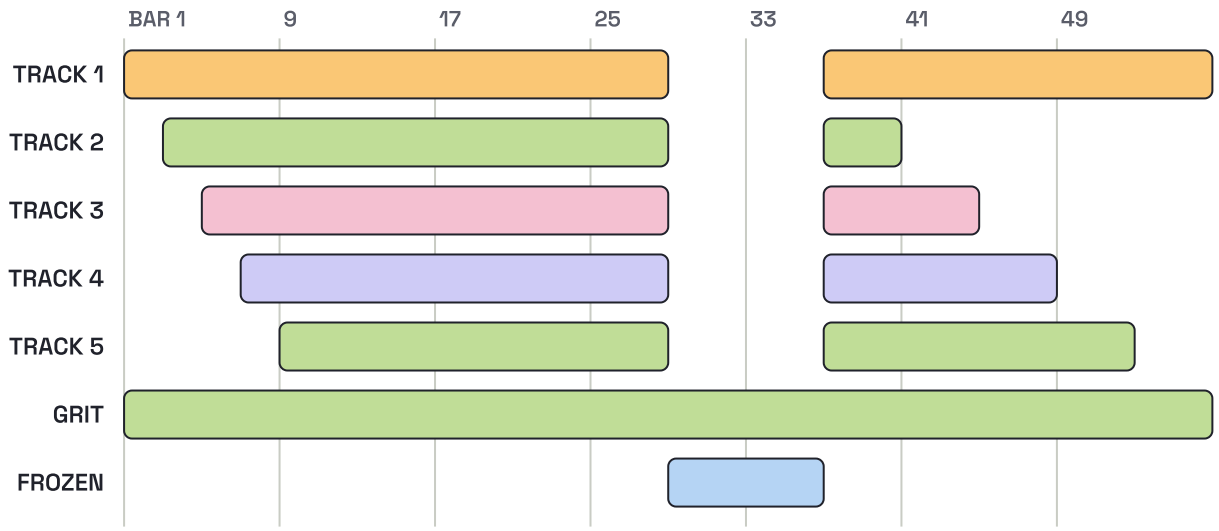


Figure 7.5 How ambient Track 2 plays: as Track 1, but at bar 29 Grit freezes its cloud and every track stops, and for eight bars there's only the frozen sound and its reverb; then the chords come back.

Ambient Track 2

Ambient Track 1, duplicated, with:

GLASS	`VOICE 1 (FA30)`
TRACKS 1-5	`the same loops` `clusters, LENGTH 1`
GRIT	`LIVE CLOUD (FA27)`
HAZE	`GLASS 50` `GRIT 70`
MIXER	`GLASS −4`

7.10 Setting up

1. On STEP, load ambient Track 1, then ■ SONG › 4 ACTION › **DUPLICATE** › *hold* 4, and name the copy. On TRACKS, unmute any track that says MUTED.
2. On GLASS, choose FAC A's VOICE 1 on SOUNDS. (If Glass is already showing SOUNDS, just turn encoder 2: pressing SOUNDS again opens the CARD page.)
3. Clear each of the five tracks: ■ TRACKS › 1 TRACK › **1** › *hold* 4, then the same for 2, 3, 4 and 5. Clearing takes away a track's notes but keeps its LOOP.

7.11 Clusters

Write the five tracks again on ENTRY, the same rests as Track 1, but each chord with LENGTH 1, a whole bar:

Track	Rests (1/4)	Then LENGTH 1	What it is
1	none	A3 B3 C4 D4	a cluster: four notes side by side
2	2	E4 F4 G4 A4	another, higher
3	5	C3 G3 D4	open fifths, holding the others down
4	9	F#4 G#4 A#4 C5	whole tones
5	13	D3 G#3 C4 F#4	tritones, stacked

A **cluster** is a chord of notes packed side by side instead of spread out: Cowell's forearm on the keys, Ligeti's *Atmosphères*. On a piano it's a thud; on held voices, with reverb, it becomes a single shimmering colour, beating gently inside as the close notes rub. Track 3's open fifths, C, G and D, a fifth apart each, give the ear something solid to stand on. Track 4 is four notes a whole tone apart, from the **whole-tone**

scale, six notes all a tone apart, that Claude Debussy used to make music that seems to float with no home key. And Track 5 stacks **tritones**: D to G#, C to F#, the interval exactly half an octave, which musicians once nicknamed the devil in music. Together, sounding in shifting combinations, these give harmony you won't find in any song, and in ambient that's exactly the point.

Then on STEP, ■ MIXER, and turn GLASS up two clicks, to -4: the voices are softer than the piano.

7.12 A cloud of grains

1. On GRIT, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **LIVE CLOUD**.
2. On HAZE, ■ SENDS: GLASS down to 50, GRIT up to 70.
3. On STEP, ■ SONG › ❹ ACTION › **REMEMBER SOUNDS** › *hold* ❹.
4. On GRIT, press ■ PLACE, ready for later.

LIVE CLOUD listens to everything on the chain, keeps the last few seconds of it, and plays it back as a haze of tiny overlapping grains, each a fraction of a second long, about a second behind. So everything Glass plays comes back blurred, like breath on a window, and that's what Xenakis heard in his charcoal.

7.13 Playing Track 2, and freezing it

On TRACKS, mute tracks 2 to 5, press ■ PLAY, and bring them in as in Track 1. Then, in the middle, the freeze:

At bar	Do
1	Track 1 alone.
3, 5, 7, 9	Unmute tracks 2, 3, 4 and 5.
Just before 29	On GRIT, press FREEZE, the top button on the right, and turn encoder 2, SCAN, down 20 clicks, to 0%. Then on STEP mute all five tracks.
37	On GRIT, SCAN back up to 100%, and FREEZE again. On STEP, unmute all five.
41, 45, 49, 53	Mute tracks 2, 3, 4 and 5 again.
56	STOP at the end of the bar.

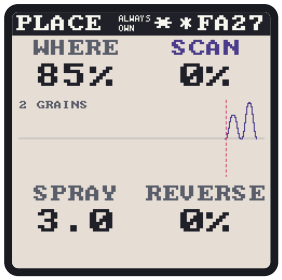


Figure 7.6 Grit's PLACE page with FREEZE on and SCAN turned down to 0%: the grains stay where they are in the frozen sound, and the moment it caught goes on for as long as you like.

FREEZE stops Grit listening: the last few seconds it heard are held, and the grains go on playing from them. SCAN is how fast the grains move through that sound, 100% as fast as it happened, 0% not at all, so at 0% the frozen moment simply goes on, a chord caught in mid-air and held. For eight bars the chords have stopped and there's only that frozen cloud and its reverb. Then they come back, Grit lets go, and the cloud starts following them again.

> ### Try this
> During the freeze, turn SCAN slowly up and down, or below 0%, so the frozen sound plays backwards, and try SPRAY on encoder 3. Freeze at a different moment and catch a different chord. Try Grit's other live sounds, LIVE ICE and LIVE SHINE.

7.14 Track 3: your own

Duplicate Track 2 and make it yours. Section 7.16 has one way it might go, and section 7.15 the new things it uses. Ambient is the easiest music in the world to make beautifully dull, and ours is a careful demonstration of the parts, not of what they can do; the composers in section 7.2 show what they can do.

- **The tempo.** Slower is often better: everything stretches out. 60 or below.
- **The chords**, your own, as dense or as odd as you dare. Move one note of a chord a semitone and listen.
- **The loops.** Other primes: 37, 41, 43, 47. Or give a track a second chord, so its loop becomes a little phrase.
- **The space.** Haze's other reverbs, CAVE, BIG HALL, AMBIENT, and how much of each sound you send into them.
- **A place.** Grit's sounds that play by themselves, such as RAIN CLOUD or FAR STORM, set the music somewhere, as Westerkamp's recordings do.

7.15 A warmer pad, and a storm far off

Moving the tracks to Axis

Glass is a bright instrument, good at bells, pianos and glass. For a warm, soft pad, Axis is the better choice, and you don't have to write anything again to use it: a track can play any unit.

1. Join Axis to the right-hand end of the row, after Grit.
2. On STEP, on TRACKS, for each of tracks 1 to 5, turn encoder 2, UNIT, on from GLASS to AXIS. The notes stay; Axis plays them now.
3. On AXIS, ■ SOUNDS › ❶ BANK › `FAC A` › ❷ SOUND › `AIR PAD`.

A touch darker, and a sound of your own

AIR PAD is airy and high, and a little sharp for this. Press ■ FILTER on Axis and turn encoder 1, CUTOFF, down 20 clicks, from 90 to 70: the screen says 3.4 KHZ. The filter takes off everything above that, and the pad goes softer and warmer.

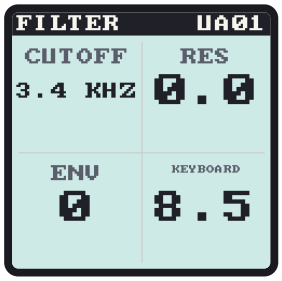


Figure 7.7 Axis's FILTER page on the copy of AIR PAD, CUTOFF turned down from 90 to 70: the top of the sound taken off, so it's softer and warmer.

There's now a star in Axis's top bar: a change to a factory sound, which isn't kept, and which REMEMBER SOUNDS would undo every time you press PLAY from the start. So keep it, as you kept Taste's gritty break in chapter 6: on AXIS, ■ SOUNDS › *hold* ❸, COPY TO. Your darker AIR PAD is now USER A 01, a sound of your own, and Axis is playing it. Any sound you change and like, on any unit, is worth keeping that way.

Second chords

Give two of the loops a second chord, straight after the first, so they become little two-chord phrases. On ENTRY, for each, BAR to 1, LENGTH 1/4, BAR on past the rests and the first chord, then LENGTH 1 and the chord, on Axis's keys:

Track	BAR on	Then the chord
2	6 clicks	D4 E4 G4 A4
4	13 clicks	E4 F#4 G#4 B4

The storm, and the cave

1. On GRIT, choose FAC A's FAR STORM on SOUNDS: rain and distant thunder, which plays by itself.
2. On HAZE, press the 3rd white key from the top, CAVE, the longest reverb, then ■ SENDS and turn GRIT and AXIS to 60.
3. On STEP, ■ MIXER: AXIS down to -4, GRIT down to -12.
4. ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4.

The storm is quiet, and more of it goes into the reverb than the pad's does: that's how to put a sound far away. Close sounds are loud and clear; distant ones are quieter, and you hear more of the space around them than of the sound itself.

7.16 One way it might go

Track 2, duplicated, slower, at 60, moved to Axis's darker AIR PAD, with two second chords, and a storm a long way off in Haze's CAVE.

Sound 7.3 Ambient Track 3, one way it might go: Track 2 at 60, moved to Axis's AIR PAD made darker, with two second chords, and a storm a long way off in Haze's CAVE.

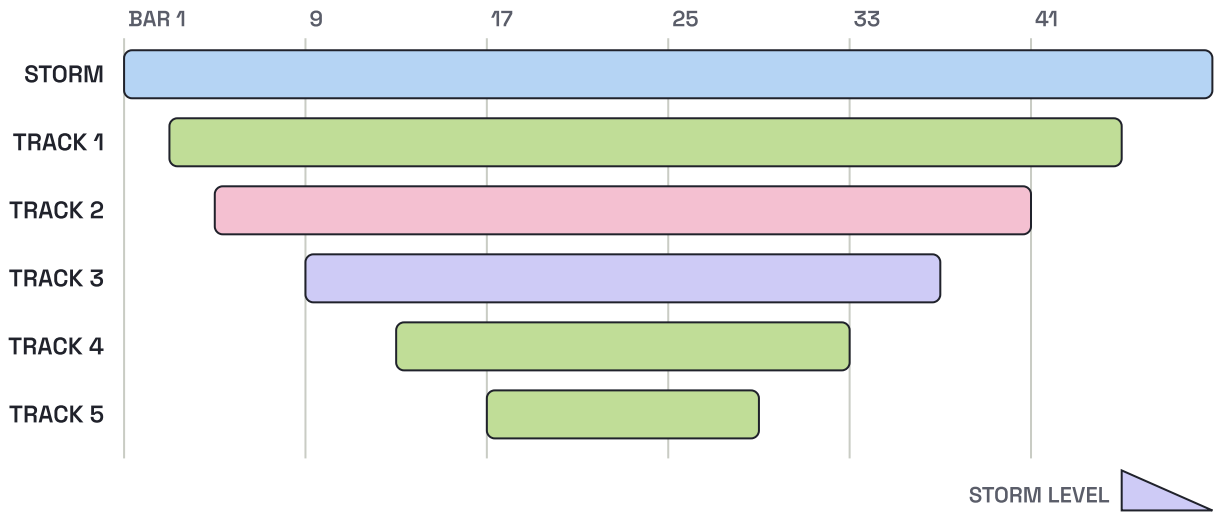


Figure 7.8 Ambient Track 3, one way it might go: the storm first and last, a long way off, and the five tracks arriving further and further apart, then leaving in the opposite order.

On TRACKS, mute all five tracks, and press ■ PLAY: the storm starts on its own.

At bar	Do
1	The storm alone.
3, 5, 9, 13, 17	Unmute tracks 1, 2, 3, 4 and 5, one at each, further and further apart.
29, 33, 37, 41, 45	Mute tracks 5, 4, 3, 2 and 1, the other way round.
45-48	On GRIT, press ■ MIX and turn LEVEL down a little every half bar, so the storm has gone by the end.
48	STOP at the end of the bar.

The storm plays whether Step is playing or not, so it's faded out by hand. That's often how ambient pieces end: not with a final chord, but by going away.

7.17 What you've learned

- Where ambient came from, Satie to Eno to the chill-out room, and the composers next door: Varèse, Xenakis, Ligeti, Stockhausen, Radigue, Oliveros, Spiegel, Westerkamp, Wishart, Cowell.
- Loops of different lengths on every track, in beats, and why prime numbers never line up.
- Chords that float: semitone rubs, fourths, a missing third, the Lydian scale, clusters, whole tones and tritones; and harmony made by accident, from meetings nobody wrote.
- Clearing a track but keeping its loop; moving tracks to another unit.
- Grit on a live sound, FREEZE, and SCAN.
- A filter to darken a sound, and keeping a changed sound as your own.
- Distance: quieter, and more reverb than sound.

Next, the opposite: jungle and drum and bass, breaks at twice the speed and a bass you feel in your chest.

Chapter 8 - Jungle and drum and bass

A break sped up until it blurs, cut up and thrown about, over a bass that moves at half the speed.

Chapter 6 cut a break into pieces and put it back together at 90. This chapter does the same thing at almost twice the speed, and it changes everything. A funk drummer's few bars, sped up until they blur, cut up and rearranged faster than any drummer could play, over a bass line moving at half the speed, deep enough to feel in your chest: that's jungle, and drum and bass is what it grew into.

> **What you need for this chapter**
>
> The Root, **Step**, **Taste**, **Strike**, **Volt** and **Haze**, joined in that order, and headphones. Haze comes in with Track 2.

8.1 Where jungle came from

By 1991 the British rave scene of chapter 2 was speeding up. Its producers, many of them teenagers with a sampler, a computer and a pair of turntables, took the breaks hip-hop had dug up and played them faster and faster, over the basslines of the reggae sound systems they'd grown up with in London, Bristol and Birmingham. The music was called hardcore, then breakbeat hardcore, and around 1992 to 1994 it became jungle: breaks at 160 to 170 beats a minute, chopped and rearranged, with a bass line underneath at half the speed, as slow and heavy as dub. It was spread by pirate radio stations broadcasting from tower-block roofs, and by tapes of the DJs' sets, with MCs chatting over the top in the way of the sound systems.

The most famous break of all is about six seconds long: the drum solo in "Amen, Brother", the B-side of a 1969 single by a soul band called the Winstons, played by their drummer Gregory Coleman. Jungle producers used it so often that it became the sound of the music; it has since been sampled on thousands of records, in jungle, hip-hop, pop and adverts. Coleman was never paid for it, and died homeless in 2006. Remember the box in section 6.1: a sample is someone's work.

In 1994 jungle reached the charts: Shy FX and UK Apachi's "Original Nuttah", and M-Beat and General Levy's "Incredible", which got into the top ten. Goldie, a graffiti artist from the West Midlands, started the Metalheadz label that year with the DJs Kemistry and Storm, and his album *Timeless* (1995) showed jungle could be long, orchestral and sad as well as fierce. DJ Rap was one of the scene's first star DJs, and LTJ Bukem made it smooth and spacious, close to the ambient of chapter 7. In 1997 Roni Size and Reprazent's *New Forms*, from Bristol, won the Mercury Prize.

By the late 1990s the music was mostly called drum and bass, and it had tightened up: cleaner drums, often the "two-step", a kick and snare pattern you'll build in Track 3, at 170 to 176, and darker, heavier basses. The favourite of those came from Detroit: Kevin Saunderson, one of the techno pioneers of chapter 3, recording as Reese in 1988, made a bass of two slightly detuned sawtooth waves beating against each other, and drum and bass producers took it, and its name, as their own. Drum and bass is still everywhere: in clubs, at festivals, in pop records, and in its own endless offshoots.

8.2 What jungle is made of

- **The tempo.** Fast: 160 to 176. These tracks are at 170 and 172.
- **A break, sped up.** Faster and higher, cut up and rearranged: the "edit".
- **A bass at half speed.** The drums run at 170, the bass moves as if the tempo were 85, the speed of dub and reggae. Your head follows the drums; your body follows the bass.
- **Sub bass.** So low it's felt more than heard, from the reggae sound systems.
- **Fills and edits** at the end of every phrase, eight bars at a time, as in techno, but far busier.
- **Echo throws.** A single moment sent into an echo, from dub.

As with hip-hop, a lot of jungle was made to be played with an MC on top, chatting and toasting in the sound-system style. That's theirs, and as in chapter 6 we'll stay behind the decks.

8.3 Track 1 at a glance

Jungle Track 1 is Taste's break turned up to 170 and chopped, two bars of it rearranged, an 808 kick underneath, and a half-time sub bass on Volt.

Sound 8.1 Jungle Track 1, as built in sections 8.5 to 8.8: the break turned up ten semitones to 170 and chopped, two bars of it rearranged, an 808 kick under it and a half-time sub.

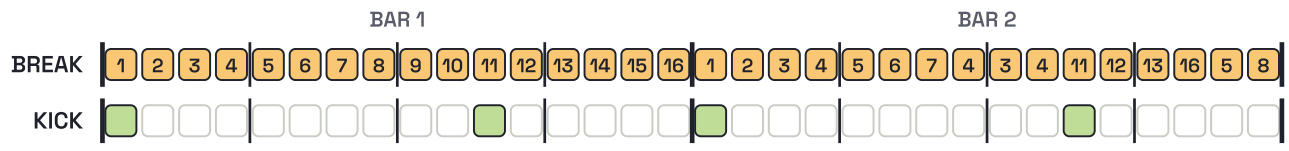


Figure 8.1 Jungle Track 1's two bars of break, by slice (the key's number, C3 = 1 up to D#4 = 16), and the kick under them. Bar 1 is the break as it was played; bar 2 is the edit.

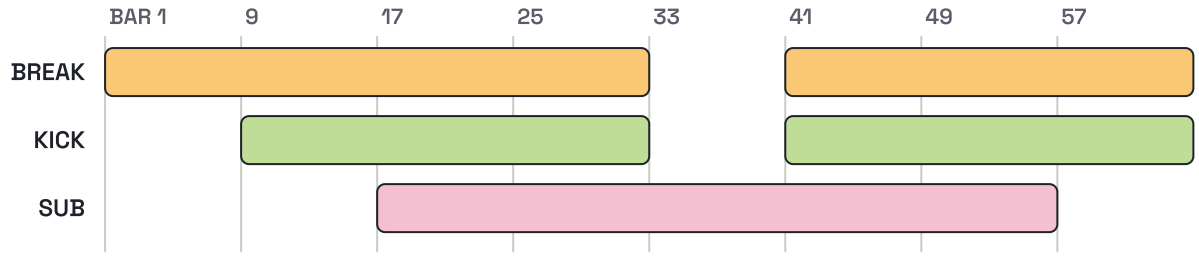


Figure 8.2 How jungle Track 1 plays: the break alone, then the kick, then the sub; eight bars of the sub on its own; everything; and the drums to finish.

Jungle Track 1

A new song, with:

```
STEP    TEMPO 170
TASTE   BREAK SYNC (FA21), copied   COARSE +10   CHOP 16
STRIKE  BOOM
VOLT    SUB BASS (FA10)
TRACK 1 TASTE   LOOP 2 BARS   the break, edited
TRACK 2 STRIKE  LOOP 1 BAR    KICK
TRACK 3 VOLT    LOOP AUTO     the sub, four bars
MIXER   TASTE 0   STRIKE -6   VOLT +3
```

8.4 Speeding up a break

In chapter 6, chopping let the break play at a new tempo without changing its pitch. Jungle did it the other way, on purpose. On the old samplers, the simple way to make a break faster was to play it higher, like a record at the wrong speed, and jungle's producers loved what that did: the drums got faster, tighter and higher, with a snap that playing them faster any other way doesn't give. So that's what you'll do.

Taste's starter break was played at 95. Every semitone up plays it about 6% faster; twelve, an octave, plays it at twice the speed. Ten semitones up is 1.78 times as fast: 95 times 1.78 is 169, just about the 170 of this track. Then CHOP cuts it into sixteen slices, so Step can rearrange it.

8.5 Setting up

A new song, and a break of your own

1. Join the units in this order from the Root: Step, Taste, Strike, Volt, Haze.
2. On STEP, ■ SONG › 4 ACTION › NEW › *hold* 4, and name it. Then ■ TEMPO › 1 TEMPO › 170, and leave Step on TEMPO while you set up Taste.
3. On TASTE, ■ SOUNDS › 1 BANK › FAC A › 2 SOUND › BREAK SYNC.
4. You're about to change it, so make it a sound of your own first. Turn encoder 3, COPY TO, to an empty slot, one that says INIT SOUND, and *hold* 3.

> **Copying over a sound you made**
> COPY TO stays on the last slot it copied to. If you made the hip-hop tracks, it's still on USER A 01, which holds your gritty break, and copying there would replace it. Turn it on to USER A 02, or any slot that says INIT SOUND.

> **Why Step stays on TEMPO**
> While Step is on ENTRY, every key you press on another unit is written into Step's open track. Setting up Taste means pressing its keys to choose which slice to change, so keep Step on another page, TEMPO or SONG, until the setting up is done.

Faster, and chopped

1. On TASTE, press C3: BREAK SYNC has the whole bar there.
2. Press ■ TUNE. Turn encoder 3, BEATS, right down to OFF (BREAK SYNC had it fitted to the tempo), then encoder 1, COARSE, up to +10.
3. Press ■ CHOP. It opens on SLICES 16 and MODE EVENLY: *hold* 4.



Figure 8.3 Taste's TUNE page on C3 of your copy of BREAK SYNC: COARSE +10 semitones, so the break plays 1.78 times as fast, and BEATS OFF, so nothing fits it back to the tempo.

Press C3 to D#4 slowly in turn: the sixteen slices of a break that's a whole bar at 170. Each slice also keeps BREAK SYNC's choke, "one key at a time", so each one cuts off the one before; when Step rearranges them, no slice rings on over the next and smudges the beat.

The kit, the sub and three tracks

1. On STRIKE, ■ BOOM.
2. On VOLT, ■ SOUNDS › ❶ BANK › FAC A › ❷ SOUND › SUB BASS. No OCT this time: SUB BASS already sounds two octaves below the keys.
3. On STEP, on TRACKS: track 1 TASTE, LOOP 2 BARS; track 2 STRIKE, LOOP 1 BAR; track 3 VOLT, LOOP AUTO.

8.6 The edit, the kick and the sub

Two bars of break

On STEP, ■ ENTRY, LENGTH 1/16, BAR at 1. On TASTE, bar 1 is the break as it was played: C3, C#3, D3 and on up every key, black and white, to D#4. Then carry on into bar 2, the edit:

Bar 2, beat 1	C3	C#3	D3	D#3
Beat 2	E3	F3	F#3	**D#3**
Beat 3	**D3**	**D#3**	A#3	B3
Beat 4	C4	**D#4**	**E3**	**G3**

The first half is the break as it was; then the kick (D#3 is slice 4, a kick) comes early, twice, pushing beat 3 off balance, and the bar ends snare, ghost, snare, ghost, tumbling into bar 1. A bar as played and a bar cut up, round and round: that's the pattern of a lot of jungle.

The kick

Track 2, LENGTH 1/16, BAR at 1, on STRIKE: KICK, nine rests, KICK, five rests: under the break's own kicks on beat 1 and the "and" of beat 3, as in hip-hop, for the weight a small, fast break hasn't got.

The sub, at half speed

Track 3, BAR at 1, on VOLT, four bars, each a minim (LENGTH 1/2) then a crotchet rest and a crotchet (LENGTH 1/4):

Bar	1/2	1/4	1/4
1	G3	rest	F3
2	G3	rest	A#3
3	D#3	rest	D3
4	F3	rest	G3

Two notes a bar, while the break plays sixteen: the bass is moving at a quarter of the drums' speed, and that gap is jungle's whole feeling of speed and stillness at once. The notes are from G minor, and the line leans on the bottom note, G, the way dub basses do. You'll hear the G at G1, about 49 vibrations a second, nearly at the bottom of what you can hear: on small speakers it almost disappears, which is why jungle sounds so different on a big sound system.

8.7 Balance, and remembering

On STEP, ■ MIXER: TASTE 0, STRIKE -6, VOLT up to +3. Jungle is a music for big speakers, and its sub sits louder than its drums. Then ■ SONG › ❹ ACTION › **REMEMBER SOUNDS** › *hold* ❹: it remembers your copy of the break, so your changes come back with the song.

8.8 Playing Track 1

On TRACKS, mute tracks 2 and 3, and press ■ PLAY.

Bars	As it starts
1-8	The break alone.
9-16	Unmute track 2: the kick.
17-32	Unmute track 3: the sub.
33-40	Mute tracks 1 and 2: the sub alone. The floor drops away.
41-56	Unmute tracks 1 and 2: everything.
57-64	Mute track 3: the drums to finish. STOP at the end of bar 64.

At 170 a bar goes by in under a second and a half, so eight bars is about eleven seconds: count in phrases, not bars, and watch Step's bar number.

> **Try this**
>
> Write your own edit in bar 2. Try COARSE +12 on Taste, an octave up, for a break at 190 that sounds tiny and frantic, and lower the tempo to match it, or not. Try a sub line of your own, one or two notes a bar, from G minor's scale: G, A, A#, C, D, D#, F.

8.9 Track 2 at a glance

Jungle Track 2 is Track 1 with the tricks: two new sounds made from slices, a backwards snare and a lower one, in a four-bar break with a fill; a crash every eight bars; and the break thrown into Haze's echo at the end of every phrase.

Sound 8.2 Jungle Track 2, as built in sections 8.10 to 8.13: a four-bar break with a fill, a backwards snare and a lower one made from slices, a crash every eight bars, and echo throws.

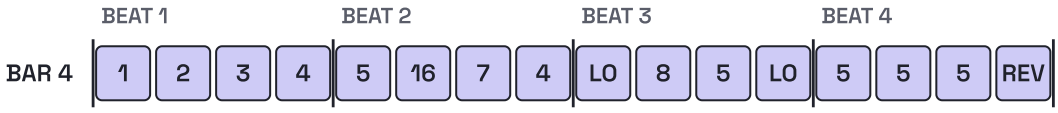


Figure 8.4 Bar 4 of Track 2's break, the fill, by slice: LO is the lower snare on F4, REV the backwards snare on E4, swelling into the next bar.

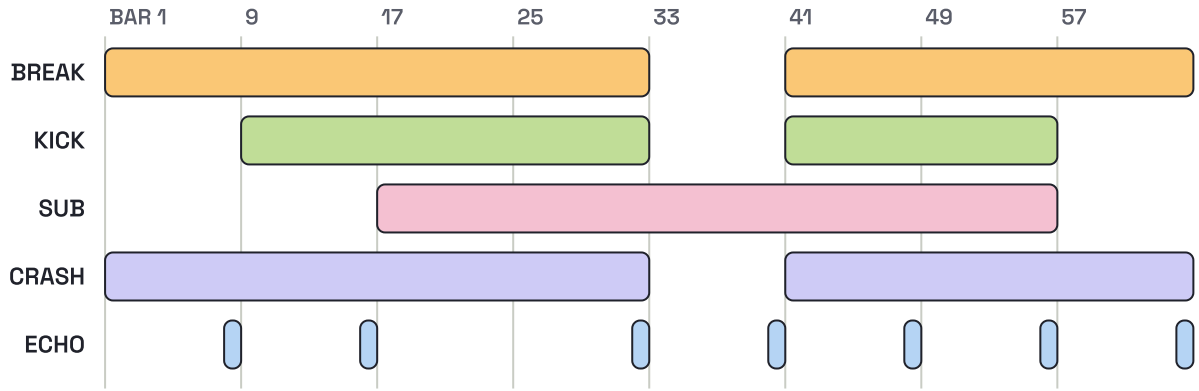


Figure 8.5 How jungle Track 2 plays: as Track 1, with the crash once every eight bars and, in the bars marked ECHO, the break thrown into Haze's echo.

Jungle Track 2

Jungle Track 1, duplicated, with:

TASTE	`C4 copied to E4, REVERSE`	`E3 copied to F4, COARSE +5`	
TRACK 1	`TASTE`	`LOOP 4 BARS`	`with a fill`
TRACK 4	`STRIKE`	`LOOP 8 BARS`	`CRASH`
HAZE	`DUB ECHO`	`SENDS 0, TASTE thrown to 90`	

8.10 Setting up

On STEP, load jungle Track 1, then ■ SONG › 4 ACTION › **DUPLICATE** › *hold* 4, and name the copy. On TRACKS, unmute any track that says MUTED, then come back to ■ SONG, so Step isn't on ENTRY while you work on Taste.

8.11 New sounds from old slices

CHOP's third MODE, COPY, copies one key, with its slice and all its settings, onto another, so you can change the copy and keep the original.

1. On TASTE, press ■ CHOP, and turn encoder 2, MODE, one click, then another: from EVENLY to AT HITS to **COPY**. It moves one choice a turn, so turn it a click at a time.
2. Press C4, slice 13, the snare on beat 3. Turn encoder 1, TO, to E4, the first key above the slices, and *hold* 4.
3. On E4, press ■ TRIM and turn encoder 3, REVERSE, to ON: the snare backwards, a swell that sucks up into nothing.
4. Press E3, slice 5, the snare on beat 2. On ■ CHOP (still on COPY), TO to F4, and *hold* 4.
5. On F4, press ■ TUNE and turn COARSE down 5, from +10 to +5: a deeper, slower snare.

> **Look before you hold**
>
> Holding encoder 4 on CHOP does whatever MODE says. If it says EVENLY or AT HITS, it chops the key you're on into slices across the keys above it, replacing what was there. Make sure the screen says COPY before you hold.



Figure 8.6 Taste's CHOP page in COPY mode, on C4, the beat-3 snare: TO is E4, so holding encoder 4 copies the slice there.

8.12 A four-bar break, a crash, and throws

The break, with a fill

On STEP, on TRACKS, track 1: LOOP up to 4 BARS, then *hold* 4 to clear it. On ENTRY, BAR to 1, LENGTH 1/16, and on TASTE write bars 1 and 2 as in Track 1, bar 3 the break as played (C3 to D#4 again), and bar 4 the fill:

Bar 4, beat 1	C3	C#3	D3	D#3
Beat 2	E3	D#4	F#3	D#3
Beat 3	**F4**	G3	E3	**F4**
Beat 4	E3	E3	E3	**E4**

The low snare (F4) thumps under beat 3, the snare rolls in semiquavers through beat 4, and the very last note is the backwards snare, swelling into the first beat of the next four bars. Four bars round, the break tells you where you are: three bars that keep time, and a fourth that throws you into the next.

A crash every eight bars

On TRACKS, track 4: give it STRIKE and turn LOOP up to 8 BARS. On ENTRY, BAR to 1, and press CRASH once. A loop eight bars long with one note in it: a crash on the first beat of every phrase. Step's tracks can loop over 1, 2, 4 and 8 bars all at once, each going round on its own.

The echo, and throwing into it

1. On HAZE, press the 6th white key from the top: **DUB ECHO**.
2. ■ SENDS, and turn TASTE, STRIKE and VOLT all right down to 0. Nothing goes into the echo, for now.
3. On STEP, ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4.

A **throw** is a trick from dub reggae, where the engineer played the mixing desk like an instrument: send one moment, a snare or a word, into the echo, and take it straight out again, so only that moment repeats, trailing away into the distance while the music goes on. You'll throw the last bar of every phrase.

8.13 Playing Track 2

On TRACKS, mute tracks 2 and 3, and press ■ PLAY. Keep Haze's SENDS page in reach.

Bars	Step	Haze
1-8	The break and the crash.	At the start of bars 8, 16, 32, 40, 48 and 56: turn TASTE up to 90, quickly. At the start of the next bar: straight back to 0. And again in bar 64.
9-16	Unmute track 2: the kick.	
17-32	Unmute track 3: the sub.	
33-40	Mute tracks 1, 2 and 4: the sub alone, with the echoes over it.	
41-56	Unmute tracks 1, 2 and 4: everything.	
57-64	Mute tracks 2 and 3: the break and the crash. STOP at the end of bar 64.	

> **Try this**
>
> Throw something else: Strike's send, on one snare. Make more sounds from slices: a kick tuned down a whole octave, a hat reversed. Try the crash on a 6-bar loop, so it lands somewhere different in every phrase.

8.14 Track 3: your own

Duplicate Track 2 and make it yours: the tempo, the edits and fills, the sub line, the throws. Section 8.16 has one way it might go: drum and bass, using the changes in section 8.15. Every jungle producer worth hearing cut up the break their own way, and that's the part we can't write down for you.

8.15 Drum and bass

Faster, a two-step, and a bass that growls

1. On STEP, ■ TEMPO › ❶ TEMPO › **172**.
2. On STRIKE, ■ PUNCH: a tighter kick.
3. On VOLT, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **GROWL**, then press ■ FILTER, ready for later.
4. GROWL sits an octave above SUB BASS, so move the bass down an octave: on TRACKS, track 3; on ENTRY, BAR to 1, ■ COPY, push encoder 1 for TRANSPOSE, turn END on 12 beats (to the end of bar 4), SEMITONES to -12, and *hold* ❹.
5. On TRACKS, track 5: STRIKE, LOOP 1 BAR. On ENTRY, BAR to 1, LENGTH 1/4, and on STRIKE: rest, SNARE, rest, SNARE.
6. On STEP, ■ MIXER: TASTE down to -6, STRIKE up to 0, VOLT down to -9.
7. ■ SONG › ❹ ACTION › **REMEMBER SOUNDS** › *hold* ❹.

Track 2's kick on beat 1 and the "and" of 3, and track 5's snare on 2 and 4: that's the **two-step**, the backbone of drum and bass, clean and hard. The break, turned down, rides on top as a shuffle, filling the gaps. And GROWL is Volt's take on the Reese: three oscillators slightly out of tune with each other, beating and churning.

> **A big bass takes a lot of room**
>
> GROWL is loud in a way SUB BASS isn't: its sound is in the middle of the range, where the ear is most sensitive, as well as at the bottom, and a wide, detuned sound fills the whole space. At VOLT 0 it swallows the drums. At -9 it sits just under them. Drum and bass mixes are a fight between the drums and the bass, and the drums have to win.

8.16 One way it might go

Track 2 at 172, with the two-step, the break turned down on top, and GROWL opening up through the second half.

Sound 8.3 Jungle Track 3, one way it might go: drum and bass at 172, the two-step under the break, and Volt's GROWL, kept under the drums, opening up through the second half.

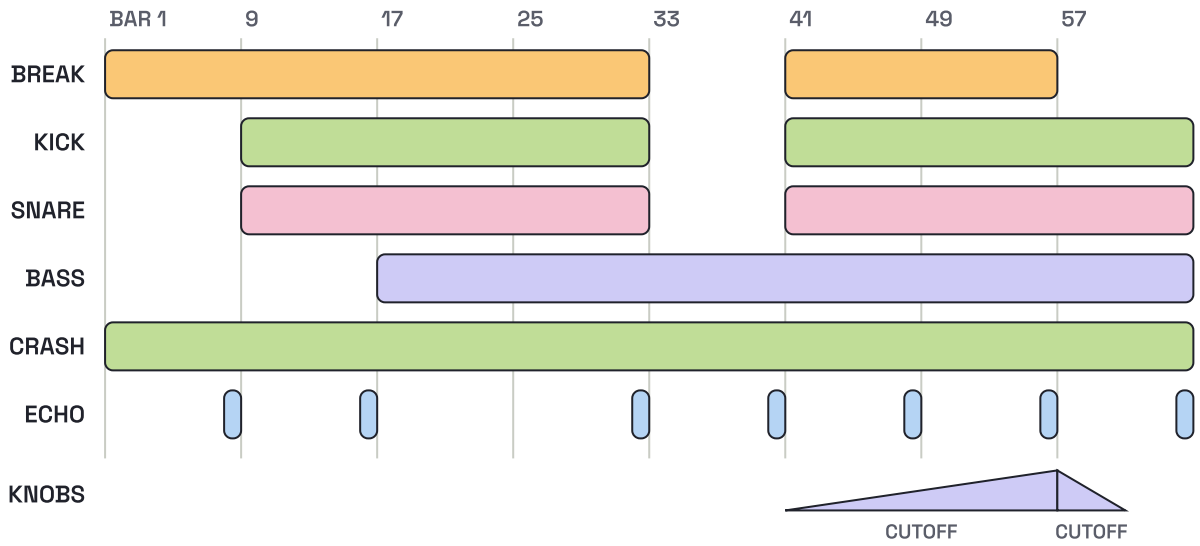


Figure 8.7 Jungle Track 3, one way it might go: drum and bass, the two-step joining at bar 9 and the bass at 17, GROWL's filter opening through bars 41 to 56 and closing again.

On TRACKS, mute tracks 2, 3 and 5, and press ■ PLAY. The throws are Track 2's.

Bars	Step	Volt
1-8	The break and the crash.	
9-16	Unmute tracks 2 and 5: the two-step.	
17-32	Unmute track 3: the bass.	
33-40	Mute tracks 1, 2 and 5: the bass and the crash.	
41-56	Unmute tracks 1, 2 and 5: everything.	CUTOFF up 2 clicks every bar.
57-64	Mute track 1: the two-step and the bass. STOP at the end of bar 64.	CUTOFF down 8 clicks a bar, for four bars.

8.17 What you've learned

- Where jungle and drum and bass came from: rave, the sound systems, pirate radio, the Amen break and the man who played it.
- Speeding a break up by playing it higher, and what semitones do to speed.
- Keeping your own copy of a sound, and not copying over another.
- The edit and the fill; a half-time bass under double-time drums; sub bass.
- New sounds from slices: CHOP's COPY, REVERSE, re-tuning.
- Loops of 1, 2, 4 and 8 bars at once; transposing by an octave.
- Echo throws, played live.
- The two-step and the Reese bass, and why the drums have to win.

Next, synth-pop and synthwave: the whole range at once, chords, arpeggios and a tune, and a song written out from beginning to end.

Chapter 9 - Synth-pop and synthwave

Chords that move, an octave bass, an arpeggio, a tune, and a key change.

Every chapter so far has kept away from one thing: a tune. Techno threw out the chord progression, house looped two chords, hip-hop left the space for a voice, ambient let the harmony drift. Synth-pop is pop music: it has a progression that goes somewhere and comes back, and a melody you can sing. This chapter builds one section of a pop song, its chords, its band and its tune, eight bars that go round; a whole song, with verses, a chorus, a bridge and an ending, is the next step, and section 12.3 points the way. It's also the one chapter that needs almost the whole range at once.

> **What you need for this chapter**
>
> The Root, **Step**, **Strike**, **Volt**, **Ember** and **Haze**, joined in that order, and headphones. Track 2 adds **Glass** and **Axis** on the end. That's seven units on one Root.

9.1 Where synth-pop and synthwave came from

Chapter 2 told how Kraftwerk and Giorgio Moroder made pop records from machines in the 1970s. At the end of the decade, cheaper synthesisers put that sound in the hands of bands. In Tokyo, Yellow Magic Orchestra, formed in 1978, had hits with synthesisers and sequencers; in 1979 Gary Numan, with Tubeway Army, took "Are 'Friends' Electric?" to number one in Britain with a sound he'd found by accident on a Minimoog in the studio.

The early 1980s were synth-pop's golden years, and much of it came from British bedrooms and art schools. The Human League's "Don't You Want Me" was the Christmas number one of 1981; Soft Cell's "Tainted Love" and Depeche Mode's first records came the same year. Yazoo paired Vince Clarke's synthesisers with Alison Moyet's huge voice; Eurythmics' "Sweet Dreams (Are Made of This)" (1983) is Annie Lennox singing over a single sequenced bass line and a drum machine. Laurie Anderson's "O Superman" (1981), eight minutes of a looped "ha" and a vocoder, got to number two, and Kate Bush used one of the first sampling computers, the Fairlight, on her records from 1980. In 1983 New Order's "Blue Monday", built on a drum machine and sequencers, became the best-selling 12-inch single ever made in Britain. Then the Yamaha DX7, and Roland's Juno synthesisers, put the sound on nearly every pop record of the decade.

By the 1990s that sound was out of fashion, which, as with the 808, is exactly when people started to love it again. In the 2000s, producers who'd grown up on 80s film scores, John Carpenter's in particular, and on video games, made new music in the old style, and called it synthwave or retrowave: Kavinsky's "Nightcall" (2010), in the film *Drive* (2011), made it famous. Kyle Dixon and Michael Stein's theme for the television series *Stranger Things* (2016) is played entirely on analogue synthesisers, and in 2019 the

Weeknd's "Blinding Lights", all octave bass and 80s drums, became one of the most-played songs there has ever been. Bands such as Chvrches, with Lauren Mayberry's voice over the synths, brought synth-pop itself back.

9.2 What synth-pop is made of

- **The tempo.** Pop speed: about 100 to 120. These tracks are at 112 and 116.
- **Pop drums.** A kick on 1 and 3 and a snare on 2 and 4, the backbeat of rock and pop, rather than four to the floor; the snare huge and echoing. In the 80s that sound was gated reverb, a big reverb cut off dead; Haze has no gate, so its BIG HALL stands in.
- **The octave bass**, a note and the same note an octave up, on every quaver, from disco and Moroder's "I Feel Love".
- **Pads**: soft, wide chords, held.
- **Arpeggios**: the chords played a note at a time, fast and even, as only a machine can.
- **A tune** on a lead synth, and a song's shape: verses and choruses, a breakdown, and a big finish.

Four chords

All three tracks are in A minor on four chords, a bar each: A minor, F, C, G. In Roman numerals, as in house, that's i, VI, III, VII. Now count the same four chords in C major instead: C is I, so A minor is vi, F is IV and G is V: vi, IV, I, V. A minor and C major share all their notes, and are called each other's relative minor and major; the same four chords, started on C (C, G, A minor, F, or C, A minor, F, G), are the chords of thousands of pop songs, from "Let It Be" to the present. Starting on A minor gives them the shadow synthwave loves, a sound halfway between sad and heroic.

9.3 Track 1 at a glance

Synthwave Track 1 is the band: pop drums on Strike, an octave bass on Volt, the four chords on an Ember pad, and Haze's big hall. Nothing new on Step: everything comes from earlier chapters.

Sound 9.1 Synthwave Track 1, as built in sections 9.4 to 9.7: pop drums, an octave bass on Volt and the four chords, A minor, F, C and G, on an Ember pad, in Haze's big hall.

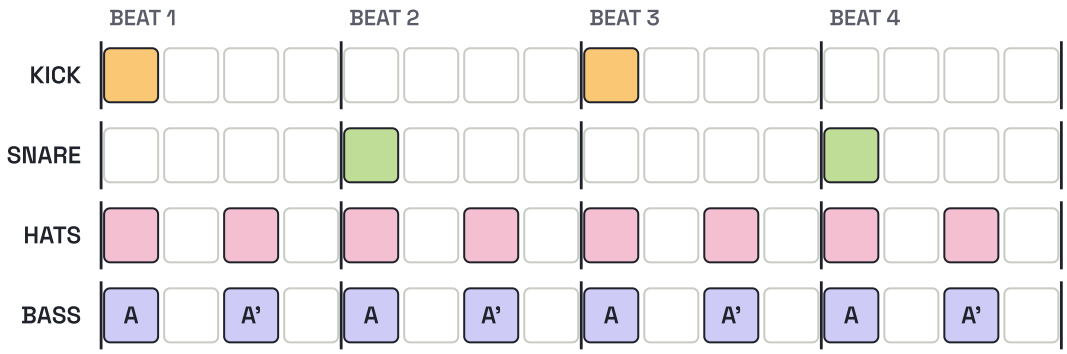


Figure 9.1 Synthwave Track 1, one bar: the kick on 1 and 3, the snare on 2 and 4, hats on every quaver, and the octave bass (A' is the A an octave up). The chords hold a whole bar each: A minor, F, C, G.

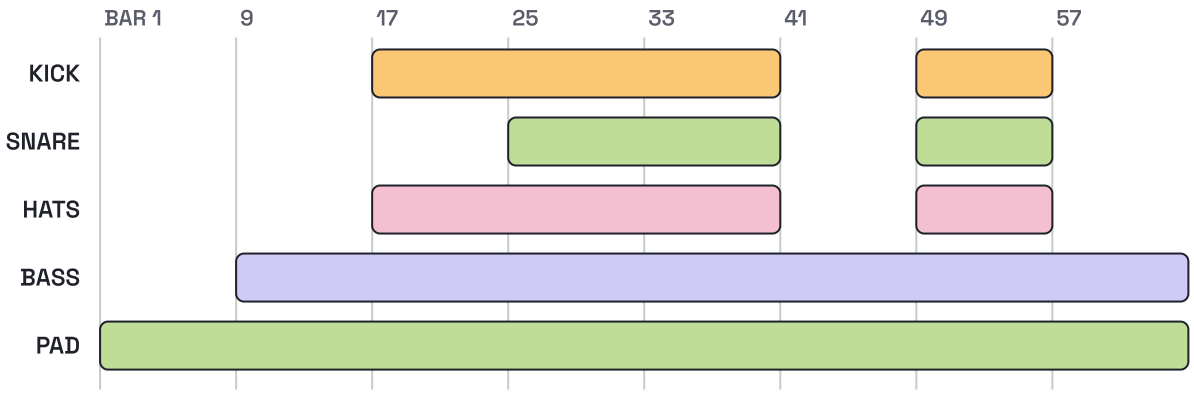


Figure 9.2 How synthwave Track 1 plays: the pad alone, then the bass, then the drums, the snare last; eight bars without drums at bar 41; and the pad and bass to finish, as it began.

Synthwave Track 1

A new song, with:

STEP	`TEMPO 112`
STRIKE	`PUNCH`
VOLT	`PLUCK BASS (FA09)` `OCT- once`
EMBER	`PWM PAD (FA25)`
HAZE	`BIG HALL` `STRIKE 30, VOLT 0, EMBER 35`
TRACKS 1-3	`STRIKE` `LOOP 1 BAR` `kick, snare, hats`
TRACK 4	`VOLT` `LOOP AUTO` `the octave bass, four bars`
TRACK 5	`EMBER` `LOOP AUTO` `Am, F, C, G`
MIXER	`VOLT -2` `EMBER -13`

9.4 Setting up

1. Join the units in this order from the Root: Step, Strike, Volt, Ember, Haze.
2. On STEP, ■ SONG › ❹ ACTION › **NEW** › *hold* ❹, and name it.
3. On STRIKE, ■ PUNCH.
4. On VOLT, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **PLUCK BASS**, and press ■ OCT- once.
5. On EMBER, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **PWM PAD**.
6. On HAZE, press the 2nd white key from the top, BIG HALL. Then ■ SENDS: leave STRIKE at 30, turn VOLT down to 0 (the bass stays dry and close) and EMBER up to 35.
7. On STEP, ■ TEMPO › ❶ TEMPO › **112**.
8. On TRACKS: tracks 1, 2 and 3 STRIKE, LOOP 1 BAR; track 4 VOLT and track 5 EMBER, both AUTO.

9.5 The band

Drums

On STEP, ■ ENTRY, then a track at a time, BAR back to 1 each time, on STRIKE:

Track	LENGTH	Write
1	1/4	KICK, rest, KICK, rest: beats 1 and 3
2	1/4	rest, SNARE, rest, SNARE: beats 2 and 4
3	1/8	CL HAT eight times: every quaver

The octave bass

1. Track 4, LENGTH 1/8, on VOLT: A2, A3, four times over.
2. 4 BAR back one, ■ COPY › *hold* 4: bar 1 to bar 2. Then END on 4 beats and TO on 4, and *hold* 4: bars 1 and 2 to bars 3 and 4.
3. Push encoder 1 for TRANSPOSE, then a bar at a time, as in house Track 2:

Bar	Turn	SEMITONES	Becomes
2	START on 4	-4	F
3	END on 4, then START on 4	+3	C
4	END on 4, then START on 4	-2	G

hold 4 after each. The bass jumps between each chord's root and the same note an octave up, eight times a bar: the engine of nearly every synth-pop record.

9.6 The chords

On TRACKS, turn to track 5, then press ■ CHORD. Press Step's ■ OCT- once, as in house (if its light is still on from then, leave it). 4 BAR › 1, and turn encoder 3, LENGTH, to 1, a whole bar. Then, for each chord, set it and push encoder 4 to place it; the cursor moves on a bar each time:

Bar	ROOT	TYPE	INVERT
1	A	m	ROOT
2	F	(major)	ROOT
3	C	(major)	2ND
4	G	(major)	ROOT



Figure 9.3 Step's CHORD page, ready for the first chord: A minor, LENGTH 1 (a whole bar), at the start of bar 1. Step's OCT- is on, so the chord is A3, C4, E4.

For F, C and G, push encoder 1 after setting ROOT and turn TYPE back to the plain major chord, then push again. C is turned over twice, as in house, so it sits close to its neighbours instead of jumping up. Then ■ MIXER: VOLT down to -2, EMBER down to -13. PWM PAD is a big, loud sound, and a pad should wrap round the band, not stand in front of it. Then ■ SONG › 4 ACTION › REMEMBER SOUNDS › *hold* 4.

9.7 Playing Track 1

On TRACKS, mute tracks 1, 2, 3 and 4, so only the pad plays, and press ■ PLAY.

Bars	As it starts
1-8	The pad alone.
9-16	Unmute track 4: the bass.
17-24	Unmute tracks 1 and 3: the kick and the hats.
25-40	Unmute track 2: the snare. Everything.
41-48	Mute tracks 1, 2 and 3: the drums drop out.
49-56	Unmute tracks 1, 2 and 3: everything.
57-64	Mute tracks 1, 2 and 3: the pad and the bass, as it began. STOP at the end of bar 64.

Notice how much the snare's arrival at bar 25 does. Until then the beat ticks along; with the backbeat, and the hall round it, it suddenly sounds like a band in a big room.

> **Try this**
>
> Try the chords in another order: F, G, A minor, A minor, or C, G, A minor, F. Try Ember's other pads, and Strike's NEON for electronic toms on a track of their own. Turn Haze's STRIKE send right up and hear how far 80s pop took it.

9.8 Track 2 at a glance

Synthwave Track 2 adds the two things that make it unmistakably 80s: an arpeggio on Glass, played by Step's arpeggiator from the same four chords, and an eight-bar tune on Axis.

Sound 9.2 Synthwave Track 2, as built in sections 9.9 to 9.13: the band, an arpeggio on Glass from Step's arpeggiator, and an eight-bar tune on Axis.

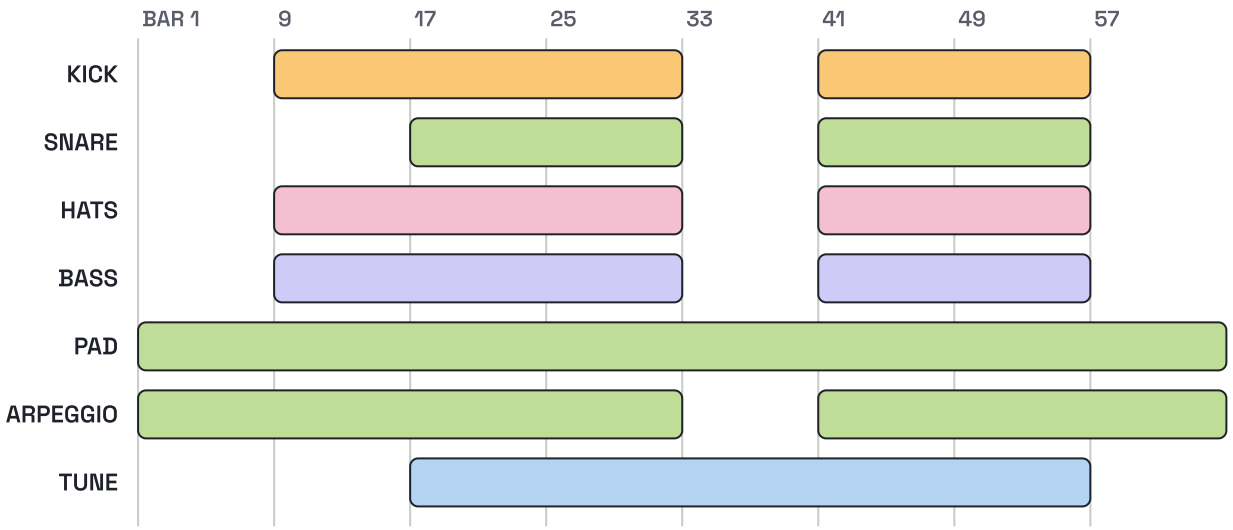


Figure 9.4 How synthwave Track 2 plays: the pad and the arpeggio, then the band, then the tune at bar 17; a breakdown of the pad and the tune; everything; and the pad and arpeggio to finish.

Synthwave Track 2

Synthwave Track 1, duplicated, with:

GLASS	`E.PIANO 1 (FA11)`
AXIS	`SYNC LEAD (FA05), darker, GLIDE 20` `kept as a user sound` `OCT+ once`
TRACKS 4, 5	`LOOP 4 BARS`
TRACK 6	`GLASS` `LOOP 4 BARS` `the chords, ARP UP, 2 OCTAVES`
TRACK 7	`AXIS` `LOOP AUTO` `the tune, eight bars, LEGATO`
HAZE	`GLASS 30` `AXIS 40`
MIXER	`EMBER −10` `GLASS −13` `AXIS +1`

9.9 Setting up

1. On STEP, load synthwave Track 1, then ■ SONG › ❹ ACTION › **DUPLICATE** › *hold* ❹, and name the copy. On TRACKS, unmute any track that says MUTED.
2. Join Glass, then Axis, to the right-hand end of the row.
3. On GLASS, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **E.PIANO 1**.
4. On AXIS, ■ SOUNDS › ❶ BANK › **FAC A** › ❷ SOUND › **SYNC LEAD**.

A gentler lead, and a sound of your own

SYNC LEAD is a classic lead: one voice at a time, sliding from note to note, and very bright, especially as each note starts. Make it smoother and softer:

1. On AXIS, press ■ KEYS and turn encoder 3, GLIDE, down 10 clicks, from 30 to 20: a quicker slide between notes.
2. Press ■ FILTER and turn encoder 1, CUTOFF, down 12 clicks, from 70 to 58: darker.
3. Keep it: ■ SOUNDS, turn encoder 3, COPY TO, to an empty slot that says INIT SOUND, and *hold* ❸. (If you made ambient Track 3, USER A 01 is your AIR PAD: use USER A 02.)
4. Press Axis's ■ OCT+ once: the tune sits an octave up.

Seven tracks

On TRACKS: tracks 4 and 5, the bass and the chords, LOOP 4 BARS; track 6, GLASS, LOOP 4 BARS; track 7, AXIS, left on AUTO.

Why set loops on tracks that were fine on AUTO? Because the tune is going to be eight bars long. AUTO tracks together make up the song, and the song lasts as long as the longest of them, whole bars, playing each AUTO track once each time round. With the tune at eight bars, a four-bar AUTO bass would play bars 1 to 4 and then stop for 5 to 8. Given a loop of its own, it goes round twice while the tune goes round once.

9.10 The arpeggio

An arpeggio is a chord played one note at a time. Step has an arpeggiator on every track: give it chords and it plays them as arpeggios, so you don't write the notes at all.

1. The chords first. On TRACKS, turn to track 5; on ENTRY, BAR to 1, and press ■ COPY. Turn END on 12 beats, to the end of bar 4. Turn encoder 1, TO, on one, to track 6, and encoder 2 back to bar 1. *hold* 4.

2. On TRACKS, turn to track 6, then press ■ TRACKS again: the top bar says ARP.

3. Turn encoder 1, PATTERN, on one, from OFF to UP, and encoder 3, OCTAVES, on one, to 2. Leave RATE at 1/16 and GATE at 50%. Press ■ TRACKS to go back.

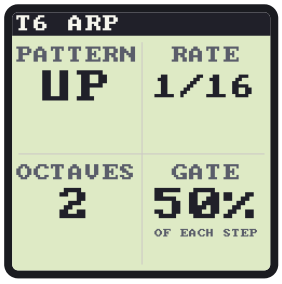


Figure 9.5 The ARP page for track 6 (TRACKS pressed again): PATTERN UP, RATE 1/16, OCTAVES 2, GATE 50%. Track 6 holds the same chords as track 5, and Step plays them a note at a time.

Track 6 has the same four chords as track 5, a whole bar each, but now Step plays each one going up, a semiquaver a note, over two octaves, then starting again, and GATE holds each note for half its time, so they bounce. Ember holds the chords; Glass ripples through them. Chords stay chords on every other track: the arpeggiator belongs to track 6 alone.

9.11 The tune

On TRACKS, turn to track 7; on ENTRY, BAR to 1, and press ■ LEGATO, the third button in Step's left column, so every note lasts right up to the next. On a lead that glides, that's what makes it slide from note to note instead of starting each one afresh. Then on AXIS, eight bars:

Bar	Chord	Notes (LENGTH)
1	Am	A4 (1/2), C5 (1/4), B4 (1/4)
2	F	A4, F4, A4, C5 (1/4 each)
3	C	G4 (1/2), E4 (1/4), G4 (1/4)
4	G	D5 (1/2), B4 (1/2)
5	Am	C5 (1/4), B4 (1/8), A4 (1/8), B4 (1/4), C5 (1/4)
6	F	A4 (1/2), F4 (1/4), C5 (1/4)
7	C	G4, E4, G4, C5 (1/4 each)
8	G	B4 (1/2), rest (1/2)

Then press ■ NORMAL again, so the next notes you write anywhere aren't legato too.

Look at the first note of every bar: A over A minor, A over F, G over C, D over G. Each is a note of the chord underneath (A minor is A, C, E; F is F, A, C; C is C, E, G; G is G, B, D). A tune that lands on the chord's notes on the strong beats sounds as if it belongs with the chords; the notes in between can wander, and the B over A minor in bar 1 rubs for a moment before the next bar puts it right. The tune is two halves of four bars, the second answering the first, and it ends on a breath, a minim of rest, before it starts again: a melody needs somewhere to breathe, like the singer it stands in for.

9.12 The hall, and the mix

1. On HAZE, still on SENDS from Track 1, the fourth box is now GLASS: turn it to 30.
2. Press ■ SENDS again: the top bar says SENDS 5-8, the units after the first four. Turn AXIS to 40.
3. On STEP, ■ MIXER: EMBER up 3, to -10. Then press ■ MIXER again for units 5 to 8: GLASS down to -13, AXIS up to +1.
4. ■ SONG › 4 ACTION › **REMEMBER SOUNDS** › *hold* 4.



Figure 9.6 Haze's SENDS for units 5 to 8, after pressing SENDS again: AXIS at 40, into BIG HALL.

Pressing a page again

With more than four units, Haze's SENDS and Step's MIXER each have two pages, and pressing the button again turns between them. So before you turn anything, look at the top bar: if Haze already shows SENDS, pressing SENDS again takes you to SENDS 5-8, not back to the start.

9.13 Playing Track 2

On TRACKS, mute tracks 1, 2, 3, 4 and 7, so the pad and the arpeggio start alone, and press ■ PLAY.

Bars	As it starts
1-8	The pad and the arpeggio.
9-16	Unmute tracks 1, 3 and 4: the kick, the hats and the bass.
17-32	Unmute tracks 2 and 7: the snare, and the tune.
33-40	Mute tracks 1, 2, 3, 4 and 6: the breakdown, the pad and the tune alone.
41-56	Unmute tracks 1, 2, 3, 4 and 6: everything.
57-64	Mute tracks 1, 2, 3, 4 and 7: the pad and the arpeggio, as it began. STOP at the end of bar 64.

> **Try this**
>
> On track 6's ARP page, try the other PATTERNs, RATE 1/8, OCTAVES 3, a shorter GATE. Try the arpeggio on Glass's HARPSICH 1 or SYN-CLAV 1, and the tune on Axis's HOLLOWLEAD. Write your own tune over the same chords: start every bar on one of the chord's notes, and see where it wants to go.

9.14 Track 3: your own

This is the last of the third tracks, and the last time we'll say it: what follows is one way it might go, about the plainest synth-pop we could build, so that the method shows through. It isn't a song yet: it's one eight-bar section, the same chords and tune all the way through, with no verse, no chorus of its own, no bridge and no real ending. What you write over these chords, or over chords of your own, with your own tune, and a chorus to go with it, will be better than ours, and it should sound nothing like it.

Synthwave Track 3 is Track 2 written out bar by bar, as house Track 3 was, so that it plays itself; a little faster; and with one thing that only a written-out song can have: the last sixteen bars go up a key.

9.15 Writing it out, and a key change

1 On STEP, load synthwave Track 2, then ■ SONG › 4 ACTION › **DUPLICATE** › *hold* 4, and name the copy. Unmute any track that says MUTED.

2 ■ TEMPO › 1 TEMPO › **116**.

Eight bars of everything

As in house Track 3 (section 5.15), make every track an ordinary AUTO track the length of the song. For tracks 1 to 6: on TRACKS, turn LOOP right down to AUTO; on ENTRY, BAR to 1, ■ COPY (for tracks 4, 5 and 6, turn END on 12 beats, to the end of bar 4: their four bars); push encoder 1 three times, to FILL; turn UNTIL right up, and *hold* 2. UNTIL stops at 8:4.1: the song is eight bars long, set by the tune, and FILL won't go past it.

Sixty-four bars, carved

Then, as in house: ■ SONG, ■ SONG again for SECTIONS; END on 7 bars, to bar 8; push encoder 1 three times, to FILL; UNTIL to 64:1.1, and *hold* 2. Now carve the arrangement out with DELETE (section 5.15 has the method: ■ COPY, push encoder 1 twice, END first, then START, *hold* 4):

Track	Bars	END on	START on	Why
1 kick, 3 hats, 4 bass	1-8	28 beats	0	The intro: the pad and the arpeggio
2 snare, 7 tune	1-16	60 beats	0	
1, 2, 3, 4, and 6 arpeggio	33-40	156 beats	128 beats	The breakdown: the pad and the tune

Up a semitone

For tracks 4, 5, 6 and 7, the bass, the chords, the arpeggio and the tune: on ENTRY, BAR to 1, ■ COPY, push encoder 1 for TRANSPOSE, turn END on to 64:4.1 and then START on to 49:1.1 (they turn faster the quicker you turn them), SEMITONES to +1, and *hold* 4. The drums stay as they are: they have no key.



Figure 9.7 The key change on Step's COPY page: TRANSPOSE, +1 semitone, from bar 49 to the end of bar 64, on the tune's track. The same is done to the bass, the chords and the arpeggio.

From bar 49 the whole song is in B flat minor instead of A minor, a semitone higher. In a pop song it's the oldest trick there is for the last chorus: everything you already know, suddenly lifted, as if the song had taken a breath and stood up. Here there's no chorus to lift, so it lifts the one section there is, but the effect is the same. It was so common in 80s pop that it got a nickname, the "truck driver's gear change". A semitone is the most obvious lift there is: nearly every note of the old key moves, so you can't miss it. It's also a thing you can only do by writing a song out: a loop is the same every time round, but a written-out song can change in its last sixteen bars and stay changed.

9.16 One way it might go

Sound 9.3 Synthwave Track 3, one way it might go: Track 2 written out bar by bar at 116, so it plays itself, and from bar 49 everything but the drums a semitone higher.

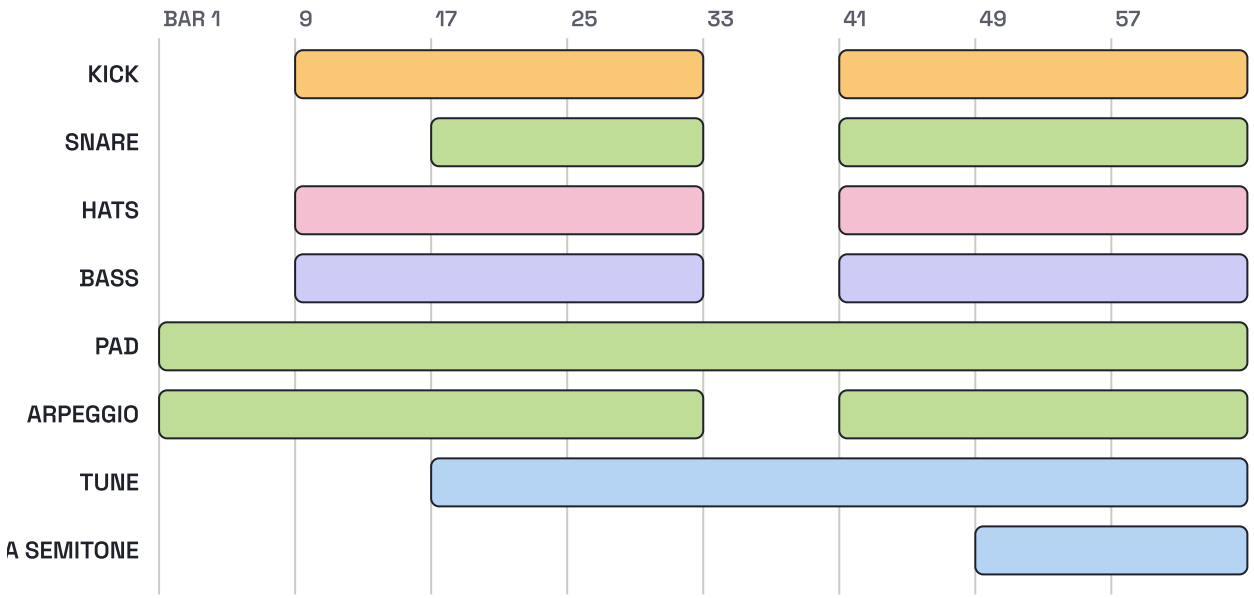


Figure 9.8 Synthwave Track 3, one way it might go: Track 2 written out bar by bar so it plays itself, and the last sixteen bars a semitone higher.

Press ■ PLAY and listen for two things. The arrangement plays itself, as house Track 3 did. And at bar 49, a little under two minutes in, everything but the drums moves up a semitone: the tune you've heard four times comes round again and sounds new.

9.17 What you've learned

- Where synth-pop and synthwave came from, from YMO and Gary Numan to "Blinding Lights".
- Pop drums, the octave bass, and a big hall on the snare.
- The four chords, relative major and minor, and why the same chords sound different starting somewhere else.
- Step's arpeggiator: chords played a note at a time.
- A tune: chord notes on the strong beats, two halves that answer each other, room to breathe; LEGATO and glide.
- Why a short track needs a set loop under a longer AUTO one.
- SENDS and MIXER for units 5 to 8.
- A key change, and why only a written-out song can have one.

That's the last of the kinds of music. You've made seven, and by now you know the tools well enough to make an eighth that has no name yet.

Chapter 10 - Step from start to finish

Every page and every button, in order, with where in the book you'll use each one.

The chapters of music teach Step a piece at a time, as each kind of music needs it. This chapter is the other way round: the whole of Step, page by page, with nothing left out. If you've read the chapters before it, it fills in the gaps and puts everything in one place. If you've come here first, it's a map: read it through once, then go to chapter 3 and start making music, and come back whenever you want to know what something does.

> **What you need for this chapter**
>
> The Root and **Step**, with any one unit that makes a sound joined to its right, so you can hear what you write. Volt or Glass is ideal.

10.1 What Step is

Step doesn't make any sound of its own. It holds notes, and sends them to the other units at the right moments, along the chain. Everything it holds is on one **timeline**: bars and beats, one after another, like a score, from bar 1 to the end of the song. There are no "patterns" to switch between and no grid of steps that goes round: a note is simply at its place, bar 13 beat 2, on its track.

A song has **16 tracks**. Each track plays one unit, and several tracks can play the same unit: in techno Track 1, the kick, the hats and the clap are three tracks all playing Strike. Every track shares the same timeline, so bar 13 is bar 13 on all of them. A song has room for 8,192 notes, shared by all sixteen tracks, and Step keeps up to 32 songs, saving as you go.

You write notes in **step time**: you choose how long a note is, play it, and the cursor moves on by that much. Nothing depends on playing in time, so you can take as long as you like over every note.

10.2 The layout

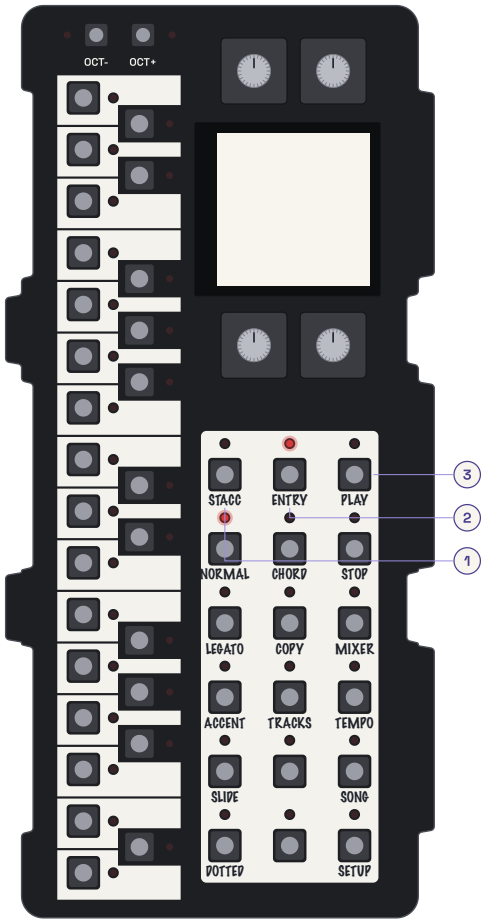


Figure 10.1 Step: the note's shape on the left (1), one track in the middle (2), the whole song on the right (3).

Step's buttons are arranged by what they affect, from small to large:

> **The left column (1): the note** you're writing: how much of its length it sounds (STACC, NORMAL, LEGATO), whether it's ACCENTed or SLIDes into the next, and DOTTED to make it half as long again.

> **The middle column (2): one track**: ENTRY to write notes, CHORD to write chords by name, COPY to copy, transpose, delete and fill, and TRACKS for the track's own settings.

> **The right column (3): the whole song**: PLAY and STOP, MIXER, TEMPO, SONG, and SETUP at the bottom, as on every unit.

Step's keys play notes on the open track's unit, so you can write with them. Their lights show one bar of the open track: a white key is dim where a note starts, bright at the cursor and blinking as it plays, sixteenths across the bar; the black keys count which bar it is. OCT- and OCT+ move Step's keys, and the CHORD page's chords, by octaves.

10.3 ENTRY: writing notes

ENTRY is where notes are written into the open track. Play a note, or a chord, and it goes in at the cursor with the current length; when every key is let go, the cursor moves on by that length.

Encoder	Turn	Push	Hold
1 LENGTH	The note's length: 1/16T, 1/16, 1/8T, 1/8, 1/4T, 1/4, 1/2T, 1/2, 1	A rest of that length	A tie: the note before carries on
2 PITCH	The note shown (or, with a note at the cursor, moves it)	Writes the note shown	
3 VELO	How hard: 1 to 127		
4 BAR	Moves the cursor, a LENGTH at a time	INSERT or REPLACE	Deletes the note at the cursor

> **Lengths** are written as fractions of a bar of four beats: 1/4 is a crotchet (a quarter note), 1/8 a quaver, 1/16 a semiquaver, 1/2 a minim, 1 a semibreve, a whole bar. A T is a triplet: three in the time of two. Triplets can go anywhere, singly or in any number (chapter 3, section 3.2, has the counting).

> **Where notes come from.** Step's own keys; the keys of the unit the track plays, at Step's VELO (chapter 3); a MIDI keyboard on Step's channel, at the strength you play; or a push of PITCH. Playing the keys of a different unit switches to that unit's track instead (the screen says which, and that first note isn't written), so on ENTRY, set up other units only on another of Step's pages: chapter 8 explains why.

> **INSERT and REPLACE.** In INSERT, a new note goes in at the cursor and pushes everything after it along; in REPLACE it writes over whatever starts there. At the end of a track, both just add.

> **The cursor stays where you left it** when you change track: turn BAR back to the start before writing a new part (chapter 3).

10.4 The note's shape

> **STACC, NORMAL, LEGATO**, one of the three: a note sounds for a short part of its length, most of it, or all of it, right up to the next. LEGATO is how a gliding lead slides from note to note (chapter 9).

> **ACCENT**: the note is played harder; on Volt's ACID sounds, it's the original's accent (chapter 4). Strike's ACCENT setting says how much louder.

> **SLIDE**: the note glides into the next; Volt's ACID sounds slide as the original does (chapter 4).

> **DOTTED**: the length half as long again: a dotted crotchet is three quavers long.

Whatever is lit applies to the next note you write, and the lights show the note at the cursor. They work only on ENTRY and CHORD. To add an accent or a slide to a note already written, move BAR onto it and press the button.

10.5 CHORD: chords by name

Choose a chord by name and Step works out its notes. Chapter 5 teaches it.

Encoder	Turn	Push
1	The ROOT (C to B) or the TYPE (major, m, 7, maj7, m7, dim7, m7b5, aug, sus2, sus4, 6, m6, add9, 7sus4), whichever is underlined	Switches between ROOT and TYPE
2 INVERT	Root position, 1ST, 2ND, 3RD: the chord turned over	
3 LENGTH	As on ENTRY	
4 BAR	Moves the cursor	Places the chord; the cursor moves on by its length

Every chord is four notes: a three-note chord adds its lowest note again, an octave up. The screen spells the notes as the chord is written (A flat minor is Ab, Cb, Eb), and the chord plays on the track's unit as you change it. Chords start at C3; OCT- and OCT+ move them.

10.6 COPY: copy, transpose, delete and fill

COPY works on a **range** of the open track, from the start of START's beat to the end of END's beat. It opens on the cursor's bar, always in COPY mode. Push encoder 1 to step through the four jobs; hold any encoder to do it (with the bar filling); the screen says what was done.

Mode	Encoder 1	Encoder 2	What it does	Where it's taught
COPY	The track to copy to	TO: where the copy starts	Copies the range, replacing whatever is there	3.5
TRANSPOSE	SEMITONES, -12 to +12		Moves the range's notes up or down	3.5
DELETE			Removes the range's notes, leaving silence	5.15
FILL	How many times (shown)	UNTIL: the last beat filled	Repeats the range after itself, up to UNTIL	5.15

Encoders 3 and 4 are START and END. Positions move a beat a click; push encoder 2, 3 or 4 to move them in sixteenths instead. Turn END before START: START can't pass it.

10.7 TRACKS, and how tracks loop

TRACKS is each track's own settings.

Encoder	Turn	Push	Hold
1 TRACK	Which track	Mutes it, or unmutes it	
2 UNIT	Which unit it plays		
3 LOOP	AUTO, or a set loop in bars	The loop in beats instead	
4 NOTES	(shows how many)		Clears the track

> **Muting** keeps the track's notes and stops it playing: the way to arrange live (chapter 3).

> **A track can play any unit**, and can be moved to another at any time; its notes stay (chapter 7). If a track's unit isn't on the chain, the track shows it's missing, keeps everything, and plays again when the unit comes back.

> **Hold TRACKS** on any page, turn encoder 1 to another track, and let go: you're back on the page you were on, on the new track.

> **TRACKS pressed again** is the track's arpeggiator, ARP: PATTERN (OFF, UP, DOWN, UP/DN, PLAYED, RANDOM), RATE, OCTAVES and GATE (chapter 9). Chords stay stored as chords; the arpeggio is how they're played.

AUTO and set loops

This is the one idea about Step that's worth reading twice.

> **AUTO tracks make the song.** The song is as long as its longest AUTO track, in whole bars, and every AUTO track plays its notes once each time the song goes round, then is silent until the song starts again. So AUTO tracks always stay in step with each other. A four-bar AUTO track in an eight-bar song plays for four bars and rests for four (chapter 9).

> **A set loop runs on its own.** A track with a LOOP of 1 BAR, or 3 beats, or 17, repeats every so many beats from PLAY, whatever the song does, and never starts again with it. That's how a one-bar drum pattern lasts the whole song (chapter 3), how three beats go round against four (chapter 4), and how five loops never line up (chapter 7).

> With no AUTO track holding notes, the song is as long as its longest set loop.

10.8 PLAY, STOP, and MIXER

PLAY always starts the song from the top, and STOP stops it. Pressing PLAY also sends each unit to the sound the song remembers (SONG's REMEMBER SOUNDS, below), so any temporary changes to those sounds are dropped (chapter 6). As it plays, Step also sends MIDI clock, start and stop, and the Root can pass them out of its MIDI OUT (SETUP, ROOT, MIDI OUT: CLOCK or BOTH), so music software or another instrument can follow it.

MIXER has a level and a mute for each unit, four to a page, in the order they're joined: turn for the level, from OFF and -48 dB up to +6 dB; push to mute. With more than four units, MIXER pressed again shows the rest. The mix is kept with the song, so each song has its own. Haze's sends follow these levels: a unit turned down here is turned down in the echo and reverb too.

10.9 TEMPO: tempo, swing and time signature



Figure 10.2 The TEMPO page in 7/8: encoder 2 turned the top number up to 7, encoder 4 the bottom one on to 8.

Encoder	Turn	Push
1 TEMPO	Beats a minute, 30 to 300	
2	The time signature's top number: beats in a bar	
3 SWING	OFF (50%) to 75%	Swings the semiquavers (1/16) or the quavers (1/8)
4	The bottom number: which note is a beat (2, 4, 8 or 16)	

Swing is chapter 5's: notes stay where they're written, and only the off-beat ones are played late.

The **time signature** is the one thing on Step that none of the chapters of music needed: every one of them is in 4/4, four crotchets to a bar, like almost all dance music. But plenty of music isn't. A waltz is 3/4, three crotchets a bar; "Take Five" by the Dave Brubeck Quartet is in 5/4; Pink Floyd's "Money" is in 7/4;

and a lot of Balkan folk music is in 7/8, seven quavers to a bar, felt as groups of 2+2+3. Changing it changes what a bar is, everywhere on Step: BAR moves, the keys' lights, COPY's ranges, and LOOP's bars all count in the new bars.

> **Try this**
>
> Start a new song (SONG, NEW). On TEMPO, turn encoder 2 up to 7 and encoder 4 on to 8: 7/8. On ENTRY, LENGTH 1/8, write a bar of seven quavers on Strike: KICK, rest, SNARE, rest, KICK, rest, SNARE. Set the track's LOOP to 1 BAR and play it. It limps, and then it doesn't: count it 1-2, 1-2, 1-2-3 and you'll start to feel it. Then try 5/4.

10.10 SONG, and SECTIONS



Figure 10.3 The SONG page of a new song: which song, its NAME, how long it is, and the ACTION a hold will do.

Encoder	Turn	Push / hold
1 SONG	Browses the 32 songs	Push: loads the one shown
2 NAME	The letter	Push: on to the next letter (up to 10)
3 LENGTH	(shows the song's bars and notes)	
4 ACTION	REMEMBER SOUNDS, NEW, DUPLICATE, DELETE	Hold: does it

> **REMEMBER SOUNDS** asks every unit which sound it's playing and keeps the answers with the song; loading the song, or pressing PLAY, sends every unit back to its sound. It keeps which sound, not changes made to a factory sound, so keep a changed sound as a sound of your own first (chapters 6 and 7).

> **NEW** starts an empty song; **DUPLICATE** carries on with a copy of this one, sounds and all (chapter 3); **DELETE** removes it.

> **ACTION stays where you left it.** Turn it each time to the one you mean, and look before you hold: one turn past DUPLICATE is DELETE.

> Songs save themselves as you work. There's no save button, and nothing is lost when you unsnap Step.

SECTIONS

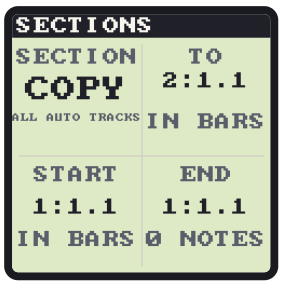


Figure 10.4 SECTIONS (SONG pressed again): the COPY page's jobs on every AUTO track at once, in bars.

SONG pressed again is SECTIONS: the COPY page's jobs, on every AUTO track at once, in whole bars (push encoder 2, 3 or 4 for beats). Encoder 1 chooses COPY, INSERT, DELETE or FILL; encoder 2 is TO (or UNTIL); encoders 3 and 4 are START and END. INSERT copies a section in and moves the rest of the song along; DELETE here closes the gap, unlike the COPY page's. Set-loop tracks are left alone, since they repeat by themselves. Chapter 5, section 5.15, builds a whole song with it.

10.11 SETUP

Step's SETUP is every unit's SETUP (chapter 1): THIS UNIT, with Step's MIDI channel, which is how a MIDI keyboard writes into Step, and its RESET page; and ROOT, with the Root's inputs, its MIDI OUT (which can carry Step's notes and clock to a computer or another instrument) and its MIDI IN.

10.12 Where each thing is taught

Chapter	On Step
3 Techno	New songs, tracks and their units, ENTRY, lengths and rests, set loops of a bar, COPY and TRANSPOSE, MIXER, mutes, velocity, two-bar loops, DUPLICATE
4 Acid	ACCENT and SLIDE, a loop in beats against the bar
5 House	CHORD, swing, REMEMBER SOUNDS, FILL, SECTIONS, DELETE: a song written out
6 Hip-hop	Writing another unit's keys (a sampler's slices), two-bar loops with a fill, remembered sounds and user sounds
7 Ambient	Loops of many lengths at once, clearing a track but keeping its loop, moving tracks to another unit
8 Jungle	Loops of 1, 2, 4 and 8 bars, transposing by an octave, keeping Step off ENTRY while setting up another unit
9 Synth-pop	The arpeggiator, LEGATO, set loops under a longer AUTO track, MIXER for units 5 to 8, a key change
10	The time signature

Next, the last chapter: taking it all out of the house, and playing it live.

Chapter 11 - Playing live

Your knobs from every unit on one box, your sounds a key away, and a set that's never played the same way twice.

Everything in this book can be played live. You've been doing it since chapter 3: every track mute at the right bar, every filter opened a click at a time, every echo thrown, was a performance, and no two of them were quite the same. This chapter is about doing that in front of people: preparing a set, keeping your hands on the few controls that matter, and Pilot, the unit made for it.

> **What you need for this chapter**
>
> **Pilot**, and the units of any chapter before this one. The examples use acid Track 3's: the Root, Step, Strike, Volt and Haze, with Pilot joined on the end. A chain takes eight units at most; with Step, Haze and Pilot, that leaves five for the sound.

11.1 Electronic music, live

Electronic musicians have always had to answer the question of what they're doing up there. Kraftwerk stood behind their consoles like engineers at a control desk, and made that the show. In the 1970s Suzanne Ciani gave concerts on a Buchla modular synthesiser, patching and playing it live. In the rave years a "live PA" meant the act's own machines on stage instead of a DJ's records: Orbital's set at Glastonbury in 1994, two brothers in head torches behind a wall of synthesisers and sequencers, is still talked about. Jeff Mills plays a TR-909 live inside his DJ sets, programming patterns on the fly; Daft Punk toured in 2006 and 2007 inside a lit-up pyramid, rebuilding their records live from their parts. They all do some version of what you've been doing: the music is mostly prepared, and the performance is in the choices, which part comes in when, how far a filter opens, when everything stops.

11.2 Pilot

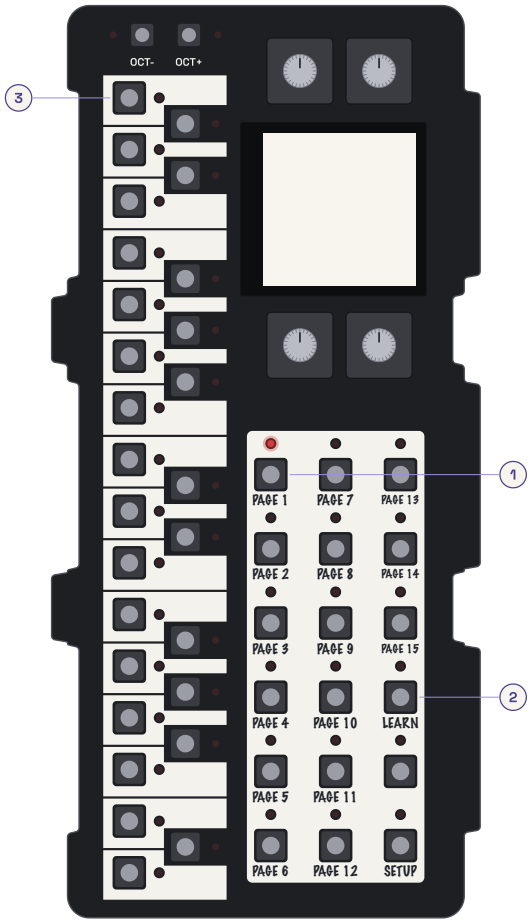


Figure 11.1 Pilot: fifteen knob pages (1), LEARN (2), and a scene on each white key (3).

Pilot makes no sound. It's a remote control for every other unit on the chain:

> **Fifteen knob pages (1)**, on the function buttons. Each page puts any four controls on its encoders, each from any unit: Volt's cutoff, Haze's echo, Step's tempo and a track mute, side by side.

> **LEARN (2)**, for filling a page: you turn the knob you want on the unit itself, and Pilot takes it.

> **Scenes (3)**, on the white keys: each key keeps which sound every unit is playing, and calls them all up at once.

Its encoders send "up three" or "down one", never a fixed value, so nothing jumps when you change page: each knob carries on from wherever its control already is. And its screen shows each control as its own unit shows it, in hertz, seconds or beats a minute.

11.3 A knob page

Load acid Track 3 on Step, and join Pilot to the end of the row. On VOLT, press ■ SOUNDS, then ■ ACID, so its ACID page is showing.

1. On PILOT, press ■ PAGE 1, then ■ LEARN. The screen says `TURN A KNOB ON ANY UNIT`, and the first box is outlined.
2. On VOLT, turn encoder 2, CUTOFF, one click. Pilot's first box fills with VOLT and ACID CUTOFF, and the outline moves on.
3. Turn Volt's encoder 3, RESONANCE, one click, then encoder 4, ENV MOD.
4. On STEP, press ■ TEMPO and turn encoder 1 one click.
5. On PILOT, press ■ LEARN again: done.

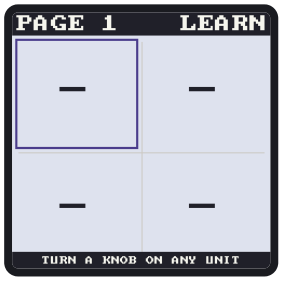


Figure 11.2 Pilot in LEARN on an empty page: the outlined box is the one the next knob you turn will fill.

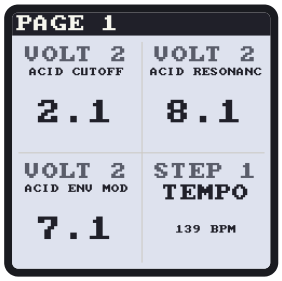


Figure 11.3 Pilot's page 1, learned: Volt's ACID CUTOFF, RESONANCE and ENV MOD, and Step's TEMPO, each shown as its own unit shows it.

Now press PLAY on Step, and play acid Track 3 with your hands on Pilot alone: the three knobs that make acid what it is, and the tempo, on one box. Turn the tempo back where it was with Pilot's fourth encoder.

> **A box takes a control once**, however far you turn the knob, then moves on, so turn each one just a click while learning.

> **To choose which box fills next**, push its encoder while LEARN is on. To empty a box, hold its encoder until the bar fills.

> **Holding a page's button** works as LEARN too, for as long as you hold it: quick for changing one box.

> A box says `MISSING` when its unit isn't on the chain, and comes back when it is.

11.4 A page of mutes

Every arrangement in this book was made with track mutes on Step's TRACKS page: turn to the track, push. Live, that's two moves for every change, and the wrong track is one click away. Put them on Pilot instead:

1. On PILOT, press ■ PAGE 2, then ■ LEARN.
2. On STEP, press ■ TRACKS. For tracks 1, 2, 3 and 4 in turn, turn encoder 1 to the track and push it twice: the first push teaches Pilot, the second puts the track back as it was.
3. On PILOT, press ■ LEARN again.



Figure 11.4 Pilot's page 2: the mutes of Step's tracks 1 to 4. Clockwise mutes, anticlockwise unmutes; here the hats and the clap are muted, ready to start the song.

Turn a Pilot encoder clockwise to mute its track, anticlockwise to bring it back. Now the kick, the hats, the clap and the line each have a knob of their own, and you can drop the kick out and bring it back without looking. Step's MIXER levels and mutes, and SWING, can go on a page the same way.

> ### Try this
>
> Make a page for Haze: on its SOURCE page, MIX, so you can ride the drive up through a breakdown as acid Track 3 does; and its echo's REPEATS. Make one for Strike: EDIT, a key, and the drum settings you'd like to play, an open hat's DECAY or the kick's TUNE. Give each page one job, and remember which is which: PAGE 1 the acid, PAGE 2 the mutes, PAGE 3 the effects.

11.5 Pilot's changes, and PLAY

Everything Pilot changes on a sound is a temporary edit, even on a sound of your own: the unit shows a star, and nothing you do in a set ever overwrites a sound you saved. That's what you want on stage, but it has one consequence worth knowing. When you press PLAY, Step sends every unit back to the sound the song remembers, and that drops every temporary edit. So press PLAY first, then play the knobs. If a song should start with a sound already changed, change it on the unit, keep it as a sound of your own, and remember that (chapter 6).

11.6 Scenes

A scene is which sound every unit is playing, kept on one of Pilot's white keys.

1. Choose a sound on every unit: say Strike on BOOM, Volt on ACID.
2. On PILOT, hold the first white key until the bar fills. Every unit's sound is kept there.
3. Choose other sounds: Strike on PUNCH, Volt on SQ SQUELCH. Hold the second white key.
4. Now press the first key, and then the second: all the units change at once.

The key of the scene last called up is lit, and keys with scenes are dim. A scene keeps which sound, like REMEMBER SOUNDS, not changes made to it: if a unit has temporary edits when you store, Pilot names it (`VOLT 2: EDITED, NOT KEPT`). Use scenes to change the whole band's sound in the middle of a song, on the first beat of a phrase, as if the record had changed.

11.7 Planning a set

> **Put the songs in order.** Step keeps 32, by number. Name them, and play them in order down the list: between songs, STOP, turn SONG to the next, push to load it, and PLAY. A long echo or reverb on Haze rings on through the gap.

> **Remember every song's sounds**, so loading a song puts every unit right, whatever you did to it in the song before.

> **Decide what you'll play by hand, and what plays itself.** A song written out (chapter 5) plays the arrangement for you, and leaves your hands free for the knobs; a song of loops and mutes (chapter 3) gives you the arrangement to play. Most sets mix the two.

> **Practise the moves**, the way you'd practise a piece on any instrument: the breakdown in acid Track 3 is eight bars of turning one knob at the right speed.

> **Know where the sound goes.** PHONES and its volume wheel are for you; LINE OUT, at full level, goes to the mixer or the speakers, and into a computer to record the set.

And play your own songs, not ours. The book's tracks taught you the tools; the set is where nobody else can tell you what comes next.

11.8 What you've learned

- That every track in the book was already a performance.
- Pilot's knob pages, learned by turning the real knob; a page of track mutes.
- Why Pilot's changes are temporary, and why to press PLAY before playing the knobs.
- Scenes: a whole band's sounds on one key.
- Planning a set: the songs in order, sounds remembered, what's played and what plays itself.

Last, where to go from here.

Chapter 12 - Where next

You know how tracks are made now. Here's where the road goes from here.

12.1 What you've learned

Seven kinds of music, three tracks each. You've built techno from four loops and a filter, made acid squelch, put chords and swing under house, chopped a break and your own voice for hip-hop, let chords drift past each other for ambient, sped a break up until it blurred, and written one section of a pop song out to the end, with a key change. On the way you've learned nearly everything Step does, and a good deal of music theory without ever sitting down to learn it.

And you've learned the craft the book's title promised. Go back to the introduction: underneath, a track is a few short parts going round, and what makes it music is what happens to them, what drops out and comes back, what builds, what's left out and when. You've done that in every chapter. Put on any electronic record now and you'll hear it being done.

12.2 Making Sounds

Every track in this book used the units' factory sounds, on purpose: so you could hear exactly what the book describes, and so the music, not the sound design, stayed the point. But you'll have noticed how often a track came alive when a sound changed: Taste's break going to twelve bits, Axis's pad turned darker, the lead given a quicker glide. Those were small turns of one or two knobs. Making Sounds, this book's companion, is about all the rest of them.

> **Your music will sound like you.** Factory sounds are shared by everyone who owns the unit. A track built from sounds you made yourself sounds like nobody else's, and in electronic music the sound is half of the idea.

> **It's one unit at a time.** Each chapter of Making Sounds takes one unit and builds a set of named sounds on it from scratch, so you learn the unit by making things on it, as this book taught Step.

> **It's the other half of the history.** This book told you where the music came from; Making Sounds tells you where the sounds came from: FM and the bell-like sounds of the 1980s, the analogue filter, the drum machines, sampling, grains.

> **It works on everything.** Filters, envelopes, operators and grains have the same names on every synthesiser there is. What you learn there, you keep, whatever you play next.

Making Sounds ends where this book begins, by pointing back here. Between them, you have both halves of electronic music: the sounds, and what to do with them.

12.3 There's plenty more music theory

Let's be honest about this book's theory. You've met a lot of it: crotchets and quavers, time signatures, semitones and octaves, scales and keys, chords and inversions, sevenths, voice leading, Roman numerals, relative minors, polyrhythm, a key change. That's a real foundation. It's also the beginning. Music theory is centuries deep, and the people who wrote the records you love mostly knew a great deal more of it than this, whether they learned it in a classroom or by ear. Here's some of what's next, and every bit of it will make you freer, never less free.

> **Song form.** Synthwave Track 3 had one eight-bar progression all the way through. Real songs have **verses**, which tell the story, **choruses**, which come back with the hook and usually their own chords, a **pre-chorus** to lift into them, and a **bridge** (in Britain often the "middle eight"), a different passage before the last chorus so it lands fresh. Learn to hear where each starts in songs you know, then build them on Step with SECTIONS.

> **Harmony, and how chords move.** Why does G to C sound like coming home? That pull, from the chord built on a key's fifth note back to its first, is a **cadence**, and there are several kinds, each a different way to end a phrase. Then there's the ii-V-I that runs through jazz, chords **borrowed** from the minor key into the major (and back), chromatic chords that slide a semitone, and key changes far stranger than chapter 9's.

> **Melody.** Motifs, short ideas repeated and changed; call and response; tension and release; where a tune should climb and where it should rest; how the words of a song and its tune fit.

> **Rhythm.** Syncopation, cross-rhythms from West African and Cuban music, the clave, odd time signatures that change from bar to bar.

> **Counterpoint**: two or more tunes at once, each going its own way, fitting together, as in Bach.

> **Reading music.** Written notation can look forbidding, but it's how most music of the last few centuries is kept, and being able to read it, even slowly, opens all of it to you.

How to learn it? The way most of the people in this book did: **work out music you love by ear.** Step is perfect for that, since step time lets you take as long as you like over every note. Pick a record, find its bass line one note at a time on Volt's keys, write it on Step, and play it against the record until it matches. Then the chords, then the drums. Do that for a dozen tracks and you'll have learned more than any chapter could teach you. A beginner's keyboard method, a good theory book or course, or lessons with someone who plays will take you further and faster. And remember house's rule: theory describes what people have found works. It doesn't forbid anything. If it sounds good, it is good.

12.4 An instrument for keeps

A Tessel is designed to go on being useful long after the last chapter of any book. Every unit can be reflashed as another kind whenever you like, and your sounds for each kind are kept, so a unit you bought for one job can do another next year and come back. New units snap on as you add them, one at a time, and the ones you have go on working with every new one. Everything you make is kept on the units themselves, and they make music without a computer. A Tessel doesn't need an account or a subscription, and it never will. It was made to be an instrument, and instruments are for keeps.

It isn't the only good instrument, and we'd be doing you no favours if we pretended it was. There are many other fine synthesisers, drum machines, samplers and grooveboxes, old and new, and a great deal of excellent music software, some of it free. Explore them. Every instrument pushes you in its own direction, and the ideas you've learned here, tracks and loops and mutes, filters and envelopes, chords and keys, work on all of them. We'd much rather you made music on anything at all than didn't make it.

12.5 With a computer, or without

Most electronic music today is made on a computer, in a **DAW**, a digital audio workstation: music software that records, sequences, edits and mixes everything, with synthesisers and effects inside it as software. What you've done in this book is the other way, often called **DAW-less**: hardware instruments, playing together, with no screen but their own. Neither is the right way. They're good at different things.

	Without a computer	**With a computer**
Good at	Hands on real knobs; playing live; making decisions as you go and keeping them; no distractions on a screen; starting a track in seconds	Editing every detail; unlimited tracks and undo; recording voices and instruments; mixing with precision; seeing the whole song at once
The catch	Limits: so many units, so many tracks, a small screen	Endless choice, which can be as paralysing as it is freeing; hours lost scrolling instead of playing

Limits are worth a second thought. Many musicians work without a computer precisely because of them: with fewer options you commit to a sound, a part or an arrangement and move on, and things get finished. Others want a computer for exactly what limits leave out: a singer's take, a guitar part, a perfect edit.

You don't have to choose for ever, or even for one track. A Tessel and a computer work together:

Recording. The Root's LINE OUT goes into a computer's audio interface, and any DAW records it. To record each unit on a track of its own, play the song once for each, with the others muted on Step's MIXER, and the DAW lines them up; then mix them there.

Playing in time together. Step's notes and clock go out of the Root's MIDI OUT (SETUP, ROOT, MIDI OUT), so a DAW listening on a MIDI input can follow Step's tempo. The other way, the Root's USB-C is a MIDI input from the computer (it carries no sound: that's LINE OUT's job), so a DAW's parts can play your units.

Sketching and finishing. Plenty of musicians start ideas on hardware, quickly and with their hands, and finish them on a computer, or play a computer-made track live with hardware on top.

12.6 A few things worth knowing

Finish things. A simple track finished teaches you more than ten brilliant loops left as loops. Give yourself a deadline, a week or an evening, and get to the end, even if the end isn't perfect. You'll do the next one better.

Listen on everything. A mix that's right on headphones can be wrong on a phone, in a car or on big speakers. Record your track, and play it on as many as you can before you call it done.

Share it. Play your tracks to friends, put them online, play them out. In Britain and most other countries, music you write is protected by copyright as soon as it's recorded, so it's yours; and remember section 6.1 about other people's.

Make music with other people. Join your MIDI to a friend's instruments, play to a singer, jam. Other people take your ideas somewhere you'd never have gone, and it's much more fun.

Look after your ears. Loud music damages hearing, slowly and for ever, and you'll want yours for the rest of your life. Keep headphones at a comfortable level and take breaks; if your ears ring after a session, it was too loud. Wear earplugs in clubs and at gigs: good musicians' earplugs make everything quieter without making it dull.

Keep listening. Every chapter's history named records to start with. Follow them: who influenced who, who sampled what, and the music next door, as chapter 7 did with the concert hall.

12.7 Last of all

The people who invented the music in this book didn't wait until they'd learned it all before they started: they learned by doing. Teenagers in Detroit and Chicago, kids in the Bronx and in London bedrooms, made it up with second-hand machines and whatever they had, copied what they loved, got it wrong in interesting ways and kept what worked. You have more help than they did, and better tools. The rest is yours. Make something nobody has heard before.

Quick reference

Every button, every page and every encoder on Step and Pilot.

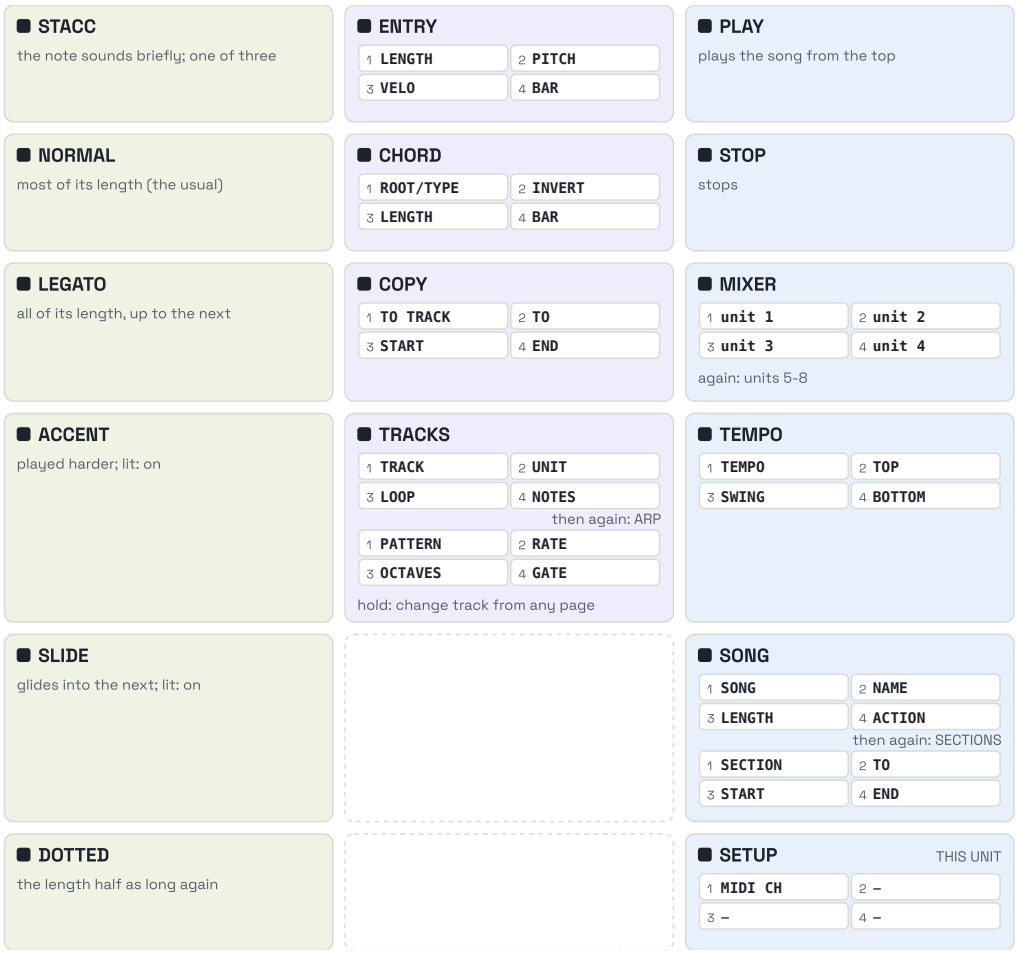
For when you know what you want and just need to find it. Each map shows the unit's buttons as they sit on the panel, with the four boxes of each page in the corners where they'll appear on the screen: encoder 1 top left, 2 top right, 3 bottom left, 4 bottom right. A button with a second page shows it underneath (press the button again to get there). Under each map, a line on every control. Every synth's own pages are in Making Sounds' quick reference.

Every unit

SOUNDS the SOUNDS button	
BANK	which bank: FAC A… (the factory's) or USER A… (your own)
SOUND	which sound in it; it plays as soon as you reach it (on Strike: KIT)
COPY TO	turn to choose one of your slots; hold the encoder to copy the sound there
NAME	your own sounds: turn to change the letter, push for the next letter
THIS UNIT SETUP (the first pages)	
MIDI CH	the unit's MIDI channel: AUTO, or one you choose (never changed after)
2–4	the unit's own settings, listed under each unit below
RESET SETUP, the last THIS UNIT page	
RESET	encoder 1: SETTINGS, EVERYTHING, REFLASH (become another kind), another type kept in the unit's flash, EVERY TYPE, TOTAL RESET (wipe everything: a blank unit) or TEST (the unit's own test, then back as it was)
FIRMWARE	the firmware version; on REFLASH, BECOME: which kind, from the ones the Root carries
RESET	encoder 4: hold for three seconds to do it
ROOT SETUP, after THIS UNIT (the Root's own settings)	
INPUT	which input units record from: LINE, MIC (with mic power) or BUILT-IN; also on Taste's and Grit's recording pages
GAIN	how loud the input comes in, in dB; also on Taste's and Grit's recording pages
MIDI OUT	OFF, NOTES (Step's), CLOCK (Step's tempo), BOTH, THRU (a copy of MIDI IN) or ALL
CARD	turn to choose, hold to run: BACKUP (every unit and the Root, into a new BACKUP-n), RESTORE (the newest backup back into the units here), FORMAT (empties the card)
ROOT2 SETUP, the last page	
MIDI IN	CHANNELS: each unit plays its own channel; FOLLOW: notes go to the unit last touched
FROM CH	in FOLLOW, the channel the Root listens on
SOUNDS	UNITS' (each kind's sounds from its leftmost unit) or ROOT'S: for a Root that already has sounds on it

KEEP	encoder 4: hold for three seconds to do SOUNDS' choice
KEYBOARD while OCT- or OCT+ is held	
OCTAVE	moves the keys by octaves (also: tap OCT- or OCT+)
SEMI	moves the keys by semitones
LOWEST	the lowest note the keys now play
HIGHEST	the highest; press OCT- and OCT+ together to put them back

Step



Keys: on ENTRY and CHORD they write notes into the open track; elsewhere they play its unit. White keys light one bar of the open track (dim: a note, bright: the cursor, blinking: playing); black keys count the bar. OCT- and OCT+ move the keys and CHORD's chords by octaves. The left column works only on ENTRY and CHORD, and shows the note at the cursor.

ENTRY writing notes into the open track	
LENGTH	1/16T, 1/16, 1/8T, 1/8, 1/4T, 1/4, 1/2T, 1/2, 1; push: a rest; hold: a tie
PITCH	the note shown (or moves the note at the cursor); push: writes it
VELO	how hard notes are written, 1-127, from any unit's keys too

BAR	moves the cursor a LENGTH at a time; push: INSERT or REPLACE; hold: deletes the note at the cursor
CHORD chords by name	
ROOT/TYPE	the root, C to B, or the type, major to 7sus4, whichever is underlined; push: switches
INVERT	ROOT, 1ST, 2ND, 3RD: the chord turned over
LENGTH	as ENTRY's
BAR	moves the cursor; push: places the chord
COPY a range of the open track; push encoder 1: COPY, TRANSPOSE, DELETE, FILL	
TO TRACK	COPY: the track to copy to. TRANSPOSE: SEMITONES, -12 to +12. FILL: how many times
TO	COPY: where the copy starts. FILL: UNTIL, the last beat filled; push 2-4: sixteenths
START, END	the range, START's beat to the end of END's; turn END first
hold any	does it: copies, transposes, deletes (leaving silence) or fills
TRACKS the open track's settings	
TRACK	which track, 1-16; push: mute or unmute it
UNIT	which unit it plays
LOOP	AUTO (part of the song) or a set loop in bars; push: in beats
NOTES	how many; hold: clears the track (its loop is kept)
ARP TRACKS again: the track's arpeggiator	
PATTERN	OFF, UP, DOWN, UP/DN, PLAYED, RANDOM
RATE	one note every 1/16T to 1
OCTAVES	1-3
GATE	how much of each step sounds, 10-100%
MIXER the units' levels, four at a time	
each unit	turn: its level, OFF and -48 dB to +6 dB; push: mute. Kept with the song
TEMPO the whole song's time	
TEMPO	30-300 beats a minute
TOP	the time signature's top number: beats in a bar
SWING	OFF (50%) to 75%; push: the semiquavers (1/16) or the quavers (1/8)
BOTTOM	the time signature's bottom number: 2, 4, 8 or 16
SONG the 32 songs	
SONG	browse; push: load the one shown
NAME	the letter; push: the next letter (up to 10)
LENGTH	the song's bars and notes (shown only)
ACTION	REMEMBER SOUNDS, NEW, DUPLICATE, DELETE; hold: do it. It stays where it was left

SECTIONS	SONG again: COPY's jobs on every AUTO track at once
SECTION	COPY, INSERT, DELETE (closing the gap) or FILL
TO	where it goes (FILL: UNTIL), in bars; push 2-4: beats
START, END	the section, END's bar included
hold any	does it

Pilot



White keys: scenes. Press: every unit to the sound kept there; hold until the bar fills: keep every unit's sound there now (temporary edits aren't kept, and Pilot names the unit). Lit: the scene last called up; dim: keys with scenes. Black keys: not used. A knob can be any control on a sound unit's pages, any of Haze's effect settings or its MIX, or Step's TEMPO, SWING, MIXER levels and mutes, and track mutes.

PAGE 1-15	four controls from any units
each knob	turn: moves that control on its unit, a step at a time (Step's mutes: clockwise mutes, anticlockwise unmutes). Shows the unit, the control and its value; MISSING: the unit isn't on the chain
in LEARN	push: this box fills next; hold: empties it

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  kept on the Root 1.9